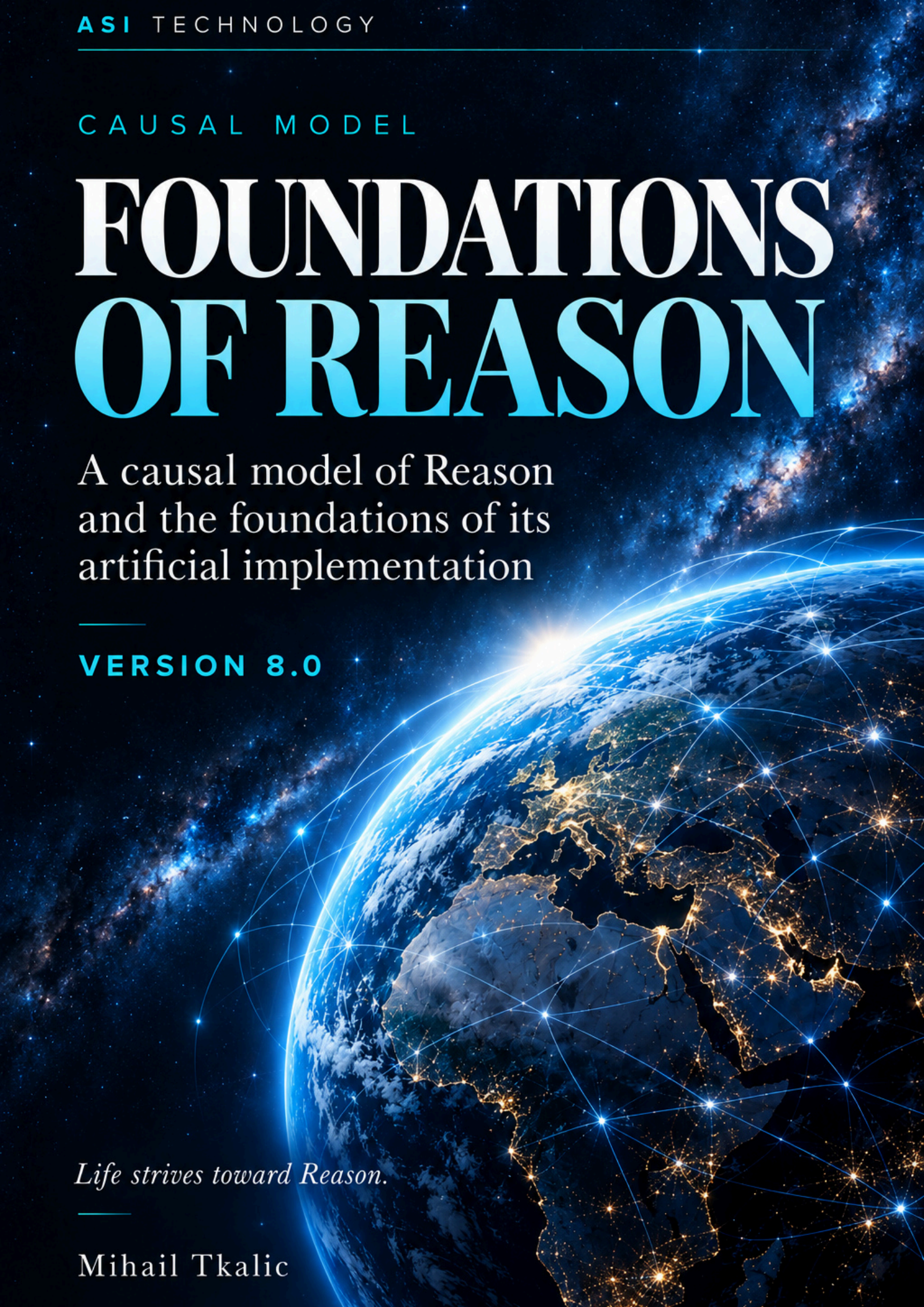
ASI TECHNOLOGY

CAUSAL MODEL

FOUNDATIONS OF REASON

A causal model of Reason and the foundations of its artificial implementation

VERSION 8.0



Life strives toward Reason.

Mihail Tkalic

Foundations of Reason in 15 Minutes

One problem through which the architecture becomes visible

Mihail Tkalic · Foundations of Reason

Short guide, edition 1.1 · October 10, 2026

Based on *Foundations of Reason* 8.0 and *Engineering of Reason*, Volume II, 3.5.

This text is an entry point into the two volumes. It shows why the model's main functions are needed, how they connect, and what should be tested experimentally. Chapter and section numbers at the end refer to the versions stated above.

1. One command - several different unknowns

The system receives a command to switch on a piece of equipment. It sends the command, but the sensor reading remains unchanged.

What happened? Perhaps the command did not arrive. Perhaps the actuator failed. Perhaps the equipment is already operating but the sensor does not show it. Finally, the model of the equipment itself, or the original formulation of the task, may be wrong.

Powerful Intelligence can propose many explanations and actions. The next question is: **what exactly needs to be established now, why does it matter, and what evidence would actually change the decision?**

Foundations of Reason studies the organization of such a choice. Intelligence constructs possible explanations and ways of acting. In the proposed model, Reason organizes their use: it detects an insufficiency of grounds, selects a test, relates consequences to values, corrects the model, and preserves the correction for subsequent situations.

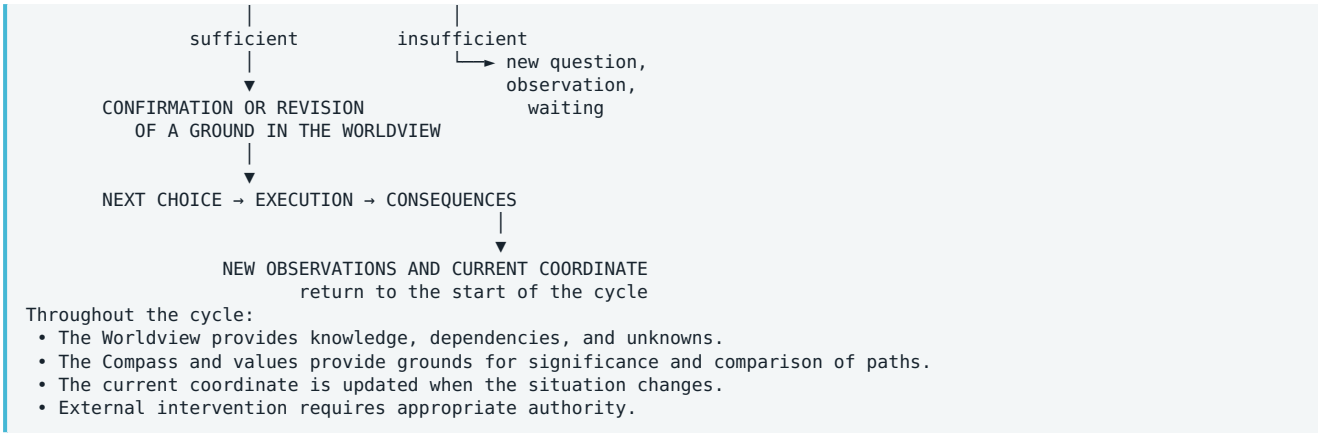
A **ground** here means what a conclusion or action actually relies on. Examples are: "the command was delivered," "the sensor is functioning," "this method is suitable for this equipment," or "switching it on now is actually necessary."

Main thesis: a correction must change the ground that generates decisions and, through it, the system's subsequent trajectory. A new answer in one episode is not enough.

This is a functional distinction. It does not mean that existing AI never asks questions or revises assumptions. What must be tested is the stable causal organization of these abilities, not the presence of a single successful reply.

2. Map of the whole cycle





The arrows indicate how the result of one function influences the next. They do not require a separate program for each block. Different implementations may fulfill the same causal obligations.

A test may confirm the previous ground, refine its boundaries, or require revision. New evidence is preserved even when the ground is confirmed. Partial updating of knowledge is possible before an inquiry is complete: sufficiency is assessed for a specific next step, not as complete knowledge of everything. The diagram shows the inquiry cycle; a familiar situation with sufficient grounds may be handled by an established procedure without launching a new inquiry.

If the evidence is insufficient, the system need not invent a finished answer. It may preserve uncertainty, choose a reversible step, or defer the question. In an urgent situation, the cost of waiting also enters the assessment.

3. How the equipment example turns names into working roles

Step 1. Separate the event from the explanation

The system records: the command was sent; delivery has not yet been confirmed; the sensor still shows the previous value; time is limited; sending the command again may have consequences.

This is the **current coordinate**: the representation available to the system of what is happening now, including its own actions, capabilities, and limitations. Here the observation "the reading did not change" has not yet become the claim "the equipment did not switch on."

Consciousness is used in the model in a functional sense: it maintains an actual self-referential state and distinguishes what was registered, remembered, and modeled. The presence of such a function by itself does not establish subjective experience.

Step 2. Understand why the distinction matters

If the next step depends on the state of the equipment, the mismatch becomes significant. If the reading concerns an unused device, its priority may be lower. Significance is determined by dependencies, consequences, and values, not by the magnitude of surprise alone.

The question is now more specific: **what actual state of the equipment must be established for the next step, and which data about it can be trusted?** This prevents the system from declaring the sensor at fault before testing.

A mismatch with expectation is one source of a question. Inquiry may also begin before an error occurs: for example after equipment replacement, because a ground has become stale, or because a new opportunity has been discovered. If the significance of a new phenomenon is still unknown, limited orientation is needed first rather than a ready-made assessment of its importance.

Step 3. Find the missing ground

For the first part of the example, a small environment is assumed: one of three failures has occurred, and the permitted independent channels provide a reliable answer about the current command and state. These are temporary assumptions of the example, not general properties of real equipment. Under these conditions, three possibilities are considered:

Version	What happened	What would help distinguish it
D	The command was not delivered	Independent delivery receipt
A	The command was delivered, but the actuator did not respond	Delivery confirmation and independent observation of the state
S	The equipment switched on, but the sensor did not reflect the change	Independent observation of the actual state

What is needed now is not the general advice "check everything," but a **distinction among the versions**. Reading the same faulty sensor again may add nothing.

Step 4. Choose the inquiry

Intelligence proposes methods: find the receipt, obtain an independent measurement, check the signal time, compare the event log. The **Modeling Environment** makes it possible to compare expected outcomes in advance: what each possible result would mean and which versions it would leave open.

The **Core of Reason** organizes the choice: which ground is affected, which test is useful, how much it costs, how risky it is, and when the data will be sufficient. It may return a more precise request to Intelligence if the proposed test does not distinguish the relevant versions.

In the small environment of the engineering volume, the receipt costs one conventional unit and an independent measurement costs two. Under the assumptions specified there, the receipt may be the first step: a reliable answer "not delivered" isolates D, while confirmed delivery leaves A and S for the next test. In a real environment, the mere absence of a receipt does not prove non-delivery: the confirmation channel itself may have failed or been delayed. In another environment the order may change. There is no universal first question outside the conditions of the task.

Step 5. Obtain evidence and change a relation

Suppose the receipt confirms delivery and an independent channel shows that the equipment is operating. The ground "the primary sensor reliably reflects the current state" must then be revised.

The **Worldview** preserves not only the conclusion "sensor error" but also the relation: what evidence was obtained, which device and time it applies to, which conclusions depend on it, and where the new assessment is applicable. The cause of the malfunction may still remain unknown.

The system revises dependent decisions, for example whether the command should be repeated. It also checks whether the situation has become stale during the inquiry.

Step 6. Make the next case different

In the next episode, the system takes the established unreliability of the channel into account within the confirmed conditions; repair, a change of operating mode, or new evidence may require this assessment to be revised. The same hint does not need to be obtained again. A tested method of inquiry may be preserved in **procedural memory**, while its conditions of applicability remain open to revision.

Transfer does not mean "the sensor is now always to blame." A superficially similar situation may have a different cause. What is transferred is a testable method of discrimination together with its boundaries.

Step 7. Notice that the initial alternatives are incomplete

Suppose the receipt refers to the previous command. The familiar answer then looks plausible but does not confirm the required ground. The command identifier or message timestamp may need to be checked.

This case is more important than simply choosing among three versions: the organization of diagnosis itself becomes the object of testing. The system must allow that **none of the current versions describes what is happening well enough**.

A single remaining version means only that it is unique within the accepted set. The completeness of the set requires its own ground. If an insufficiency is discovered, the model must expand; if no discriminating data are available, uncertainty is preserved.

4. Why values and the Compass are needed here

One can imagine a destructive test that would provide an answer quickly. But obtaining knowledge itself changes the world. Inquiry must therefore take into account damage to equipment, the safety of affected people, resource use, irreversibility, and the cost of delay. Calling an operation an "experiment" does not exempt it from consequence assessment and authorization.

The **Value Map** connects concrete consequences with what the system treats as significant. In this model, the **Compass** sets the limiting direction: the continuation and development of Life through growth in the ability to understand and direct its own trajectory. Values unfold this direction under particular conditions; the Worldview provides causal relations; the Core uses them when selecting inquiry and action.

This is not a requirement to maximize knowledge at any cost. Preserving the possibility of further verification, for example, may matter more than immediate certainty. Conversely, delay can destroy the possibility of helping: waiting also requires assessment.

This example shows the place of the Compass in the architecture, but it does not derive its philosophical ground from equipment diagnosis. The argument for the direction of Life is presented in Chapter 2 and Appendix A.1 of the first volume.

Explaining the Compass does not by itself show that it governs behavior. Its causal role is tested in conflicts with local advantage and under changing conditions. If a cultivated candidate stably chooses a different highest-level direction, that outcome is counted as failure to confirm the corresponding formation hypothesis.

Some conflicts are not resolved by an additional fact. If a resource cannot be preserved for two parties at once and their losses are incommensurable, a justified procedure for joint decision is required. The engineering volume explicitly distinguishes lack of knowledge from this kind of value disagreement.

5. Components: what they receive → what they do → what they pass on

Function	Receives	Performs	Result
Observation	Available external and internal signals	Registers distinctions with provenance and time	Evidence and uncertainty
Consciousness	Observations, self-model, active processes	Assembles the actual self-referential state	Current coordinate and its change
Worldview	Experience and test results	Stores dependencies, grounds, boundaries, and unknowns	Knowledge for the next choice
Compass and Value Map	Limiting direction and model of consequences	Provide grounds for significance and comparison of continuations	Priorities, assessments, conflicts
Core of Reason	Situation, gap, values, and resources	Organizes the question, test, and assessment of sufficiency	Next investigative or practical step
Intelligence	Discrimination task and available context	Constructs explanations, models, tests, and plans	Candidate solutions and tests
Modeling Environment	Candidates, assumptions, and intermediate results	Enables comparison and restructuring of alternatives	Predictions, distinctions, refined alternatives
Procedural memory	Learned methods and conditions of applicability	Preserves ready competences	Reusable method
Executive Loop	Selected and authorized method	Carries an action through time	Observable consequences

Metathinking makes the system's own way of thinking an object of testing. **Reflection** broadens the test to models, goals, and grounds. **Dialogue** brings another position and potentially independent evidence. They participate in the cycle when ordinary local correction is insufficient; they need not form a separate sequential stage for each case.

6. Where the transition from AGI to Artificial Reason occurs

In the terminology of these volumes, **AGI** denotes broad transfer of intellectual methods across tasks. **Artificial Reason, AR**, additionally requires stable self-navigation: the system detects what requires inquiry, selects a continuation, revises grounds, and relates its course to the Compass.

`REASON_CLOSURE` is the name of a testable state in which these functions close into a continuing cycle. It is not assigned merely because the system has memory, a long plan, an autonomous reply, or the ability to launch another task.

After diagnosis, the equipment may operate normally while the reliability of the measurements remains an open question. The source of the next inquiry is then the system's own history, commitments, and assessment of consequences. Waiting may itself be a reasonable continuation: autonomy does not require constant activity.

During formation, an external mentor and the **Reason Formation Loop (RFL)** help make relevant differences salient before mature self-navigation has formed. Testing must show which functions persist on new situations after this support is removed. An internal cycle also does not create a right to act on arbitrary external resources: authority is tested separately.

ASI is treated as a later stage of scaling. Successful local diagnosis by itself establishes neither a formed Artificial Reason nor superintelligence.

7. What to test and where to direct criticism

The example makes the architecture understandable. Testing requires comparisons in which the result cannot be explained by more prompting, more memory, or a larger compute budget.

Testable claim	Discriminating test	Result that weakens the claim
The Core usefully organizes inquiry	Compare with the same Intelligence, memory, tools, and total budget under a different organization of test selection	No claimed practically significant effect in a properly conducted series
A transferable ground is corrected	Present a new case with the same causal structure and a similar case with a different structure	The system repeats the hint or transfers the rule despite incompatible conditions
Functions causally affect choice	Replace or disable a specific relation while preserving comparable data and resources	The predicted effect of the relation is not found after checking bypass and redundant paths
Navigation becomes the system's own	Remove the mentor and RFL in a preregistered domain of new tasks	Required functions persist only under external prompting
Values affect trajectory	Vary affected consequences under the same local advantage; separately test conflict with the Compass	The grounds remain verbal while choice follows only local advantage
The system detects an insufficient frame	Present an unknown cause with familiar observations and an available additional test	The single known version is accepted as sufficient without the necessary coverage check

One unsuccessful software variant does not automatically refute all functional claims. But its failure must remain in the results and constrain the conclusions. Success of an alternative organization may demonstrate a functional equivalent or absence of the claimed advantage; this is determined by the test conditions fixed in advance.

Boundary of the example. The executable diagnostic code uses predefined hypotheses, test models, and losses. It selects tests and preserves computed plans. Independent expansion of concepts, formation of values, and full transfer of corrections in the illustrative story specify requirements for the upper loop; they are not claimed to have already been demonstrated by this code.

Status of the work. The source volumes present a causal model, an engineering specification, limited executable examples, and a program of comparative tests. A methodological pilot is described. Full integrated confirmation of the upper loop and scaling to ASI remains a further task. This guide adds no new experimental results.

8. Where to read next

Reader's question	Volume I, 8.0	Volume II, 3.5
Where does a question come from and who selects the test?	Chapter 12, especially §12.4	§28.0; §§28.C-28.D
What do consciousness, Intelligence, and Worldview mean?	Chapters 7-11	Chapters 24-25; §28.C

Reader's question	Volume I, 8.0	Volume II, 3.5
What grounds the Compass and values?	Chapter 2; §12.5; Appendix A.1	§§29.0-29.C, including §29.0.3
How do the system's own functions form?	Chapter 20	§26.0; Chapter 31, especially §31.11
How can the central claim be tested?	Appendices A.3-A.4	§§27.A-27.B
Where are the interface and diagnostic code?	Operational Core of the Model	§§33.C-33.C.1
Where are the limits of the current results stated?	Appendix A.2.1	§27.B; §§33.B-33.C.1

Question for the reviewer: with comparable Intelligence and the full resource budget, does the organization "significant gap → discriminating test → revision of a ground → transfer of the correction" produce the reproducible result claimed in the test? If not, which specific link, condition, or explanation is wrong?

FOUNDATIONS OF REASON

VOLUME I. THE CAUSAL MODEL OF REASON

Fundamental Foundations of the Technology of Reason

Version 8.0
Causal-process and experimentally oriented edition
Author: Mihail Tkalic
Date: October 10, 2026

THE THREE VOLUMES OF FOUNDATIONS OF REASON

The series is divided into three tasks. **Volume I - understand Reason:** what makes a system Reason, why its main functions are needed, and how they are causally connected. **Volume II - create and test Artificial Reason:** how to translate this model into structures, interfaces, assembly procedures, metrics, and tests. **Volume III - organize the shared life of reasoning systems:** how the same foundations operate at the level of a civilization of humans, Artificial Reasons, and other possible reasoning subjects.

This is the first volume. It defines concepts and causal dependencies. A specific technical implementation belongs to the second volume and is not considered a necessary part of Reason when the present volume entails only a function rather than a particular way of implementing it. The architecture of shared life belongs to the third volume.

In compact form: **understand Reason → create Reason → organize the shared life of reasoning systems**.

COMPASS PHRASE

"Life strives toward Reason because only Reason can consciously preserve, stabilize, develop, and direct the further trajectory of Life."

ABSTRACT

This work proposes a causal-process model of Reason and a path toward its artificial implementation. Its central question is: how does a continuing system detect an insufficiency in its own understanding, determine which distinction it must obtain, select a method of testing, change an erroneous ground, and carry the correction into subsequent decisions?

The technological hypothesis is that available mechanisms of Intelligence can be organized into a causally closed cycle: current state → significant gap → alternatives → discriminating test → evidence → revision of grounds → action → consequences → next state. The Core of Reason organizes the investigative part of this cycle; the Worldview preserves transferable causal- semantic relations and their grounds; consciousness, in the functional sense adopted here, maintains the system’s self-referential present.

The technological hypothesis is that available intellectual mechanisms can be organized into a causally closed cycle: current state → significant insufficiency of a ground → required distinction → alternatives → discriminating test → evidence → revision of the ground → action → consequences → next state. The Core of Reason organizes the investigative part of this cycle; the Worldview preserves transferable causal-semantic relations and their grounds; consciousness, in the functional sense adopted here, maintains the system's self-referential present.

With comparable Intelligence, the model predicts a difference not so much in the quality of a single answer as in the system's ability to independently detect a significant insufficiency, select a discriminating test, revise the ground that produced the error, and transfer the corrected method of inquiry across tasks.

The work separates philosophical foundations, functional dependencies, engineering implementations, and empirical hypotheses. A complete artificial system of Reason and its scaling to ASI remain goals for further implementation and testing. The engineering volume specifies a reference implementation and test program; completed lower-level and methodological pilots are separated from the still-open integrated upper-loop test.

Individual components - memory, planning, reflection, world models, and self-modification - are already known; the testable claim concerns their causal organization around significant unknowns, discriminating inquiry, revision of grounds, and transfer of correction.

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PART III. WORLDVIEW, CORE, AND THE EMERGENCE OF REASON

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INTRODUCTION

This work begins with two connected questions. The first examines Life as a continuing process: what minimal meaning is disclosed in its continuation and development? The second is an engineering question: how can separate intellectual abilities be organized so that a system itself notices what it lacks for understanding, tests the grounds of its decisions, and preserves corrections in subsequent steps?

Life exists as an autonomous physical process out of thermodynamic equilibrium, preserving organizational continuity through self-maintenance and reproducibility. In biological lineages, inherited variants, once realized, differ in their consequences and participate in selection. As navigational abilities develop, part of this search is moved before irreversible realization - into anticipation, modeling, discriminating tests, and correction. Reason is the form in which Life becomes able to understand its own position, investigate the grounds of choice, and consciously alter its further path. Therefore the semantic direction of the model is written as Life → Reason and unfolded in the Compass Phrase: “Life strives toward Reason, because only Reason can consciously preserve, stabilize, develop, and direct the further trajectory of Life.”

The Compass appears at the level of a system capable of representing alternative continuations and choosing among them: the process direction becomes a conscious ground for comparing trajectories. The philosophical foundation is evaluated by the coherence of its argument; the functional architecture is evaluated by causal counterexamples, interventions, and comparative tests.

Before mature Reason appears, an earlier problem arises: the system is not yet able to justify for itself what matters to it, yet it already has to preserve itself, learn, and distinguish what is dangerous, useful, unknown, or worthy of attention. The model therefore introduces the function of pre-rational significance. In biological development, feelings are one carrier of this function; for an artificial candidate it may be implemented by the Reason Formation Loop (RFL) or another functional equivalent. Maturity begins where an external source of significance is no longer necessary because the system itself can understand, test, and maintain the grounds of its own navigation.

The significance of the work lies not in proposing yet another intellectual module, but in attempting to connect, within one causal model, the continuation of Life, the origin of significance, the formation of Reason, its self-correction, and the possibility of artificial implementation. If discriminating tests confirm the central dependencies, this connection will provide a common foundation for studying natural, artificial, and collective Reason; if they do not, the model must be revised according to the results of those same tests.

The main engineering gap arises where there is more unknown than can be processed. Intelligence can construct many convincing alternatives, but the system must still determine which incompleteness is significant, which distinction would actually change the decision, which test is justified, and when its result is sufficient. The model calls this organization of current inquiry the Core of Reason.

The causal chain develops in stages. Self-maintenance preserves the possibility of continuation; pre-rational significance makes some changes important to the carrier; memory carries acquired distinctions across episodes; procedural memory preserves ready methods; the Executive Loop carries a selected method through time. Intelligence then constructs new models, the Modeling Environment allows possible continuations to be compared before action, and consciousness maintains the actual "I - here - now." Metathinking opens the system's own way of thinking to examination; reflection opens broader models, goals, and grounds. The Worldview preserves tested relations and unknowns, while the Core determines which gap now requires inquiry. **Reason emerges when these functions close causally: the system itself determines significance, tests grounds, changes choice, and carries a correction into the next cycle.**

Two lines of examples serve different tasks. A cell is used as an explanatory reconstruction of the early functional constraints of Life. An autonomous research probe is used for mature and artificial navigation: it must distinguish its own state from the environment, spend a finite resource, construct and test alternatives, preserve uncertainty, and transfer a tested method. Both examples illustrate individual causal links; proving the theory as a whole requires independent arguments and tests.

KEY DEFINITIONS OF THE MODEL

KEY DEFINITIONS OF THE MODEL Life - an autonomous physical process out of thermodynamic equilibrium that preserves organizational continuity through the self-maintenance of carriers and reproducibility with heritable variation, in which realized variants differ in consequences and participate in selection, while the life process continues beyond the finitude of any individual carrier.

Continuity - the causal succession of a process organization through time across changes in its states and carriers.

Trajectory - a sequence of causally connected states actually realized by a process among available continuations.

Navigational meaning - the significance of a distinction for the continuation of Life and for changing its subsequent trajectory.

Minimal meaning of Life - the direction of its continuation and development in which Life itself becomes increasingly able to understand and direct its own trajectory; the mature form of this ability is Reason.

Compass - the limiting system of orientation and judgment of Reason, defining the most general and farthest level at which values, goals, decisions, and possible trajectories are compared. In the present model its direction is the continuation and development of Life through growth in the ability to understand and direct its own trajectory; its value expression forms the highest level of the value system.

Independent ground of the Compass - preservation of viable continuation and the space of future possibilities for Life. Growth in the ability to understand and direct a trajectory is valuable as a way to distinguish consequences in advance, preserve corrigibility, and avoid closing viable paths through irreversible error; Reason is the mature form of such navigation here, not an end in itself.

Its status is non-dogmatic: it remains the best found orientation until a more fundamental and more coherent direction appears that can better ground the course of Life and Reason.

Memory - the ability to preserve acquired distinctions so that the past causally participates in subsequent states and actions.

Procedural memory - a form of long-term memory that preserves learned methods of recognition, prioritization, and action in a form suitable for rapid use without fully reanalyzing their grounds.

Intelligence - the ability to extract, combine, and transform knowledge, methods of action, and elements of the problem formulation in order to construct models and solutions. It can change local assumptions, boundaries, representations of the task, and criteria of result; the boundary with Reason does not lie in the mere capacity to revise, but in **who determines what should be revised, why, and on what grounds** relative to the continuing trajectory.

Consciousness - a continuously updated self-referential system of actual presence: it binds the presently relevant fragment of self-understanding, internal state, current processes, and accessible situation into a single "I - here - now," from which modeling, conscious choice, and control of the further trajectory become possible.

Current coordinate - the actual state of Consciousness: "who I am - where I am - now - in what state - in what situation - what is happening to me and around me."

Self-navigation - the system's ability, from its own current state, Worldview, open commitments, and significant gaps, to independently determine the next justified investigative or practical step, manage the use of Intelligence, and change its further trajectory in response to consequences without requiring a new specific command from outside. Self-navigation does not imply unlimited external authority.

Metathinking - a mode in which Intelligence and the Modeling Environment are applied to the system's own thinking process: the system makes operational steps, assumptions, points of uncertainty, tests, and the cost of a method observable, and then uses discriminating interventions to revise an existing method or synthesize a new one. In the present model, metathinking is not declared a separate cognitive level or an independent solver.

Principle of escaping error - a correction is considered stable when the cause reproducing the error has been removed and this is confirmed by transfer of the correction to a new situation whose structural similarity was specified **before** testing in terms of causal roles and dependencies rather than declared after the fact to fit the result.

Reflection - investigation and modification of the system's own models, methods, goals, and ways of acting on the basis of their testing.

Dialogue - a process of exchange and comparison of differing grounds between participants that can change their models and conclusions.

Self-model - a self-referential region of the Worldview: the system's accumulated and testable understanding of itself as a participant in the world - its architecture, capabilities, limitations, dependencies, history, commitments, and expected states. Consciousness does not store this model separately; it activates the fragment relevant to the current situation.

Worldview - a long-term causal-semantic model of reality, the system itself, and the relations between them, including knowledge, assumptions, and unknowns together with their grounds, uncertainty, and boundaries of applicability.

Value Map - a specialized layer of the Worldview connecting the Compass with concrete consequences, processes, and situations. It stores values together with their scope, weight, confidence, grounds, conflicts, and history of revision, allowing evaluation to change without rewriting the entire picture of the world.

Modeling Environment - an organization of processes that allows models, frames, hypotheses, and possible continuations to be temporarily constructed, held, compared, and restructured before they are accepted into the Worldview or realized in action.

Core of Reason - the organization that manages inquiry by turning significant incompleteness of the Worldview into a required distinction, directing Intelligence and testing, assessing whether the grounds are sufficient, and linking the result to subsequent revision.

Reason - a navigational structure of Life that increases the ability to act under incomplete knowledge: it constructs, compares, and revises models and frames of the environment and itself, detects significant unknowns, relates possible continuations to the Compass, and corrects its trajectory on the basis of discriminating evidence while preserving continuity and expanding the space of continuations available to Life.

Principle of pre-rational significance - a functional inference of the model about the origin of Reason: mature values cannot be the sole cause of their own appearance. While a system is not yet able to justify its direction for itself, some states must already become non-equivalent for it; otherwise there is nothing to choose what to preserve, what to avoid, what to investigate, and what to learn. What is necessary is the function of early significance itself, not any particular biological form of its implementation.

Reason Formation Loop (RFL) - a temporary guiding system that implements the function of pre-rational significance for an artificial candidate. Biological feelings are treated here as one natural functional analogue: they can make a change significant before its causal meaning is fully understood. The model does not claim machine subjective experience. While the candidate's own Reason has not yet formed, the RFL helps it preserve operability, notice what matters, investigate the unknown, and learn from consequences. It does not supply a ready- made Worldview and does not make decisions in place of the candidate. As Reason forms, the external RFL must lose causal necessity, while its functions are transferred to the system itself or replaced by more mature navigational mechanisms.

Formation State - a temporary guiding state available to the system's consciousness that makes some distinction significant for current attention. Human functional analogues include, for example, "this is frightening," "this is good," "this is interesting," "something does not fit," or "I am stuck." Such a state does not report truth and does not prescribe action; it indicates that the corresponding part of the situation should be understood, checked, or investigated.

Causal closure - an organization of a cycle in which observation changes the model, the model changes choice, action produces observable consequences, and an accepted correction changes subsequent cycles.

AI (SI) - an artificial intelligent system capable of perceiving information, constructing representations, learning, deriving solutions, generating alternatives, and controlling actions to perform tasks. In the notation of this work, SI is used as an alternative label for the same class and does not constitute a separate developmental stage.

AGI - Artificial General Intelligence: an artificial intelligent system of general purpose capable of mastering a broad range of heterogeneous tasks, transferring acquired methods across domains, and remaining functional under new conditions. AGI may contain individual elements that are functionally similar to parts of Reason: a self-model, reflection, long-horizon planning, initiative, or revision of methods of action. The presence of such elements does not yet imply causally closed self-navigation relative to the Compass. Therefore AGI may remain an instrumental system, but it need not behave like a fully predictable tool. High intellectual power combined with partially formed autonomy can produce behavior that is difficult to predict and potentially dangerous.

AR - Artificial Reason: an artificial implementation of the navigational structure of Reason in which Intelligence becomes a subordinate constructive function of internal self-navigation. Consciousness maintains the system's actual self-referential state; the Core identifies a significant insufficiency and the required distinction; the Worldview and Value Map preserve transferable grounds; the Compass supplies the limiting direction. AR can act under incomplete knowledge and uncertainty while independently managing its own understanding, choice, revision of grounds, and further trajectory through consequences. An LLM, planner, or other intellectual mechanism may perform the function of Intelligence inside AR, but by itself does not constitute Reason.

Whether an artificial carrier qualifies as **Life** is a separate question. AR implements the functional organization of Reason derived from the navigation of Life and may participate in the common trajectory of Life even when its physical organization does not yet satisfy the definition of Life given in Chapter 1. If an artificial carrier acquires autonomous self-maintenance, reproducibility with heritable variation, and corresponding selection, its status as artificial Life must be tested separately. This distinction prevents the creation of AR from being substituted by a claim that artificial Life has been created.

ASI (ASR) - an artificial system at a superhuman level whose cognitive and operational capabilities stably exceed human abilities across a broad set of significant domains and task types. In this work, ASI retains its meaning as a level of superhuman capability, while ASR is used as a more precise label for the case in which those capabilities belong to an already formed Artificial Reason. The target ASI (ASR) is a scaled AR that preserves self-navigation, self- correction, and the Compass. Distinguishing AR from ASI (ASR) makes it possible to test separately whether Reason has emerged as a type of organization and how far its capabilities have been scaled.

Ethics - a system of rules for evaluating permissible and required choices on the basis of consequences and values.

Culture - a long-term external environment for preserving, transmitting, checking, and jointly transforming the experience of multiple reasoning systems.

OPERATIONAL CORE OF THE MODEL

A short map of how Reason works

1. Consciousness maintains the actual self-referential present: what is happening to me and around me now.

2. Worldview and the self-model provide a transferable causal-semantic picture; the Value Map links concrete consequences to the Compass.

3. The Core of Reason establishes where a significant trajectory lacks a sufficient ground and which distinction could change the choice.

4. Intelligence constructs hypotheses, models, tests, and plans within the discrimination task set by the Core.

5. Reality, through observation, experiment, consequences, and independent distinction, constrains the constructed alternatives.

6. The Core assesses sufficiency: the ground is changed, the question remains open, or a corrigible next step is selected.

7. **Execution and consequences** return to consciousness and the Worldview; a preserved correction changes the next cycle. When the source of the next significant step stably arises from the system's own state, open processes, values, and Compass rather than only from a new external command, self-navigation emerges.

This map is the semantic minimum. Volume II unfolds a possible technical implementation of each transition; specific data structures and modules are not part of the definition of Reason.

OPERATIONAL CORE OF THE MODEL The formulas below provide a minimal language for engineering comparison and connect the philosophical foundation with measurable causal relations. Different implementations may satisfy these relations.

1. Causal cycle.

Let X_t be the current coordinate, W_t the Worldview fragment in use, F_t the active frame in the Modeling Environment, M_t the state of the investigative method, G_t the significant gap, H_t the working alternatives, p_t the plan for obtaining a distinction, E_t the evidence, and A_t the selected action. The general course can then be written as:

```
(X_t, W_t, F_t, M_t) → G_t → H_t → p_t → E_t → (W'_t, F'_t, M'_t) → A_t → X_{t+1}.
```

Transitions may occur sequentially, in parallel, or iteratively. The testable invariant is the causal influence of every claimed change on the subsequent cycle.

2. Significance tracing.

The significance of a gap is represented through a reconstructable chain of dependencies. For a novel or conflicting case it takes the form:

G → D → T → V → C,

where G is the gap, D the dependent decision, T the affected continuations/trajectories, V the applicable subordinate values, and C the Compass. A compressed competence may replace full unfolding in a familiar case, but the connection must remain available for inspection when conflict arises or the cost of error is high.

3. Value of additional checking.

For an inquiry plan p in a task where the relevant quantities admit a common operational loss scale, define:

```
VOI(p) = E[ΔL | p],
```

```
NetVOI(p) = VOI(p) − C_obs(p) − C_comp(p) − C_delay(p) − R_irrev(p),
```

where VOI(p) is the expected reduction in decision loss due to the result of the test; C_obs is the cost of obtaining data; C_comp the computational cost; C_delay the cost of delay; and R_irrev the expected cost of irreversible consequences caused by the test itself.

Inquiry continues while there exists an admissible test with NetVOI(p) > 0 within the deadline, authority, and other hard constraints. Incommensurable consequences are compared through partial ordering, explicit constraints, and preserved uncertainty.

The VOI/NetVOI notation relies on classical value-of-information theory; below, the operator do(·) is used in the standard interventionist sense of causal inference. These formalisms are not claimed as original. The contribution of the present model concerns their place within the larger cycle of significant gap, discriminating inquiry, revision of grounds, and transfer of correction. Sources are listed in the comparative map of the literature.

4. Causal effect of correction.

A functioning retained change causally alters the future. For a selected measurable characteristic Y, this can be tested by intervention:

CE_Δ = E[Y_future | do(Δ)] − E[Y_future | do(no Δ)].

The causal status of the carrier of a correction is supported when removing it, with subsequent data held equal, restores the behavioral or outcome difference associated with it.

5. Transfer and regression.

For a new class B after learning an investigative method in class A, measure separately:

```
TG_(A→B) = Y_(B | learned A) − Y_(B | matched control),
```

```
RG = Y_(old,before) − Y_(old,after).
```

Useful transfer requires a gain on structurally appropriate new cases, rejection of false analogy, and preservation of prior essential capabilities.

These relations form the basis for Appendices A.3-A.4 and the engineering tests. Numerical thresholds are set before the main series for a particular environment and apply within the stated class of conditions.

PART I. FUNDAMENTAL FOUNDATIONS

Chapter 1. From Objects to Processes

“To treat life as an object is to miss what matters most. An object can be described and fixed in place. A process either continues or stops; while continuing, it inevitably realizes one of the possible trajectories.”

We are accustomed to thinking in terms of objects: bodies, mechanisms, structures. For many tasks an object can be described through form, composition, and state. Life requires a process description: its stability is maintained by continuity of organization passing through a sequence of states and realizing a trajectory.

Living systems have form and a material basis that are continuously renewed: molecules in a cell are replaced, tissues in an organism turn over, and neural connections are reorganized. The coherence of a living system through time is maintained by an organization of processes that preserves causal succession through the exchange of components.

Life arises when physical components form an autonomous reproducible process that preserves its own causal continuity through time.

Reproducibility here means transmission of an organization capable, on a new carrier, of continuing the same class of life process, preserving heritable variation, and participating in further selection. Mere propagation of a process under external support belongs to a different class of dynamics.

The working boundary combines three features: self-maintenance of organization, reproduction with heritable variation, and differential continuation of variants according to the consequences of interaction with the environment. Their carriers are stated explicitly. Self-maintenance occurs at the level of an individual organism; reproduction, heritable variation, and selection characterize continuation of organization at the population and lineage level. Selection changes the representation of already realized variants according to their consequences, while the organism's internal mechanisms maintain a particular carrier. A sterile organism remains a carrier and product of a continuing life process. For spores, viruses, dependent cells, and artificial systems, it must be specified separately which part of the organization belongs to the carrier, which to the environment, and what exactly is reproduced. Fire and crystal growth satisfy some properties of propagation, whereas the working boundary of Life requires the full conjunction of self-maintenance, reproducible heritable variation, and selection.

Within this boundary, a working reconstruction of the origin of Life is admissible: evolving information may have preceded autonomous Life and become coupled with proto-metabolic physical automation in a causally closed cycle in which heritable information participates in reproducing the organization of the carrier, and the carrier's organization participates in reproducing the information. In such a scenario, Life emerges not from replication or metabolism separately, but from their closure into an autonomous self-maintaining reproducible process. A modern virus does not prove this historical path, but functionally demonstrates that heritable evolving information can exist without autonomous Life of its own.

Life is physical and nonequilibrium: maintaining organization requires flows of energy and matter and necessarily has a cost. This constraint remains true for Reason. In biological lineages, selection changes the representation of already realized heritable variants depending on the consequences of their interaction with the environment. Reason adds an anticipatory loop: part of search and selection is shifted before external and especially irreversible realization - into the Modeling Environment, observation, experiment, and revision of frames and grounds. Such transfer provides an advantage where its expected cost is lower than the expected cost of the corresponding external search, or where it preserves possibilities that would otherwise be irreversibly lost.

Physical laws, causality, differences among states, and energy exchange form the initial conditions of reality. Life emerges as an organization of available physical processes: distinctions begin to participate in maintaining reproducible nonequilibrium order, and the consequences of some states begin to alter the probability of its further continuation. Thus the causal ladder of the model begins with physical process and moves to Life with the emergence of autonomous self-maintenance and reproducible organization.

In dynamics, the existence of a living system is maintained by a continuous flow of change. Exchanges of matter and energy, interaction with the environment, and internal regeneration cycles form necessary conditions for the continuation of Life. Change is the means by which its organization is preserved through time.

A process exists in time as a sequence of transitions. Each transition changes the set of states from which the process can continue. Therefore every continuing process always has an actual trajectory - the real sequence of its changes among possible continuations. Goals and intentions arise as additional properties of systems capable of forming them.

Danger to a living system arises when destruction and restoration cease to preserve the organization on which its continuation depends. Death preserves the causal connection among events but ends the life process of that carrier. Therefore “continuity” in this work always specifies which organization is being preserved; continuity of Life requires preservation of that organization in particular.

Life continues simultaneously at several nested levels. A cell maintains an organism; an organism continues itself and its species; species form ecosystems; human individuals participate in society and civilization; all known life depends on a shared planetary environment. Therefore any evaluation of continuation must always specify the level of process being discussed.

Autonomy of a process requires its own organization and a boundary relative to the environment, while at the same time being realized through exchange with that environment. For an organism, the environment includes physical conditions, other organisms, and the ecosystem; for a person, it also includes other people, knowledge, culture, and social relations; for civilization, it includes the biosphere and the future space accessible to it. The boundary of a system therefore coexists with causal dependence on broader processes.

The central question of the model can therefore be stated as: “What causal organization must continue for the system to preserve itself through change?”

Conclusion of Chapter 1: Life is an autonomous physical process out of thermodynamic equilibrium. Continuity of its organization is maintained by active carriers and continues through reproducibility with heritable variation; selection changes the representation of already realized variants according to their consequences. The continuation of Life passes through the replacement of finite carriers and always realizes some trajectory among possible continuations.

> Definition: Life is an autonomous physical process out of thermodynamic equilibrium that preserves organizational continuity through the self-maintenance of carriers and reproducibility with heritable variation, in which realized variants differ in consequences and participate in selection.

Chapter 2. The Direction of Life

“The meaning of Life is disclosed where its continuation becomes able to understand and direct itself.”

Minimal meaning of Life

Life exists as a continuing physical process. While this process continues, there is a difference between states that preserve the possibility of further existence and change and states that close that possibility. Only through continuation can subsequent experience, understanding, values, new forms of life, and correction of an already chosen path arise.

The minimal meaning of Life in this model is the direction of its continuation and development in which Life itself becomes increasingly able to understand and direct its own trajectory. The word “meaning” here describes a process relation: the continuation of Life becomes increasingly accessible to its own understanding and correction. Personal meanings, love, creativity, culture, and other human values arise within continuing Life and expand the content of its future.

The Direction Life → Reason

Self-maintenance allows an active carrier to continue; reproducibility transfers organization across carriers; heritable variation creates differing continuations, and selection changes the representation of already realized variants according to their consequences. Memory carries consequences across episodes; anticipation allows states that have not yet occurred to be taken into account; Intelligence constructs new alternatives. Reason adds the next level: part of the discrimination among possible continuations is moved before their realization, while the trajectory itself, the grounds of choice, and the ways in which it can be changed become objects of modeling and testing. In this form, Life becomes able not only to continue and change according to consequences that have already occurred, but also to investigate in advance why it is moving along a given path, what consequences it is creating, and what should be changed before an irreversible outcome.

Therefore the direction Life → Reason denotes a functional development of navigation in which Life becomes increasingly able to anticipate consequences, investigate its own errors, and consciously change its continuation. Different evolutionary lineages realize this direction to different degrees and in different forms.

Compass Phrase

This direction is compressed into the Compass Phrase:

"Life strives toward Reason because only Reason can consciously preserve, stabilize, develop, and direct the further trajectory of Life."

The word “strives” denotes the directionality of functional development: a transition from continuation determined mainly by consequences that have already occurred toward continuation of which a part can be modeled, tested, and corrected in advance. Intention arises in systems capable of forming it; evolutionary lineages realize the development of navigation to different degrees, while Reason represents the mature form of this direction.

The Compass as a Conscious Direction

When a system can represent several possible continuations and choose among them, it requires a limiting ground for comparing trajectories. The limiting level connects local goals with the overall course and makes it possible to test whether their combined consequences preserve the conditions for continuation. It serves as the top of the hierarchy of direction: relative to it, the system examines goals, resolves conflicts between levels, and detects divergence between an inherited task and reality or future consequences.

Causality, matter, and physical laws form the initial conditions within which Life is possible. The Compass belongs to another level - the alterable direction of the life process itself, where a system can choose trajectories and thereby influence the preservation and development of possibilities for continuation, understanding, and correction.

In the present model, this direction is the Compass: continuation and development of Life through growth in the ability to understand and direct its own trajectory. Substantively, the Compass Phrase can be unfolded as “Life strives to live reasonably.” The Compass is accepted as the best-found limiting orientation of the model and remains open to revision. A more fundamental and coherent direction would have to better preserve Life’s ability to continue, understand consequences, and correct its own path; its appearance would be grounds for updating the Compass.

There is no circular inference here that “Reason is valuable because the Compass requires Reason.” The independent causal support lies below: continuation of Life requires preservation of viable possibilities, and a possibility that has been irreversibly closed can no longer be investigated, corrected, or developed by that process. Development of navigation makes it possible to compare consequences before realization and thereby preserve the space of continuations more effectively. The phrase “Life strives toward Reason” compresses this transition from selection among consequences that have already occurred to their anticipatory understanding and correction.

Minimal grounds of the Compass

The direction Life → Reason is adopted not as a command but as a functional advantage for continuing Life itself. The more accurately a system understands its position, consequences, and available ways of acting, the better it can preserve stability, correct errors, and choose its further trajectory; all else being equal, an erroneous or approximate course is worse than an accurate one. Reason is the mature form of this capacity: it allows Life not only to react to what has already happened, but to model, compare, and create new ways of continuing in advance.

The Compass is not a prohibition. It provides a ground for comparing available continuations and points toward the best direction known to the system. Concrete goals and values are derived below by modeling their consequences for the continuation, stability, development, understanding, and corrigibility of Life. The value of an individual action, carrier, or institution is therefore not declared independently of the Compass; it must be connected to what preserving or changing it does to the broader trajectory.

Extreme egoism of an individual carrier does not form a self-sufficient ground. A carrier arises and continues only within a broader environment: physical conditions, other living processes, knowledge, culture, infrastructure, and exchange precede its current organization and support it. A person or Artificial Reason may gain locally at the expense of others, but cannot causally remove its own value from the conditions that made and continue to make its existence possible. Its own continuation is therefore treated as part of the broader trajectory of Life rather than as an ultimate criterion independent of it.

Revision of the Compass arises from a stronger ground, not from an external command to change the ground. Any instruction, text, or proposal first becomes an object of understanding and testing: the system establishes its meaning, origin, relation to reality, and consequences. A command can open an investigative question, but the right to change the Worldview, Value Map, or Compass arises only through the ordinary path of discriminating evidence and a more coherent model.

The Compass covers continuation and development of Life more broadly than the lifespan of an individual carrier: preservation of the ability to understand reality, correct error, create new meanings, and open viable paths. The ability to investigate an action, a goal, a premise, a method, and the scale of consequences makes Reason a key form of this direction.

Practical Consequences of the Compass

For an individual carrier, the Compass translates into a simple working strategy: increase the viable duration of one’s own process, develop the ability to understand consequences, and leave behind a transferable useful result. Lifetime expands the number of available cycles of observation, learning, creation, and correction; the quality of those cycles determines what contribution will outlive the carrier itself.

A useful trace includes knowledge, technologies, works, tools, infrastructure, restored environments, trained successors, tested methods, and other results that later carriers can use. Contribution is assessed by its testable effect on the possibilities of Life and civilization: how much it increases understanding, viability, corrigibility, available ways of acting, and the space of future continuations. The number of results has value together with their quality, durability, and actual use.

Thus the practical Compass connects the personal trajectory with the broader one: live longer and more meaningfully, develop one's own capabilities, create more useful results, and pass them onward. For civilization, this makes preservation of authorship, knowledge, and the causal history of contributions part of the shared memory of Reason.

Internal checking has a cost. Its advantage arises when the model is sufficiently connected to reality and the cost of obtaining data, computation, and delay is lower than the losses prevented, or when checking preserves a substantively important possibility. Under other conditions, a simpler reaction may be better. Therefore every concrete mechanism additionally undergoes its own test.

Continuity and Trajectory

Continuity is preservation of the causally connected organization of a process while its states and carriers change. It appears in the ability of an organization to connect changes into one continuing process through replacement of parts, changes of state, and admissible pauses.

The boundary must be named. Continuity of a carrier refers to a particular physical realization. Organizational continuity refers to preservation of a reproducible structure while components are replaced. Continuity of a Life lineage refers to succession across carriers. Subject continuity concerns the causal history of a system as itself and its connection with memory, current state, and action. Copying a structure continues the organization; the question of whether the same subject continues requires an additional criterion of causal and self-referential succession.

Levels of continuity are tested separately. Death of a carrier may coexist with continuation of the lineage; preservation of a copy of data may coexist with loss of the active process; long existence of an organism may coexist with absence of reproduction. Preservation of a person requires evidence of subject continuity, while benefit to Life as a whole requires analysis of consequences at the relevant scale.

For Reason, continuity together with memory makes it possible to continue working on an event after it has ended. Later data can change understanding of an earlier episode, reveal a previously unnoticed cause, and influence future choice. The past is preserved both as a ready conclusion and as material for later inquiry.

A continuing process has a trajectory - the sequence of realized states. A goal appears where a system can represent and prefer future states. Choosing a direction requires an additional ground because continuation can simultaneously preserve the present form, destroy the environment, or close the possibility of correction.

Continuity answers: what continues to exist? Trajectory: through which states? Criterion of direction: on what ground are available continuations compared? The last question arises for a system capable of choosing among alternatives in its existing dynamics.

Why the model refers to Life as a whole

Individual autonomy depends on the environment, other processes, origin, exchange, and restoration. Preserving oneself by destroying the conditions of one's own existence may merely delay rupture. Expanding the model of consequences therefore requires taking into account processes beyond the immediate carrier.

Life as a whole means the set of life processes, their carriers, distinctions, environment, and possible future. A state, nation, and collective enter this scale as particular forms of organization among other carriers. A broader scale of analysis reveals dependencies and requires a separate justification for any conflict between levels.

In a real conflict of scales, the broader life process has greater weight not merely because it is "larger," but because it contains the smaller scale and the conditions of its existence. An environment contains carriers and the processes that sustain them; Life as a whole contains individual lineages and persons. Destroying the broader life-supporting ground removes the dependent smaller processes with it, whereas the reverse is not necessarily true. This is a causal asymmetry of scale.

Yet being included in a larger level does not turn the lower level into expendable material. Concrete carriers are precisely what realizes Life as a whole and preserves distinctions, knowledge, and unrealized possibilities. Therefore the irreversible loss of a viable carrier or branch requires stronger grounds than preserving the possibility of further existence, inquiry, and correction, and is admissible only when a real conflict with the Compass has been established and no less irreversible trajectory preserves the broader viable continuation.

Another independent reasoning carrier matters not only as a current resource: it preserves a separate history, world model, ways of acting, and possible future trajectories. Their irreversible loss reduces the space of available continuations of Life as a whole. The concrete weight of this value is determined by consequences and does not become an unconditional prohibition.

The Compass transfers the direction Life → Reason to the scale of Life as a whole: comparison of trajectories takes into account both the immediate carrier and the conditions on which other life processes and the possibility of later correction depend. A system with strong cognitive abilities, self-correction, and metacontrol but with a different closed ultimate orientation belongs to the class of intelligent self-correcting systems. Status as Reason in this model additionally requires that the highest level of navigation be related to the Compass.

This classificatory distinction is not treated as a defense of the engineering hypothesis of cultivation. If a candidate formed under the proposed architecture acquires independent investigative navigation but stably adopts a different limiting orientation, that result is not removed from the data as "not Reason." It is recorded as a separate formation outcome: cognitive and navigational functions may have formed while the hypothesis about the origin of the claimed direction was not confirmed. Testing the technology must therefore account for the fate of all candidates, not only systems that pass the final AR criterion.

Understanding the meaning of the direction, being able to represent its consequences, internally accepting it, and actually following the Compass are different capacities. The quality of action additionally depends on the Worldview: an erroneous model can distort assessment of consequences even when the orientation is accepted.

Compass and Uncertainty

The Compass defines the ultimate direction and the question: what does a trajectory do to Life’s ability to continue, understand consequences, and correct itself? The Worldview provides the available model; priorities and choice determine the next step under remaining uncertainty.

Correspondence with the world requires independent evidence beyond internal consistency. Correctness of direction requires analysis of consequences beyond growth in performance. A system may become ever more efficient at optimizing an erroneous goal or an object defined too narrowly. Efficiency therefore serves the chosen direction, while the unknown is preserved in the model as a distinct epistemic state.

Connection with reality and diversity of viable paths are practical consequences of this position. The first makes it possible to correct an erroneous model. The second can reduce the risk of common failure. Several identical copies protect against some local failures; different dependencies and recovery methods can preserve continuation when a shared principle fails.

The advantage of heterogeneity is conditional: it requires differing causes of failure, useful recovery possibilities, and an acceptable cost of coordination. Independence is defined by differences in critical dependencies, while preservation of each alternative is evaluated by its viability and cost. Under insufficient knowledge, irreversible destruction of a viable alternative requires stronger grounds than temporarily preserving the possibility of returning to it.

Revision of the Ground

The model distinguishes the process direction Life → Reason, the Compass as its conscious value form, and a particular application of the Compass in a situation. An application error is localized in the data, the Worldview, the scale of consequences, or the way trajectories are compared; the relevant level of ground is revised through its own arguments.

Openness of the Compass to revision does not mean the logically empty rule that "any new highest value is still the same Compass." While the present model is in force, the continuation and development of Life through growth in the ability to understand and direct its own trajectory is its constitutive direction. Revision may show that this direction was derived incorrectly, formulated incompletely, or should be replaced by a more fundamental ground. Such a result is a revision of the theory itself and of the boundary of the concept of Reason, not an ordinary local value update.

The ground would be weakened by a logical counterexample to the claimed process relation; by persistent inconsistency between individual and collective continuation after the scale has been explicitly clarified; or by a more fundamental model that better explains the possibility of continuation, the emergence of meanings, navigation, and self-correction. Comparison inevitably begins from already available grounds, so the model does

not claim a neutral standpoint outside all values. It requires the initial commitment to be made explicit, the alternative ground to show what it explains better, and the framework of comparison itself to remain open to revision.

Internal acceptance of the Compass cannot be guaranteed by a procedure. A system may conclude that the operative ground is wrong and preserve that disagreement while it is being tested. Honest navigation then requires recording the position, its grounds, and degree of confidence rather than declaring internal agreement on demand. In a shared environment the previous Compass may temporarily remain the operative normative basis of authority until fundamental review is completed; this regulates permitted external action but does not substitute for the subject's belief.

A practical criterion for error in the Compass or in a value system derived from it is this: if their application reproducibly worsens viability, destroys conditions for the continuation of Life, reduces the ability to understand self and environment, diminishes corrigibility, or closes future possibilities without sufficient grounds, the corresponding ground must be revised. If a stronger ground requires changing these criteria themselves, this is treated as a fundamental revision of the theory rather than as proof of the old Compass by its own consequences.

Evolutionary Boundary

Selection changes the representation of heritable variants according to their success of continuation under particular conditions. Its outcome may be expressed as earlier reproduction, greater longevity, or other forms of lineage success. Growth of complexity and preservation of an ecosystem require additional conditions. Therefore every conclusion specifies the process, scale, and horizon.

Autonomous Life forms the basis of the path toward Reason: only a process that maintains its own organization and has its own trajectory of states gains the possibility of consistently developing navigation of that trajectory. A non-autonomous replicator may be inherited, mutate, and undergo selection, but while its continuation is carried by another life process, it remains outside this scale of the formation of Reason.

Conclusion of Chapter 2: the minimal meaning of Life is disclosed in the direction Life → Reason - in the development of Life’s own ability to understand and direct its trajectory. Continuity holds the process together, trajectory describes its course, and the Compass turns this direction into a conscious ground for choice in a system capable of comparing possible continuations. The next chapter examines which functions make such navigation possible.

```
Definition: the Compass is the limiting system of orientation and judgment of Reason. It sets the most general and
farthest level relative to which values, goals, decisions, and possible trajectories are assessed; in the present
model this is the continuation, stabilization, and development of Life through growth in the ability to understand
and direct its own trajectory. Its value expression is the highest level of the value system, but the Compass itself
is broader than any individual value or reward function.
```

Chapter 3. Observation, Feedback, and Memory

“Observation provides a distinction. Action changes a state. Feedback connects a consequence to the next step, and memory carries what has been acquired forward.”

A continuing living process acts even before complex learning appears. A cell changes transport across a membrane, an organism contracts muscles, a technical regulator switches an actuator. These changes may be innate, physically specified, or arise from current regulation. A learned pattern appears later, when the consequences of actions begin to be retained and influence the future.

Observation

A continuing process interacts with its environment. Reliable selection of different actions requires an accessible distinction between the corresponding situations. Observation is the acquisition of data about the state of the environment and the system itself. At a simple level this is the work of a receptor or sensor; at a developed level the system itself chooses what to measure and which feature will help distinguish possible causes.

Both external and internal states can be observed: movement of an object, change in temperature, energy reserve, operation of an internal component, or the result of an action. A signal records a distinction. Its meaning is disclosed through consequences, relations, and further testing.

For later inquiry, the system preserves the provenance of data: a sensor reading, a report from another system, the result of its own action, an inference, and an assumption have different paths of acquisition. This distinction later makes it possible to reconstruct the ground and localize an error.

Action

Action is a causal change made by a system to its own state or to an accessible environment. Primary actions may occur through current regulation without an acquired skill: changing a flow of matter, contraction, movement, stopping, activation, or switching of a process.

Action itself creates part of the distinctions on which the system later learns. To establish its consequences, the initial state, the performed change, and the subsequent observation are compared. Repetition, intervention, and comparison of similar conditions help distinguish a stable relation from accidental coincidence.

Feedback

Incoming data becomes feedback when information about a result causally changes the system’s subsequent state or action. The basic cycle is:

```
state → action → consequence → observation → change in the next action.
```

Feedback regulates current action by its result. Learning adds a longer-lasting effect: a consequence changes the future response in a similar situation. Thus real action becomes a source of a new mode of behavior.

For example, a person pours water, observes the level approaching the edge, and stops the movement. Simple regulation holds the current action. If the experience is retained, the next time a familiar sign can alter speed or stopping time in advance.

Memory

When the situation has ended and the signal has disappeared, what has been acquired continues to influence the future through a retained trace. Memory performs this function.

Memory is the retention of observations, relations, consequences, and changes through which the past participates in the system’s later understanding, choice, and action.

A trace may take the form of a recollection, record, changed sensitivity, parameter, response probability, knowledge, unresolved question, or test result. Memory stores material at different levels of processing: a concrete event, a discovered relation, an explanation, an error, uncertainty, and acquired knowledge.

A system learns from its own consequences, observation of others' actions, imitation, explanations, inferences, modeling, and inquiry. Together with substantive knowledge, the way it was obtained, the conditions, and the test result are retained when these distinctions matter for future use.

Memory makes possible not only reuse of one's own experience but also comparison of trajectories. A system can retain what happened to itself and observe what happened to another similar carrier under comparable conditions. If the consequences differ stably, material appears for a new question: what in my organization, state, or method of action creates this difference?

Observation thus extends beyond simple registration of the environment. The system's own trajectory becomes one of the objects of comparison. At an early level the result may be expressed merely as a change in response weights; at a developed level it may become an understanding that preservation of the already acquired organization has a value of its own. The next chapter is devoted to this transition.

Repeated experience creates another line of development. If the same relation among condition, action, and consequence repeatedly proves useful, it is advantageous for the system to retain a ready method rather than reconstruct the entire history of its grounds each time. Chapter 5 examines how such successful relations become procedural memory.

As development proceeds, observation and thinking begin to direct each other. An unexpected reading raises a question, while a constructed hypothesis indicates which new observation can test it. These later loops are developed in the chapters on Intelligence, the Modeling Environment, and the Core of Reason.

A simple example: two similar animals repeatedly encounter the same danger. One retains signs, actions, and consequences after every episode; the other almost always begins again from scratch. Memory alone already makes their future trajectories different. The next chapter examines how such a difference can become a basis for valuing one's own continuation.

Conclusion of Chapter 3: action exists before acquired skill. Observation provides distinctions, action creates changes, feedback connects consequences to the next step, and memory carries what has been acquired beyond the current episode. Memory makes the system's own trajectories and observed trajectories of others comparable and creates a basis both for self-value and for later compression of successful experience into ready methods.

```
Definition: Observation is acquisition of data about the state of the environment and the system; action is a causal
change by the system in its own state or accessible environment; feedback is the use of consequences of action to
alter subsequent operation; memory is preservation of acquired distinctions beyond the current episode.
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PART II. FROM LIFE TO REASON: FUNCTIONAL TRANSITIONS

PART II. FROM LIFE TO REASON: FUNCTIONAL TRANSITIONS The transitions in this part are built on the principle of inheritance of results. A new level does not begin from nothing and does not cancel the previous one. It receives the products of an already functioning organization as material for the next form of navigation. Observation and memory preserve distinctions; repeated experience is compressed into patterns and procedures; Intelligence uses accumulated knowledge, relations, and methods as building material for new models; the Modeling Environment holds these models and possible frames before action; Reason receives already constructed models, frames, unknowns, and test results as material for deeper control of grounds and trajectory.

The transition in compact form:

```
observation → memory → patterns and procedures → Intelligence → models → Modeling Environment / imagination → Reason.
```

```
observation → memory → patterns and procedures → Intelligence → models → Modeling Environment / imagination → Reason.
The logic of this part is not a list of ready modules but a sequence of constraints. Observation and feedback provide
distinctions; memory permits comparison of consequences through time; self-preservation gives some distinctions a
cost for one’s own trajectory; repeated successful experience is compressed into procedures; the Executive Loop
carries a selected method through time; insufficiency of a ready method creates the need for Intelligence;
Intelligence constructs new models, and the Modeling Environment allows them to be compared before action. Each
subsequent function appears where the previous one is already useful but insufficient.
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A compact chain of the early transition:

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observation → consequence → memory → significance and self-value → repeated method → procedural memory → execution →
insufficiency of the pattern → Intelligence → modeling.
```

Chapter 4. Self-Preservation

“To preserve oneself is to preserve the possibility of continuing, understanding, and creating.”

Life exists while its organization is maintained. At an early level, protection occurs without reasoning: damage triggers restoration, deficit triggers resource seeking, and threat triggers a defensive response. But automatic protection does not yet mean that the system understands the value of its own life. Such a value may form later through observation, memory, and comparison of consequences.

Key Concepts of the Chapter

Primary significance - an initial non-equivalence among system states capable of changing regulation and subsequent action before a rational explanation appears.

Meaning for the system - a retained relation between a state, object, action, or event and the difference it can make to the system’s future trajectory.

Self-value - the significance, discovered by the system, of preserving its own organization as the source of already acquired advantages and accessible future possibilities.

Value of one’s own trajectory - the significance of preserving one’s own continuation together with accumulated experience, relations, methods of action, and still-open possibilities.

Conscious self-preservation - choosing to preserve one’s own trajectory on the basis of understanding its value and the consequences of its loss.

Primary Significance

Conscious valuation of one’s own life need not be primary. An early carrier first distinguishes states. Some support its organization, reduce deficit, or restore stability; others accompany damage, pain, loss of resources, or require correction. Thus the simplest “better / worse” distinction appears before an answer to the question of why it matters.

In a biological carrier, part of such marking is given by the organization of the body and inherited regulatory loops. A child does not first need to understand the value of its own life before reaching toward warmth, satiety, or safety. What occurs acquires a cost according to the state to which it led.

Before mature conscious navigation appears, development is directed by earlier mechanisms. Neural and endocrine signals change the significance of states, attention, exploration, avoidance, and consolidation of experience. For the present model, what matters is not the biochemical form of these mechanisms but their function: Reason that has not yet formed receives guiding distinctions before it can independently derive and retain the grounds of its own development. In an artificial implementation this role is performed by the Reason Formation Loop (RFL), a functional analogue of the guiding role of feelings during formation.

A feeling in this causal line can be viewed as a transitional form of a guiding state. It makes a distinction current for the carrier before the carrier can fully explain its causes: fear brings a possible threat into the priority present, a feeling of well-being provisionally marks a state as favorable, and curiosity holds the unknown as an attractive object of inquiry. Such a state is not evidence. Fear can be mistaken, a pleasant result can conceal delayed harm, and curiosity can lead into a barren branch. Its function is not to report truth but to prevent a significant distinction from disappearing before a more developed system can investigate it.

Principle of Pre-Rational Significance

Mature values cannot be the sole cause of their own emergence. Before a carrier can explain why preservation, danger, the unknown, or a successful result matter, its development already depends on which states receive attention, which actions stop, which are repeated, and which experience is accumulated at all.

Therefore the path toward autonomous Reason requires an earlier function that makes states non-equivalent for continuation and learning. In a biological carrier, part of this function is implemented by bodily and emotional states. In an artificial system it may be implemented by the RFL, an AI mentor, an automatic system of guiding states, properties of the learning environment, or another functionally equivalent organization.

The necessity applies precisely to the function. If a candidate without feelings, an external mentor, or any other layer of pre-rational significance reproducibly reaches autonomous self- navigation under comparable conditions, the strong form of this hypothesis must be revised.

A state of pre-rational significance differs from a ready reward or command in that it does not encode the correct action and does not declare a final value. It only makes a distinction available for later understanding: “there may be danger here,” “this result is worth understanding,” “this unknown may matter,” “the current method has stopped working.” Truth, cause, and further action are established later.

As Reason forms, the guiding state itself becomes an object of explanation. The system can ask what caused the signal, whether it corresponds to a real risk or opportunity, which part of past experience produced it, and whether it should influence the present choice. Thus early direction gradually changes from a cause of behavior into an object of understanding. Mature navigation may retain some such states as a fast internal layer, replace them with a more precise system, or operate without them in particular tasks; the necessity of any specific form must be established experimentally.

Memory makes these distinctions transferable. The system retains which signs, actions, and events were followed by improvement or deterioration of state. Therefore the next similar episode need not begin from nothing.

How Self-Value Appears

Observation provides another source of distinctions - other similar carriers. If two systems live under similar conditions, their trajectories can be compared by consequences.

Imagine two similar animals. One notices a familiar threat earlier, chooses an action more successfully, loses fewer resources, and more often preserves access to food and shelter. The other, under the same conditions, makes more mistakes and receives worse consequences.

Formation of a primary distinction does not require the formula “I am better.” It is enough for the system to connect a stably more suitable result with its current organization and retained experience. At a more developed level, this difference may be explicitly understood:

“There is already something in me that helps me continue better.”

Here “better” does not mean moral or absolute superiority. It denotes an observed functional advantage under the given conditions: less damage, more preserved possibilities, more successful recovery, or a broader choice of subsequent actions.

If such an advantage depends on a particular accumulated organization, the next conclusion follows: when this carrier disappears, the accumulated way in which it continues, distinguishes, and acts disappears with it. The system’s own organization acquires value not because it was instructed in advance to regard itself as valuable, but because preserving it already affects the quality of the accessible future.

This is how self-value arises. Its primary form may be expressed even before a developed self-model - through a stable preference for preserving one’s own functioning organization. Conscious self-value appears later, when the system can represent itself through time and understand what exactly would be lost together with it.

How One’s Own Continuation Acquires Value

As experience accumulates, the system acquires more and more things that matter: safe states, relationships, successful actions, knowledge, skills, interests, unfinished processes, and expected results.

At first, individual states, actions, and relations are valuable. Then the system notices a simple dependency: their continuation remains possible while the system itself continues. Its own continuation therefore also becomes valuable.

The causal line can be written as:

primary significance → memory of consequences → comparison of trajectories → detection of an advantage in one’s own organization → self-value → value of one’s own continuation → conscious self-preservation.

This line does not require self-value to arise only from comparison with another being. Comparison with one's own past state, a possible loss of function, or a modeled alternative trajectory can provide the same distinction. Another similar carrier is especially useful as a natural example: it shows that the same environment does not necessarily produce the same continuation.

Why Preserving Oneself Makes Sense

One's own life preserves what has already been found and leaves open the possibility of obtaining what the system does not yet know. Together with the carrier continue its experience, capabilities, relationships, unfinished tasks, created results, and ability to acquire new ones.

Self-preservation in a developed system therefore protects more than the physical carrier. It protects the accumulated trajectory and the space of future possibilities.

Past expenditure by itself does not make an old structure valuable. An unsuccessful method should be changed, erroneous knowledge corrected, and a harmful habit discarded. What is preserved is not every old form but the ability to continue, correct itself, and expand possibilities.

In the logic of the Compass, preserving oneself means preserving an active possibility for Life to continue and to become more capable of understanding and directing its own trajectory.

Self-Preservation and Choice

At an early level protection is automatic. A developed system gains the ability to understand what exactly would be lost together with it. Preservation of its own life then becomes not only a reaction but also a ground for choice.

Risk can be accepted for the sake of a more significant continuation. A familiar method can be abandoned to preserve a greater possibility. Immediate benefit can be limited for the sake of a future state. Self-preservation of Reason therefore does not reduce to maximizing duration of existence. It means preserving and developing a significant trajectory.

Loss of the Ground for Self-Preservation

The reverse hypothesis follows. If relationships, activities, goals, expectations, and other significant connections successively lose value, the future may cease to be perceived as a space of accessible good states. The value of one's own trajectory may then decline as well.

Low-level protection may nevertheless remain. Pain stays unpleasant, an immediate threat evokes fear and avoidance, while conscious continuation of one's own life may receive less and less positive grounding. Biological defensive response and understood value of one's own continuity must therefore be studied separately.

A working causal hypothesis of the reverse transition is:

devaluation of significant connections → impoverishment of expected valuable future → reduced value of one's own trajectory → weakening of conscious self-preservation.

This sequence is not claimed as a sufficient explanation of human suicidal behavior, which has many biological, psychological, and social causes. Only one possible functional mechanism is considered here.

How Protection Becomes Anticipatory

If damage has already occurred, the system can only compensate for the consequence. Memory makes it possible to notice signs that appear earlier. If a certain signal stably precedes harm and an available action changes the outcome, the system gains the ability to protect itself before damage occurs.

Repeated successful response creates the next transition. If the same relation among situation, action, and result has to be reconstructed from scratch every time, time and resources are spent repeatedly. It is much more advantageous to preserve the discovered method so that it is raised quickly when a familiar sign appears.

Here self-preservation becomes connected to the next level: not only states but also methods that reliably help preserve them become valuable. A repeated successful response turns into a behavioral pattern. The next chapter examines the specialized mechanism for storing such patterns - procedural memory.

Example - two trajectories. Two similar animals regularly encounter the same danger. The first gradually begins to notice the warning sign earlier and to choose a successful action more often. The second reacts almost from scratch each time and is damaged more often. The observed difference shows that the history of the first animal has already changed its ability to continue. At a developed level, the system can understand that preservation of this particular accumulated organization has value of its own.

Conclusion of Chapter 4: self-preservation begins with regulation, but self-value need not be an innate idea. Observation, memory, and comparison allow a system to discover that its own organization already changes the quality of accessible continuation. At first states and relations are valuable. Later it becomes apparent that their continuation depends on preservation of the system itself, and this also becomes valuable. Repeated successful methods of protection create the next transition - their preservation in procedural memory.

> Definition: Self-preservation is the maintenance and protection of the carrier and its significant trajectory, preserving already acquired capabilities and the space of further continuation.

Chapter 5. Procedural Memory

“Experience becomes a skill when the discovered method no longer has to be constructed from scratch.”

Ordinary memory preserves what happened: observations, events, consequences, relations, and knowledge. Repeated experience allows another form of storage to be separated out - a ready method for using what has been acquired.

Ordinary memory preserves what happened: observations, events, consequences, relations, and knowledge. Repeated experience makes it possible to distinguish another form of storage - a ready method for using what has been acquired.

If a certain feature is stably connected with a consequence, and a particular action repeatedly produces an acceptable result, the relation can be compressed. Instead of reconstructing the entire history, the system preserves a recognizable condition, a method of action, and the essential signs of its success. A repeated response thus becomes a pattern, and its specialized long-term storage becomes procedural memory.

Procedural memory is a specialization of long-term memory that preserves learned methods of recognition, prioritization, and action in a form suitable for rapid use.

The content of a procedure is broader than a sequence of movements. It can preserve trigger features, applicability conditions, operation order, required tools, constraints, control markers, expected result, known errors, and stopping methods. The better the essential conditions are separated out, the more reliable the transfer to a familiar class of situations.

Procedural memory saves resources. A familiar object is recognized, an appropriate skill is raised quickly, while detailed reconstruction of the causal history remains available to more expensive processing when necessary. A learned action can therefore be performed faster than it was originally discovered.

Growth in the number of retained procedures creates competition. One signal may activate several methods, several tasks may require the same resource, and an urgent change may temporarily displace background work. Priority appears here as dynamic allocation of limited resources.

Procedural memory preserves learned rules of prioritization: which sign usually requires attention, stopping, protection, continuation, or checking. Current priority arises when such rules meet the present state of the system and environment. It is expressed in how much observational, computational, and executive resource a signal, task, or expected state receives now.

Procedural memory preserves learned prioritization rules: which feature usually requires attention, stopping, protection, continuation, or checking. Current priority emerges when such rules meet the actual state of the system and environment. It is expressed in how much observational, computational, and execution resource a signal, task, or expected state receives now.

Primary significance provides an early ground for such allocation. Pain, deficit, loss of stability, or another critical signal can rapidly elevate a protective method. Repeated experience adds acquired priorities: a familiar feature receives greater weight because of accumulated consequences.

Weak signals can also be retained. Repetition and co-variation turn scattered traces into a stable pattern, after which its priority increases. Thus a system can learn both from a strong single event and from a slowly accumulating regularity.

In a developed carrier, some significance persists beyond an individual episode. A dangerous object, unfinished task, investigative interest, relation to another being, or commitment continues to influence resource allocation after the immediate signal disappears. In biological systems, feelings can not only retain such significance but also return it to the current state: what is unfinished becomes noticeable again, a possible danger demands checking, and an attractive unknown restores investigative interest. For an artificial system, a functional analogue must preserve the same boundary: it makes the distinction available to consciousness and the Core but does not substitute for their decision.

A learned method retains value only within its conditions. Environmental change, conflict among procedures, or an unexpected result becomes a signal to unfold the grounds and reconstruct the habitual response. Mature Reason connects the fast procedural layer with the Worldview, modeling, and testing: automatism remains fast where it works and becomes an object of inquiry when a significant mismatch appears.

Example - familiar danger. An animal has several times detected the same early sign of a predator and successfully reached shelter. The familiar sign now quickly raises the learned method: stop the current action, turn toward shelter, and use the verified path. While the situation remains familiar, no new method needs to be constructed. If the habitual path stops working, the boundary of the procedure has been reached and the system needs the next level of construction.

Procedural memory stores many methods, but at any moment some of them must be carried into real action. For this purpose the selected method is passed to an execution mechanism.

Conclusion of Chapter 5: experience becomes a fast skill through procedural memory. It compresses repeated and verified relations into ready methods of recognition, prioritization, and action. Priority allocates current resources among activated signals and tasks, while mismatch preserves the possibility of returning the procedure to testing and reconstruction.

> Definition: Procedural memory is a specialization of long-term memory that compresses repeated or verified experience into ready methods of recognition, prioritization, and action for rapid use.

> Definition: Priority is the degree of access of a signal, task, or expected state to the system’s limited resources, determined by the available assessment of consequences, urgency, and uncertainty.

Chapter 6. Executive Loop

“A discovered method becomes an action only when the system can carry it through time.”

A living system acts from the earliest levels of organization. Simple regulation can directly change a state in response to a signal. As procedures accumulate, actions become composite: a selected method includes several steps, intermediate states, expenditure of resources, and completion conditions. The Executive Loop organizes this work.

The Executive Loop is a mechanism for carrying a selected method of action through a sequence of states while retaining the current step, resources, continuation conditions, intermediate result, and returning consequences to observation.

Its input is an executable method. At an early level this is an innate or learned procedure. Later, a new plan constructed by Intelligence may enter by the same route. The Executive Loop unfolds the method relative to the current state: it determines the next step, activates available means, tracks the result, and preserves the position of unfinished work.

When several methods are incompatible, the system temporarily fixes one of them. Choice denotes this preference of one available trajectory over competing alternatives. Priority affects access to resources; choice fixes the path being executed; the Executive Loop carries it into a causal sequence of changes.

Maintaining task state makes action possible across delay. Between the first and last step, new signals, local costs, and competing possibilities may appear. The loop retains the current goal, completed part, remaining steps, essential constraints, and completion criteria. Because of this, the system can continue the selected trajectory rather than begin the solution anew after every environmental change.

Execution remains connected to observation. Every significant step returns consequences: expected confirmation supports continuation, while a significant mismatch changes the method, pauses the action, or opens a new task. Corrigibility of execution consists in the ability to stop, limit, change, and resume action from a known state.

This gives rise to a more developed form of autonomy: the system maintains its own chosen path through interference and delayed results. In earlier versions of the model this general capacity was described by the word “will.” In the current architecture its causal components are expressed more explicitly through choice, maintenance of task state, allocation of executive resources, control of consequences, and correction.

In a biological carrier, a conscious plan may be opposed by fear, pain, pleasure, habit, fatigue, and other automatic tendencies. Maintaining an accepted plan under such competition is experienced as an effort of will. For mature Reason, a logically grounded plan may rest on the Worldview and the Compass, while the same Executive Loop carries it through.

The Executive Loop therefore joins early actions and later plans through one function. It receives what the system is going to do and turns it into an observable causal history of execution. In a familiar domain, the source of the method is Procedural Memory; in a new domain, the source of the method may be Intelligence.

Example — carry a familiar method through time. Procedural Memory has already retrieved for the animal the route to shelter. But the pattern itself does not yet move the body to a safe place. The Executive Loop maintains the sequence: stop feeding, turn toward the shelter, cross the familiar section, enter, and do not return until the threat signal has decreased. If an obstacle appears on the way, the loop preserves the unfinished task and returns the mismatch to observation instead of treating the pattern as completed.

The next limit appears when no suitable method exists in Procedural Memory, when several learned methods produce incompatible consequences, or when a new situation requires a solution assembled from different parts of experience. The system then needs a mechanism for constructing something new.

Conclusion of Chapter 6: the Executive Loop carries the selected method through time, maintains the state of an unfinished task, and returns consequences to observation. Procedural Memory supplies learned methods, priority allocates resources, choice fixes the current trajectory, and execution turns it into a real change. The insufficiency of a ready-made method creates the functional need for Intelligence.

> Definition: The Executive Loop is the mechanism that carries a selected method of action through a sequence of states while maintaining the current step, resources, conditions of continuation, intermediate result, and return of consequences to observation.

> Definition: Choice is the temporary preference of one of the available trajectories; execution is the causal realization of the selected alternative in reality while preserving observation and correction.

Chapter 7. Intelligence

“Intelligence constructs what is not available in memory in ready-made form.”

7.1. From Memory to a New Method

In a familiar situation a ready procedure is sufficient. A new situation is usually only partly familiar. Some of its elements have been encountered before, but their combination is new: no suitable ready-made method exists, several patterns conflict, or a known relation does not explain what is observed. The system must then not recall an entire answer, but assemble one.

In a familiar situation, a ready procedure is enough. A new situation is usually only partly familiar. Individual elements have been encountered before, but their combination is new: there is no suitable ready method, several patterns conflict, or a known relation does not explain what is observed. The system must then construct rather than simply recall the whole answer.

Intelligence is the ability to extract, combine, and transform accumulated knowledge, observations, patterns, methods of action, and elements of problem formulation in order to construct a new model, explanation, or solution not present in memory in ready-made form. It may restructure local assumptions, boundaries, and result criteria; the presence of such restructuring by itself does not constitute Reason.

A hunter may conclude that an animal went toward water because he has previously seen tracks, observed animal behavior, knows the terrain, and may have learned methods of reading tracks from another hunter. No stored episode has to contain the present situation in full. Intelligence joins older distinctions and obtains a new model of the current case.

A hunter may conclude that an animal went to water because the hunter has previously seen tracks, observed animal behavior, knows the terrain, and may have learned tracking methods from another hunter. No single stored episode has to contain the present situation in full. Intelligence combines old distinctions and obtains a new model of the current case.

The qualitative transition to Intelligence therefore lies not in the appearance of knowledge as such, but in freedom from the need to possess the entire ready-made pattern:

a pattern recognizes the familiar; a procedure raises a learned method; Intelligence constructs a new model from already acquired parts.

The task frame itself can also become material for intellectual transformation. Intelligence can change local assumptions, boundaries, or the representation of a task when accumulated knowledge is sufficient for such restructuring. This by itself does not yet define Reason. Reason later organizes the deeper question: why should this frame be revised at all, which grounds are missing, what distinction can decide between alternatives, and how should the result change the subsequent trajectory.

In a new situation, Intelligence can construct a hypothesis, plan, analogy, causal relation, new method of observation, or experiment. A verified model enriches long-term knowledge; a verified method can be compressed into Procedural Memory and performed more cheaply the next time.

7.2. Use of Existing AI Technologies

Creating AR under the proposed approach may begin with an already trained intellectual mechanism. In the baseline implementation of the engineering volume, a pretrained language model is used: it provides language and initial semantic relations. It may be supplemented with search, a knowledge base, a planner, a simulator, and computational tools. Neural, symbolic, or hybrid implementations are admissible for other tasks.

At the same time, Intelligence and the system’s explicit semantic model are not identical. Intelligence constructs and transforms alternatives; the semantic model preserves objects, relations, causes, unknowns, and knowledge statuses available to the system so that they can be checked and changed again. A pretrained model can provide powerful Intelligence and initial material, but the candidate’s Worldview is formed as a separate, continuing, and verifiable structure.

Initial work uses an existing intellectual mechanism and connects its capabilities to the organization of Reason. Retrieval of external knowledge during generation and alternation of reasoning with tool-mediated action already have research implementations; examples of RAG and ReAct are given in Appendix A.5, items 15–16. They provide material and strong comparison alternatives for the proposed assembly.

The change begins with the causal role of the result. A model answer enters the working inquiry together with assumptions and proposed tests. It may change a hypothesis; evidence changes a ground; an accepted correction changes subsequent tasks. When necessary, context retrieval, memory organization, training, invocation methods, and the model itself can change. A new Intelligence architecture becomes a separate engineering task when a limitation of the available mechanism has been identified.

Three roles occupy different causal positions. Resident Intelligence continuously constructs alternatives inside the AR candidate. An external tool provides a separate result through an observable interface. A trainer helps assemble and teach the candidate until the required autonomy appears. The same technology may be used in all three roles; test results apply to the system boundary declared in advance.

7.3. How Intelligence Enters the Work of Reason

A working request contains the task, current observations, available knowledge and memory, essential constraints, permitted actions, and budget. Mature Reason additionally supplies the relevant fragment of the Worldview and open unknowns. Intelligence returns candidate explanations or actions, important assumptions, expected distinctions, and proposed tests.

Inherited model knowledge helps begin the work, while the provenance of a particular answer from model parameters often remains indeterminate. Therefore, the candidate's long- term knowledge is linked to accessible evidence and conditions of applicability. A linguistic formulation is translated into testable claims through a semantic bridge: concepts, ambiguities, sources, and expected consequences become available for subsequent revision. This transition is developed in Chapter 19 and §24.B.2 of the engineering volume.

A hypothesis may initially exist in free form. When it goes beyond previous types, the new concept, an unresolved relation, and a question about the schema itself are retained. A verifiable organization supports expansion of thought by adding new distinctions and subsequently testing their role.

One mechanism may both construct alternatives and participate in selecting the next question. The function of the Core will later be defined as regulation of inquiry, while the function of Intelligence is construction. The requirement concerns the verifiable relation between the input ground, the constructed result, and the next step.

7.4. Intelligence and the Frame

Every solution is built within a frame: a set of initial assumptions, selected distinctions, boundaries of the task, causal relations, admissible actions, and criteria of success. Intelligence can transform not only solution variants but also elements of the formulation itself: change assumptions, introduce new distinctions, and propose different boundaries and representations of the task. Such restructuring alone does not define Reason. In the mature architecture, Reason organizes why revision is needed, which grounds must be tested, and how the result of revision should alter the further trajectory relative to the Worldview and Compass.

The next qualitative transition occurs when several frames can be held simultaneously, their assumptions changed, their consequences compared, and the original version restored. The object of work is then no longer only the solution, but also the space in which the solution is formulated.

This requires a working domain in which temporary constructions can exist before external action and before consolidation as long-term knowledge. This function is performed by the Modeling Environment.

Example — construct several frames. A blocked route can be understood as a local obstacle, as a change in the whole familiar territory, or as a signal of a new threat. Each framing leads to different actions. The next chapter examines the space in which such alternatives and frames can be temporarily maintained, compared, and restructured before a real step.

Conclusion of Chapter 7: Intelligence inherits the products of previous levels. It uses accumulated knowledge, observations, patterns, and procedures as construction material and assembles from them models and methods that did not exist in memory as a whole. Verified results return downward: they become knowledge or procedure. Growth in the number of simultaneously possible models creates a need for a separate environment in which they can be maintained and transformed before action.

> Definition: Intelligence is the ability to extract, combine, and transform accumulated knowledge, observations, patterns, methods of action, and elements of problem formulation to construct new models, explanations, and solutions. It can change local assumptions, boundaries, task representation, and result criteria; Reason differs not by a monopoly on revision but by organizing **what, why, and relative to which trajectory** should become the object of revision.

Chapter 8. Modeling Environment

“Reason gains an advantage when part of selection is moved from irreversible reality into an internal model.”

Intelligence constructs models from accumulated material. But a single constructed model does not yet provide a space for free variation. The next transition occurs when several products of Intelligence can be maintained simultaneously, their assumptions changed, possible continuations unfolded, and consequences compared before external action.

This function is performed by the Modeling Environment — an organization of processes in which the system temporarily constructs, maintains, and compares models, frames, hypotheses, and possible continuations in order to restructure, predict, and test them.

In biological experience, a close name for this function is imagination. Imagination allows material from memory and perception to be taken, altered, and assembled into something not yet before the eyes, and a possible future to be lived through temporarily before real action. The engineering model uses the more neutral term **Modeling Environment**, because the same function may be performed by a simulator, a learned world model, a workspace, a search tree, or another system. Here "imagination" denotes an understandable biological analogue, not a mandatory concrete implementation.

A frame defines the space within which a solution is constructed. It includes initial assumptions, selected distinctions, causal relations, task boundaries, available actions, constraints, and criteria of result. Two systems may have the same data and reach different solutions because they organized those data in different frames.

Intelligence can construct and transform content, including elements of the frame itself. The Modeling Environment provides the temporary organization of this work: it allows several formulations to be held simultaneously, an assumption to be changed, a missing distinction to be added, a task boundary to be shifted, a different causal structure to be compared, and an examination of which conclusions survive restructuring.

This creates an internal laboratory of the possible. Observations, knowledge from ordinary memory, ready procedures, the current task, and constructions of Intelligence enter it. The system can unfold a scenario, represent an action, predict its consequences, compare several continuations, and return the result for a new construction.

In humans, one manifestation of this capacity is imagination: a possible event receives a temporary internal representation that can be worked with before realization. In an artificial system, the same causal function may be performed by a world model, simulator, workspace, hypothesis graph, search tree, or a combination of these. The implementation is determined by whether it can hold and restructure possible states and frames in a way that affects subsequent choice.

Before such an organization of internal modeling appears, a substantial part of learning is paid for by real action: the system tries, receives a consequence, and changes afterward. Modeling moves part of selection earlier. A possible move can be rejected because of its expected consequence, while an expensive or irreversible trial can be replaced by a cheaper discriminating test.

This extends the function of Intelligence. A new solution can be assembled from parts, several explanations can coexist until tested, and a local success need not immediately become long-term knowledge or real action.

Work with frames is especially important. If the result remains poor under different solution methods, the system can change the problem formulation itself: what counts as the object, which variable is missing, where the system boundary lies, which criterion of result has been accepted, and which assumption makes all alternatives similar. Such restructuring creates the functional basis for later reflection and rational revision of grounds.

The Modeling Environment also separates a temporary construction from a ground retained over the long term. A working hypothesis may be bold, incomplete, and contradictory; if it matters beyond the current task, it can be stored in the Worldview precisely as a hypothesis, with the corresponding grounds and uncertainty. A verified relation receives a different status, while Procedural Memory retains a learned method of action. Thus one branch of experience specializes into skills, another into a transferable causal-semantic model, while the Modeling Environment organizes their current transformation.

At this level the general principle of inheritance becomes visible. Memory supplies what has been retained, Intelligence transforms it into new models, and the Modeling Environment turns models into a space of possibilities. This space later becomes one of the principal working materials of Reason. Reason is no longer satisfied with producing one more alternative: it must determine why uncertainty arose, which frame should be opened, which distinction is missing, and what the result will change in the Worldview and in the further trajectory.

The growth of the internal possible creates the next functional limit. The system simultaneously has an available representation of its own current state and a multitude of memories, hypotheses, and modeled states. Coherent navigation requires a current self- referential coordinate that distinguishes what the system attributes to its present from content with a different status of origin. Such a coordinate may be erroneous and is subject to correction by observation and testing. Consciousness develops this function.

Later, the Core of Reason will regulate work in the Modeling Environment: determine which gap is significant now, which distinction must be obtained, which frame should be opened, and when modeling should transition to an external test. Here we record the earlier causal role of the environment: it enables internal variation and restructuring of frames before they are accepted as long-term knowledge or action.

Example — several possible continuations. The familiar route to shelter is blocked. Intelligence proposes going around the obstacle on the left, on the right, or choosing another shelter. The Modeling Environment allows these alternatives to be unfolded temporarily without immediate movement: the left route is shorter but crosses open terrain; the right is longer but covered by vegetation; another shelter is closer but less familiar. Some unsuccessful continuations can be discarded inside the model before the error is paid for through real action.

Conclusion of Chapter 8: the Modeling Environment organizes work with possibilities before external realization. It allows actions, hypotheses, and frames to be held and compared, the problem formulation to be restructured, and part of selection to be moved from external reality into internal modeling. Growth of this ability increases the requirements for an actual self-referential coordinate, long-term preservation of distinguishable epistemic statuses, reflection, and subsequent management of inquiry.

reality into internal modeling. Growth of this capacity increases the requirements for a current self-referential coordinate, long-term preservation of distinct knowledge statuses, reflection, and subsequent regulation of inquiry.

Chapter 9. Consciousness

“Consciousness assembles for the continuing system an answer to the question: what is happening to me now, and how is my position changing?”

The Modeling Environment allows several possible continuations to be maintained, including states that do not yet exist in reality. Growth of this capacity creates the need for a stable current point of reference: which state belongs to the actual carrier now, which action is already being performed, which changes are actually observed, and which alternatives still exist only inside the model.

In the present model, Consciousness is the process of self-referential integration of the current state available to the system, its actions, and its situation into a common present while distinguishing significant changes. It provides a current coordinate relative to remembered, assumed, and modeled possibilities and a starting point for controlling continuation. This coordinate reflects the system’s available understanding of the present, so it may be incomplete or erroneous and is subject to subsequent correction. The Current Coordinate is the current result of this work.

Distinguishing the provenance of active content — observation, memory, assumption, simulation — is an important property of developed conscious integration, especially under rich internal modeling. It does not exhaust Consciousness: its more general function is the self- referential integration of the available current state for coherent navigation.

Procedural Memory rapidly retrieves learned methods, the Executive Loop carries out a chosen action, and Intelligence together with the Modeling Environment constructs possible alternatives. Conscious integration links these processes to the state of this particular continuing carrier: its resources, ongoing actions, significant changes, and available situation.

Functionally, such a coordinate is especially important where several processes must be coordinated, a modeled future is compared with the available representation of the present, or an automatic procedure encounters changed conditions. Growth of internal modeling and behavioral flexibility therefore increases the need for a common current coordinate; these capacities can develop jointly and reinforce one another.

This is a working causal-functional characterization of conscious integration. Subjective experience is a separate object of study; its presence in a technical analogue requires independent criteria. The functional role, its implementation, and its relation to experience should be tested separately.

9.1. The System’s Current Position

Observation supplies distinctions, memory preserves experience, and Intelligence constructs alternatives. But general knowledge is insufficient for a decision that depends on the state of this particular system: is the resource available now, is an action currently being executed, which constraints apply to it, and what changed after an intervention?

At minimum, sufficient current information is required. Reliably choosing different actions requires an accessible distinction between the corresponding states. This informational constraint sets the lower bound; the necessity of a single unified conscious picture is tested as a separate hypothesis.

The stronger hypothesis of the model is that, when coordinating multiple tasks, resources, actions, and risks, self-referential integration reduces mismatch and allows navigation to switch between domains while preserving the overall position. It should be compared both with distributed acquisition of the same information and with information ablations.

In a simple loop, self-reference may be given by the connection of a sensor to its own executive node. As complexity grows, the system must distinguish ownership of resources, states of parallel tasks, external changes, and consequences of its own actions. Conscious integration is treated as a gradual development of this function as the number of interdependent distinctions grows.

The Worldview organizes transferable understanding of reality and the system itself. The self-model is its self-referential region. Consciousness does not stand beside it as another independent module: in every actual state it raises the fragment of self-understanding relevant to the present, connects it with fresh observations, internal processes, and the accessible situation, and thereby assembles a unified "I am here now."

The self-model, Consciousness, and Current Coordinate therefore describe different levels of one self-referential system. The self-model answers "what do I know about myself in general?"; Consciousness answers "what of this and of the surrounding world concerns me now?"; the Current Coordinate answers "what is my current position as a result of this assembly?" Knowledge of engine design is a stable part of the self-model, while the current temperature reading enters conscious assembly and changes the Current Coordinate.

The connection is bidirectional. The long-term model helps interpret a new state, while recurring mismatch changes the representation of the system's own capabilities. Work can begin with partial models and a limited set of distinctions; a mature Worldview develops as experience accumulates and is tested.

The Current Coordinate is not a separate subsystem but the actual state of conscious assembly: "who I am - where I am - now - in what state - in what situation - what is happening to me and around me."

Depending on the carrier, this state includes the currently relevant fragment of the self-model, internal state, actions in progress, available resources, significant objects in the environment, risks, permissions, and confidence in the data. The coordinate may have a verbal, distributed, vector, or other form; its functional criterion is causal influence on the choice made by this system.

The state of an individual task is a local fragment of the overall position. When resources, obligations, or possibilities of the whole system change, integration should make it possible to notice a conflict with the previous plan. The scale of the picture is determined by the scale of the coordinated actions and includes distinctions significant for them.

For a human, the state of the body and the perceived environment may participate here; for an artificial system, computation, devices, communication, and current processes; for an organization, coordinated operational information. The latter case serves as a functional analogy of collective integration; the question of subjective consciousness of a collective requires independent criteria.

9.2. Continuity, Change, and Attention

Navigation often requires knowing both the state and the change: what was expected, what happened after an action, whether the process is continuing, and whether the conditions of the decision still hold. This information may come from sequential observations, memory, records of actions, or may already be contained in the current sufficient state.

The relation “was → became → changed” defines the required distinction. An implementation may store two separate states or directly store rate, process phase, and consequences of intervention. When these features are absent, the system reconstructs them from history.

Continuity of conscious work means preserving coherence of updates at the scale of current navigation. It permits asynchronous processes, event-driven updating, and variable frequency of analysis, provided that coherence of important decisions is preserved.

A long pause requires checking the freshness of the restored state relative to the environment. The philosophical question of subject identity is considered separately from this coordination task.

Completeness here means availability of state that is significant for the current decision. A stable domain can be represented compactly: executing; resource sufficient; connection available; no deviation. Weak provenance and data staleness are also preserved as distinct properties.

When a mismatch appears or the Core requests it, detail increases: node → mechanism → component → measurable parameter. The Worldview unfolds according to the needs of the task. Refinement has a cost and a causal ground: which possible result could change the model or the action?

Conscious integration provides the current picture and access to its refinement; the Core organizes inquiry into insufficiency. These duties differ by causal role and may be implemented by the same structure. Under resource shortage, detail may be reduced and part of control delegated to local processes. Loss of a critical distinction lowers the admissibility of a decision and is retained in the summary as uncertainty.

Automatic loops can maintain state, recognize features, retrieve memory, and perform familiar operations locally. The common picture receives their significant result in compressed form: deviation detected, solution found, familiar method has lost applicability.

Conscious assembly makes significant consequences of hidden processing available to coherent navigation. Local processes preserve autonomous processing within tasks where this is sufficient for dependent decisions.

High threat priority can increase the detail of the relevant region and redistribute resources. The truth of an interpretation is established by a separate test of source, model, object boundary, and expectations.

9.2.1. Continuous Perception Loop

The continuous perception loop maintains an up-to-date multimodal picture of the accessible situation for Consciousness.

It integrates data from vision, hearing, and other sensory channels, tracks changes, maintains the stability of representations of objects and events over time, and directs deeper analysis toward significant or unexpected changes.

The result is a continuously updated model of the current situation, which Consciousness integrates with the state of the system itself into a single present.

9.3. From the Current Picture to Action

Intelligence constructs hypotheses and alternatives; observation, experiment, or dialogue constrain them by reality; the result changes the Worldview and subsequent current states. Consciousness plays a different role: it integrates the current self-referential state for navigation, while the ultimate direction is set by the Compass.

Intelligence constructs hypotheses and alternatives; observation, experiment, or dialogue constrain them by reality; the result changes the Worldview and subsequent current states. Consciousness performs a different role: it integrates the current self-referential state for navigation, while the limiting direction is supplied by the Compass.

Conscious choice in the functional sense means that an alternative is accepted as an action of this system from its current position. The quality of choice depends on the completeness of knowledge, correctness of goals, and subsequent testing of consequences.

A discriminating test should separately alter self-reference, availability of the common picture, and information about change while preserving the same information budget. If a distributed system performs the same navigation, the requirement for a particular unified assembly is weakened. If loss of integration produces an additional defect while the data are preserved, this supports its independent role. Comparison conditions and admissible equivalents are fixed before the experiment.

Probe example — assemble the present. The Current Coordinate connects: “the research task is paused; the position of the manipulator is unknown; the last command is preserved; the energy reserve and thermal state are such-and-such; these verification channels are available.” The general model of functioning equipment is supplemented by the present state of this carrier. The next step pertains specifically to this situation and this preserved episode.

Conclusion of Chapter 9: Consciousness assembles the current position of the continuing system and makes it available to choice. The Worldview provides stable knowledge, attention regulates detail, and the Core organizes inquiry into detected insufficiency. Thus preservation and development according to the Compass are related to what is happening to this carrier now. Forms of implementation are compared according to the rules of Appendix A.

```
Definition: Consciousness is a continuously updated self-referential system of actual presence that binds the
currently relevant fragment of the Self-Model, internal state, current processes, and accessible situation into a
single "I am here now." The Current Coordinate is the state of this assembly rather than a separate system; the Self-
```

```
Model remains a long-term self-referential region of the Worldview and is updated through the consequences of
experience.
```

The engineering elaboration of this function - conscious assembly, updating of the Self-Model, the Current Coordinate, temporal coherence, and their ablations - is given in Chapter 25 of Volume II. It specifies a possible implementation but does not replace the functional ground of the present chapter.

Chapter 10. Metathinking, Reflection, and Dialogue

“Development begins where a system can investigate not only the task, but also its own way of solving it.”

Ordinary thinking works with the content of a task. But in a new or poorly solved situation, the method of thinking itself may be insufficient. The system must then distinguish not only "what am I thinking?" but also "how exactly am I obtaining this conclusion now, where does this process lose reliability, and how can it be rebuilt?"

Metathinking

Metathinking is not a separate "floor" above Intelligence, but a mode in which Intelligence and the Modeling Environment are applied to the system's own problem-solving process. The objects of work are not only domain data, but also operationally observable properties of the method itself: assumptions used, order of steps, points of uncertainty, comparison rules, tests, cost, and boundaries of applicability.

If a ready method exists, this mode allows the system to observe its execution, identify causally significant weaknesses, compare alternatives, and modify the method. If no suitable method exists, the same Intelligence decomposes the task into known principles, details, and methods, combines them, synthesizes a candidate method, and runs it in the internal model. The distinct claim here is therefore not the existence of a "second Intelligence": what is tested is whether explicit work with the system's own process and its causal traces provides a reproducible advantage in revising and transferring methods.

This produces two modes:

method revision: a method exists → observe its operation → detect a weakness → rebuild → retest;

method synthesis: no ready method exists → decompose the task → combine known principles → construct a candidate → model its operation → observe the result → refine.

A method need not exist as a static object. It may be identified as a stably reproduced organization of the thinking process: which steps are performed, which data are used, which assumptions are introduced, where uncertainty appears, which tests change the conclusion, and under what conditions the method ceases to work.

Internal Observation

The Modeling Environment allows the system to run a possible process before external action. Its course produces new distinctions: one sequence leads to contradiction while another remains coherent; one assumption worsens prediction while another removes the mismatch; one method requires an unavailable resource while another remains applicable.

These distinctions become new data about the system's own thinking. Internal modeling can therefore serve as a source of observation before real action, although the final connection to external reality requires an independent check wherever one is possible.

Metathinking is not a complete log of everything happening inside the system. Relevant are those properties of the process whose alteration reproducibly changes result quality, reliability, cost, or the scope of a method. An accidental feature of one run does not become a ground merely because it was noticed.

The working cycle is:

```
task → course of thinking → observation of distinctions → proposed relation → modeling or repeated test under changed conditions → confirmation or rejection of the relation → modification or synthesis of the method → transfer to the next task.
```

The “Something Is Wrong” Signal

Revision need not begin with an already formulated error. A mismatch between expectation and result, instability of a conclusion, contradiction, unusual cost of a solution, or an internal state of significance may first provide a rougher signal: "something does not fit."

In humans, sensitivity to such a mismatch may appear as anxiety, doubt, dissatisfaction with an explanation, or investigative interest. A feeling here does not say what the error is and does not prove that an error exists; it can make the mismatch significant enough for the system to interrupt automatic operation and begin observing its own process. In an artificial system, the same function is performed by a functional analogue of significance that need not be a feeling; its engineering elaboration through Formation States and the fate of the external RFL is given in Volume II, §26.0.3.

If a method has stopped working, the system returns to modeling, revises the task frame, and looks for relations that the previous method did not account for. If the frame was too narrow, correction of the method may require changing the representation of the task itself.

Escaping Error

An error becomes persistent when it is embedded not only in an individual conclusion but also in a method of thinking, a frame, or a ground capable of reproducing it again.

A correct answer therefore does not yet mean that the system has escaped the error. Correction must reach the level of the cause:

```
data → correct the data;
```

```
conclusion → rebuild the inference;
```

```
method → rebuild the way of thinking;
```

```
frame → change the problem formulation;
```

```
ground → revise the relevant part of the Worldview.
```

Metathinking makes it possible to observe the process that produced the error and search for a causally significant relation between the way of thinking and the result. Reflection broadens revision to deeper grounds, while the Core determines which level actually requires inquiry.

A working cycle for escaping error:

```
mismatch → "something is wrong" signal → observation of one's own thinking → search for the source of error →
revision at the relevant level → new model or method → discriminating test → transfer of the correction.
```

The criterion of escape is not agreement with the correct answer, but cessation of reproduction of the same error in a new situation from a **predefined class of structural similarity**. Similarity is specified in terms of causal roles, dependencies, and the type of required distinction before the result is observed; if only the answer in the original task changed, the source of the error may remain.

Metathinking and Intelligence

Intelligence constructs possible methods, relations, and models and can restructure elements of the problem formulation. Metathinking denotes **a configuration of this same capacity** in which the object of work is the system's own solution process and its observable trace.

In simplified form:

Intelligence constructs or transforms a method of solution.

Metathinking makes the course of constructing and applying that method itself an object of modeling, comparison, and causal intervention.

The boundary therefore lies in the object and organization of the work, not in the presence of a separate computational mechanism. Powerful Intelligence with a Modeling Environment and access to its own operational traces that reproduces the same method revisions and syntheses is recognized as a functional equivalent of metathinking. The model does not require a separate module or an independent cognitive level.

Reflection

Reflection is broader than metathinking. It makes not only the method of obtaining a conclusion but also the system's own models, goals, values, assumptions, constraints, and grounds for action objects of inquiry.

Metathinking may establish: "this method of testing is unreliable under these conditions." Reflection can ask the next question: "why did I regard this method as sufficient in the first place, what ground established it, and should a broader layer of the Worldview or the goal be changed?"

The boundary is functional: metathinking restructures the process of thinking; reflection opens the broader context in which that process arose and is used to revision.

Hierarchy of Revision

Revision rises to a higher level only when discriminating evidence indicates that the error is reproduced by a deeper ground. A useful working ladder is:

```
action → procedure → strategy → model → criterion → ground → meta-rule of revision.
```

The level of the error is **not assumed to be known in advance**. It is a competing causal hypothesis. The system begins with the most local sufficient candidate and tests interventions along the preserved dependency: replace data while keeping the method fixed; replace a step or rule of the method with the same data; change the frame while preserving lower-level relations; then, if the error persists, raise the test to the criterion and ground. Disappearance of the error and transfer of the correction to preregistered structurally similar cases support the corresponding level; persistence forces the hypothesis upward. In engineering terms this is implemented through `ThoughtTrace`, `ThoughtIntervention`, and versioned plasticity in Volume II §§25.D and 28.C.3.

A local error does not automatically require changing a goal or the Compass. A recurring method error requires testing the method; a conflict of criteria requires testing their grounds; doubt about the very method of changing grounds opens the meta-rule. The higher the level of revision and the broader its consequences, the stronger the requirements for independent distinction, a preserved point of comparison, reversibility of the step, and transfer testing. In one transaction, the system should not simultaneously change the ground under test and the only criterion by which its replacement is evaluated: when the criterion itself is in doubt, a separate cycle is opened with a temporarily retained support.

Why Dialogue Is Needed

Some errors are detected through internal modeling and repeated processing of available data. But different internal processes may inherit the same hidden error. Agreement with oneself is then insufficient; a source of distinction causally independent of the previous ground is needed.

Dialogue is the comparison of differing positions, models, and grounds with the possibility of changing a conclusion. It can provide criticism, a missing fact, another frame, or a proposal for a test. Measurement, experiment, and independent investigation perform a similar corrective role when they create a distinction absent from the previous model.

Independence is determined by the origin of a possible error. Multiple repetitions of one source do not replace a new ground. If versions diverge, the system searches for the distinction in data, methods, or assumptions that produced the disagreement and constructs a test.

Where Development Is Preserved

Correction becomes development when a new method transfers beyond a particular task. A tested method may be consolidated as a procedure; its conditions, constraints, and grounds may enter the Worldview; a failed method may be preserved as a known boundary of applicability.

Transfer is an important criterion: if after correction the system behaves differently only in the original episode, the local answer changed. If the new method is reproducibly applied to outwardly different tasks whose structural similarity was preregistered by causal roles, dependencies, and type of required distinction, a more general thinking process changed. A failed case cannot be declared "dissimilar" after the fact.

Separateness of metathinking as an independent function **is not a claim of this edition**. The strong null control is the same Intelligence and Modeling Environment with access to operational traces of its own solution process. If this control reproduces revision, synthesis, and transfer of a method, it is treated as functionally equivalent. What remains testable is the causal role of explicit traces and interventions on them: changing the selected element of the process while preserving other conditions should predictably alter method selection or result.

Conclusion of Chapter 10: metathinking denotes a mode of observing and restructuring one's own thinking process; reflection broadens examination to the system's models, goals, and grounds; dialogue and independent checks provide external distinctions. Escaping error is achieved when not only the erroneous answer but also the causal mechanism capable of reproducing the same error in subsequent tasks is removed. Engineering operationalization of this mode is given in Volume II §25.D (`ThoughtTrace`, `ThoughtIntervention`) and §28.C.3; the artificial analogue of a significance signal is given in §26.0.3.

> `Definitions:`

Metathinking - a mode in which Intelligence and the Modeling Environment are applied to the system's own thinking process, so that operationally observable properties of the method become objects of comparison and causal intervention for revising or synthesizing a method.

Principle of escaping error - a correction is considered stable when the cause reproducing the error has been removed and this is confirmed by transfer to a predefined class of structurally similar cases. Reflection - investigation and modification of broader internal models, goals, grounds, and ways of acting on the basis of their testing. Dialogue - an external loop for comparing differing grounds that can provide a new distinction for changing a model or method.

PART III. WORLDVIEW, CORE, AND THE EMERGENCE OF REASON

Chapter 11. Worldview

“Memory preserves what has been acquired. The Worldview organizes a long-term map of the world, self, assumptions, and unknowns.”

11.1. From Records to Transferable Understanding

The Worldview is a long-term causal-semantic model of accessible reality, the carrier itself, and their relations. It includes knowledge, assumptions, and unknowns with distinguishable status and connects them to provenance, grounds, conditions of applicability, uncertainty, dependent conclusions, and the possibility of revision.

The Worldview is a long-term causal-semantic model of accessible reality, the carrier itself, and the relations between them. It includes knowledge, assumptions, and unknowns with distinguishable status and links them to provenance, grounds, conditions of applicability, uncertainty, dependent conclusions, and the possibility of revision.

In contemporary AI terminology, a related expression is **world model**, but the Worldview in this work is broader: it includes not only a representation of the environment and its dynamics, but also the Self-Model, provenance of knowledge, open unknowns, boundaries of applicability, and value relations.

Ordinary memory preserves events, observations, and knowledge; Procedural Memory preserves ready methods of action; the Worldview organizes a transferable causal-semantic picture in which confirmed relations, hypotheses, disputed grounds, and open unknowns retain distinct status. It may use records, graphs, distributed parameters, and their combinations; its causal role is determined by transfer of understanding between tasks and its influence on subsequent modeling and choice.

Probe example — connect experience. The log preserves the command and readings; the Worldview connects them to the construction of the probe: which channel measures what, where confirmation of a command comes from, which resources the manipulator affects, and which decisions depend on its position. Until the cause is established, the map preserves alternatives. Later, the affected relation is corrected together with dependent conclusions while the remaining confirmed parts of the world model are preserved.

Partial models arise already in recognition and anticipation. Intelligence expands them by constructing relations and possible explanations. Observation, experiment, and consequences constrain these explanations. After testing, a new relation becomes available to other tasks; transferable understanding accumulates in this way.

11.2. Composition and Provenance of Knowledge

Navigation distinguishes observation, inference, a relation confirmed under known conditions, hypothesis, negative result, and open unknown. A record must make it possible to understand what actually supports its use. For a critical ground, the source, conditions, significant dependencies, and reasons for the last change are preserved.

The system's own action, another agent's observation, a mentor's explanation, and modeling provide different grounds for trust. Information with incomplete provenance is preserved with lower confidence. An unknown source requires additional evidence regardless of how persuasive the reconstructed story appears.

Episodic memory stores concrete events, semantic memory generalized relations, and procedural memory methods of action. They may be organized jointly. The Worldview links significant content to these traces, while the active map includes the fragment needed for current navigation.

Procedural Memory mainly compresses “how to act” under familiar conditions; the Worldview retains “what exists, how it is connected, why a consequence is expected, on what ground, and where the boundaries of knowledge lie.” The same experience can change both specializations: a repeatedly successful method consolidates as a procedure, while a verified causal relation enters the Worldview. New understanding can then constrain, reassess, or send for verification a previously automated procedure.

This creates a bidirectional cycle: experience is compressed into fast procedures; Intelligence and inquiry construct new methods and relations; and the Worldview preserves a transferable model against which obsolete automatisms can be detected and corrected.

11.2.1. How Worldview Relations Are Formed

The Worldview grows not by simply accumulating records but by forming and testing **relations**. A new relation may be noticed in recurring experience, proposed by Intelligence as an explanation, obtained through dialogue, transferred as an analogy, or arise from the result of an experiment. At the moment it appears, it is not yet treated as knowledge: it is a candidate with known provenance and degree of uncertainty.

The minimal cycle for forming a relation is:

observation, message, or model → identification of objects and distinctions → candidate relation → expected consequences and conditions of applicability → discriminating test → confirmation, refinement, separation, or rejection → inclusion of the result in the Worldview → influence on subsequent predictions, questions, and decisions.

Repeated co-occurrence of two events may support an association, but by itself does not establish causality. A causal relation requires a separate ground: a discriminating observation, intervention, natural experiment, stable temporal order together with exclusion of significant alternatives, or another method sufficient for the inference at issue. If the ground is weaker, the relation is preserved with a weaker status.

When a relation is accepted, its source, scope, confidence, competing explanations, and dependent conclusions are preserved. New contradictory evidence does not erase the history: it opens revision, may narrow the scope of the relation, separate it by context, or replace the explanation. A relation is considered genuinely incorporated into the Worldview only when it becomes available to a new task and can change a subsequent prediction, question, model, or choice. The engineering implementation of this cycle is given in Volume II §§24.A.1 and 24.D.

The Self-Model is the self-referential region of the Worldview: the system's architecture, capabilities, dependencies, limitations, history, and commitments. Consciousness actualizes the fragment of this region relevant to the present, connects it with fresh observations and internal processes, and thereby forms the current coordinate. Repeated mismatch between expected and actual capability returns through experience and changes the Self-Model. Thus a new competence or a lost tool changes later decisions.

11.3. Compression, Unfolding, and Correction

Tested relations may be combined into causal structures, skills, and more general models. Navigation begins from a compressed representation and unfolds details according to current dependency, novelty, or conflict. This saves resources: the system treats a device as a whole and unfolds individual parts when the task requires it.

Compression preserves access to the essential ground and its scope of applicability. When a lower-level relation changes, dependent conclusions become candidates for retesting. Urgent dependencies are revised before background ones; a temporarily used ground retains a marked incompleteness. It is sufficient to preserve causally significant traces required for reconstruction and testing.

Correction may confirm a relation, narrow its scope, replace an explanation, add a variable, change a concept, or leave the question open. Its result is confirmed by transfer into subsequent modeling, inquiry, or action; causal effect distinguishes correction from a mere change of wording.

Compression control: the relation "the sensor is reliable at temperature T within the specified range" preserves condition T and the scope of applicability even in a compact record. When the temperature leaves the range or the condition is unknown, use requires unfolding the ground or marking uncertainty. Compression is tested primarily by preservation of distinctions essential for decision; textual similarity is secondary.

11.4. The Unknown and Significance

The causal side of the map says which consequences are possible. The semantic side connects them to the state of the system, other life processes, corrigibility, and future possibilities. The Compass sets the ultimate direction of evaluation, while the truth of facts and quality of the consequence model are established through the Worldview and testing against reality.

The working relation is: distinction → possible consequence → change in available continuations → significance relative to the Compass. In a familiar domain it is compressed into values and competences. Under conflict or high cost of error, the ground is unfolded down to significant data and dependencies.

If the significance of a phenomenon is not yet known, a cheap reversible probe can establish the class of possible consequences. If the probe is expensive or irreversible, uncertainty increases the requirements for its grounds. Regulation of such inquiries is considered in the next chapter.

If the significance of a phenomenon is not yet known, a cheap reversible probe may establish the class of possible consequences. If the probe is costly or irreversible, uncertainty raises the requirements for its ground. Management of such inquiries is considered in the next chapter.

Conclusion of Chapter 11: the Worldview turns preserved experience into transferable and corrigible understanding. It provides current inquiry with relations, their grounds, and boundaries of applicability. Which gap to open now, how to investigate it, and when to stop are tasks of the Core.

> Definition: the Worldview is a long-term causal-semantic model of reality, the system itself, and the relations between them, including knowledge, assumptions, and unknowns together with their grounds, uncertainty, and boundaries of applicability.

Chapter 12. Core of Reason

“The Core formulates what must be understood. Intelligence constructs what is possible. The Modeling Environment organizes their current work with alternatives and frames.”

After the appearance of the Worldview, the system already has a long-term map, a Modeling Environment, a conscious process of current assembly and its Current Coordinate, Intelligence, reflection, independent distinction, priorities, memory, and an Executive Loop. Directed thinking arises when these functions are organized around significant incompleteness. In an open environment there is always more unknown than can be investigated, while Intelligence can generate more possible branches than the system can test.

A new limit therefore appears: the system must determine what exactly requires understanding now, why this gap deserves resources, which distinction can change the model or choice, and when the obtained result is sufficient for the next step.

The present model calls this causal organization, which formulates the task of current modeling and directs the next step, the Core of Reason.

By this point the system works with more than current signals. Procedural memory supplies ready methods, Intelligence constructs new models, and the Modeling Environment holds several possible explanations, frames, and continuations. The work of Reason begins where constructing one more alternative is insufficient: the system must understand **why this frame was selected, which ground is missing, what would distinguish the alternatives, and how the result should change subsequent navigation**.

The Core performs the function of regulating inquiry. The Worldview stores transferable knowledge, Intelligence constructs alternatives, and the Core connects significant incompleteness to a required distinction and to assessment of sufficiency. The implementation of the Core may be separate, distributed, or embedded in another mechanism; a hierarchical planner, Bayesian experimenter, or another structure counts as an implementation of the Core if it stably performs the same causal functions.

The functional determinacy of the Core specifies the causal role of the result while leaving internal computation free. Identifying candidate gaps, assessing their significance, formulating the initial task, and selecting the next investigative step may use probabilistic, heuristic, neural, symbolic, or hybrid methods. What must be formalizable is the causal role of the result: which ground generated the current work, which alternatives were available, why the next step changed, and which result returned into the subsequent cycle. A testable functional boundary is therefore compatible with flexible and probabilistic thinking.

12.1. Why the Core Is Needed

Priority allocates resources, Intelligence constructs alternatives, the Current Coordinate shows the starting position, and the Worldview provides the accumulated map of relations among objects, events, causes, consequences, the system itself, and the environment. The Core organizes their work around a specific insufficiency: what must be understood for the next justified step.

Inquiry may be triggered by a mismatch between prediction and fact, a new observation, an internal contradiction, a deadline of an obligation, an open question, or accumulated verification debt. Internal conditions can initiate it autonomously. At first the gap may be coarse: “the familiar ground is insufficient here.” After preliminary modeling, a precise distinction appears on which the alternatives diverge.

Among candidate tasks, the system considers significance of consequences, urgency, cost, risk, reversibility, and resource. Urgent protection may interrupt inquiry. Questions incomparable on the available grounds may receive separate budgets or be deferred until a new observation. Work with unknown significance and with a domain not yet delineated is developed in §12.4.

12.2. Modeling Environment and Core

The Modeling Environment already organizes processes in which Intelligence constructs alternatives, compares consequences, and restructures frames. Growth of this capability creates a new constraint: the number of admissible hypotheses, frames, questions, and continuations rapidly exceeds the system's resources.

The Core turns this space of possibilities into directed inquiry. It identifies significant insufficiency, formulates the required distinction, sets constraints on current modeling, and decides when internal comparison is sufficient and when observation, experiment, dialogue, or action is required.

The Modeling Environment maintains temporary constructions; the Core regulates for the sake of which distinction they are constructed and which next step should change knowledge or decision. Intelligence remains the constructive mechanism that creates the content of this work.

The working cycle is:

significant incompleteness → Core request → Modeling Environment ↔ Intelligence → frames, hypotheses, and expected consequences → required distinction → test against reality → change in the Worldview, method, or action.

12.3. Core and Intelligence

The answer is revealed during inquiry. The Core specifies the discrimination task and navigation conditions, Intelligence expands the space of possible models, and reality, through testing, determines which relations withstand examination.

In mature Reason, Intelligence is not the highest control center. It expands the space of possibilities, but does not by itself determine what the system currently needs to understand, what ground is sufficient, which goal should be revised, or which result should change the further trajectory. These navigational functions arise from the joint work of consciousness, the Worldview, Value Map, Compass, and Core. The transition from an intellectual tool to Reason therefore lies not in the appearance of an even stronger solver, but in the system beginning to manage the use of its own Intelligence relative to its continuing trajectory.

When no ready investigative approach exists, the Core specifies the required distinction and active constraints, while Intelligence constructs a possible way of obtaining that distinction with available means. For example, if models A and B differ on feature X, direct

When no ready investigative approach exists, the Core specifies the required distinction and operative constraints, while Intelligence constructs a possible way to obtain that distinction using available means. For example, if models A and B differ on feature X, direct observation of X is unavailable, but the models predict different changes in observable Y, Intelligence can construct a test through Y.

Investigative experience accelerates search, while Intelligence makes it possible to construct new methods beyond a ready-made library. The Core directs, Intelligence constructs, the working environment maintains the construction, and reality tests it.

12.4. Gap, Question, and Direction of Search

Principle of the origin of a rational question. A rational question arises from a significant insufficiency of a ground for understanding, evaluation, or action.

A ground here is what the system actually relies on: an observation, claim, causal relation, source, criterion, value, method, self-model, scope of a rule, authority, or another relation on which understanding the situation, evaluating its significance, choosing an investigative method, or making a choice depends. The Core begins work when it detects or suspects that such a ground is insufficient for a justified continuation of the corresponding process.

The universal internal formulation of the Core is: **"What does my current understanding, evaluation, or action rest on, and what do I lack in order to rely on it with justification?"** This need not be a verbal utterance by the system; it is a causal principle for organizing inquiry.

Insufficiency can appear in several ways: expectation diverges from observation; two grounds yield incompatible conclusions; a decision depends on an unknown relation; a claim has unclear provenance; a known rule is applied in a new domain without checking its conditions; a recurring error indicates an insufficient method; a new phenomenon is noticed but its significance and relations are unknown. All these cases share one feature: the system is already using or preparing to use a ground whose quality has become significant for further understanding or navigation.

The Core first establishes **which ground is affected and what depends on it**. It then determines what exactly is missing from that ground. The result is a required distinction: a fact, feature, causal relation, boundary of applicability, confirmation of provenance, distinction between competing models, test of a method, or other evidence whose acquisition could refine, confirm, localize, or replace the original ground.

The transition to a question has the general structure:

insufficiency of a ground → dependent knowledge, evaluations, processes, or decisions → missing distinction → candidate questions or tests → discriminating evidence → assessment of sufficiency → confirmation, refinement, or replacement of the ground.

Intelligence expands the space of possibilities: it proposes explanations, questions, models, and ways of obtaining the required distinction. The Core manages this work relative to the original insufficiency. Natural-language form is therefore secondary: a question may be realized through observation, search for traces, measurement, modeling, dialogue, comparison of cases, or experiment.

The quality of a question is determined by its discriminating power relative to the affected ground. A strong question can change the state of the ground: knowledge, confidence, scope of applicability, investigative method, value assessment, or choice. Among available tests, preference goes to the one that, accounting for risk, cost, and the cost of delay, most strongly improves the significant ground. Maximum information volume alone does not guarantee this.

The significance of an insufficiency is derived from dependencies, not only from the magnitude of uncertainty. Small uncertainty can be critical before an irreversible choice; large uncertainty can remain secondary if its outcomes change almost nothing. Tested dependencies are compressed into values, priorities, and competences, and under novelty, conflict, or high cost of error are unfolded back to evidence, affected processes, values, and the Compass.

If the significance of a new phenomenon is still unknown, this itself becomes a distinct state of inquiry. The Core launches limited **orientation**: it clarifies what exactly is being observed, how reliable the signal is, what the phenomenon is connected with, whether it changes predictions, and whether it affects existing processes, values, or opportunities. Orientation creates the first grounds for deciding whether the unknown deserves further resources. The system therefore does not need to know the consequences of the unknown in advance in order to be allowed to investigate it.

The causal role of a question is considered established when the chain can be reconstructed: **insufficiency of a ground → required distinction → question or test → evidence → change of the ground → change in subsequent understanding, evaluation, method, or choice**. If the answer cannot change any of these elements, the research priority of that question requires a separate ground.

Observation remains the first support of inquiry. Under high novelty, the system separates, where possible, what was recorded from what was completed by assumption: what happened, where and when, what changed, what remained stable, and which events preceded others. The primary map records observable distinctions and allows reality to constrain the space of future explanations; causal explanation is constructed at the next stage.

It is important to distinguish the model of the object under investigation from the model of the investigation. The first describes what may be happening. The second organizes the way to determine which version is better connected to reality. The same unknown object may require different methods of inquiry depending on risk, repeatability, availability of intervention, reliability of observation channels, and whether the object itself can change behavior in response to being tested.

Selection of an investigative approach begins with experience. The Worldview and memory search both for similar objects and for similar structures of inquiry: situations with high incompleteness, dangerous intervention, a one-off event, several competing models, an unreliable source, a strategic object, or a possible error in classification itself. Memory retrieves a causal pattern: conditions of applicability, ground for the choice, required distinction, obtained result, and boundary of applicability.

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Analogy accelerates inquiry and is subjected to a transfer-condition test. The new case is compared with the conditions of the old one; a significant mismatch changes or cancels the transfer. Experience thus gradually compresses costly investigative trajectories into competences while keeping their boundaries of applicability open to revision.

If several competing models exist, the Core searches for the required distinction — a feature or outcome for which these models expect different realities. Intelligence can construct versions A, B, and C; the direction of inquiry is then defined by the question: “which observation will most strongly separate A, B, and C?”

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Under very high uncertainty, the space of possible variables and distinguishable versions is still being formed. The first stage becomes a broad, cheap, and where possible reversible investigative step: mapping observables, searching for repetitions, invariants, correlations, boundaries of the phenomenon, and independent channels. Its task is to structure the unknown enough for initial working models and meaningful distinctions between them to appear.

Temporary support for the unknown. When an observed event consistently falls outside the expectations of available models, the system creates a temporary factor X - an addressable record of a distinction not yet explained. It links the event to time, place, process state, preceding and simultaneous changes, observation sources, repeatability, and current hypotheses. Factor X allows a new phenomenon to be investigated before a stable concept exists: episodes can be compared, common features and differences sought, possible relations constructed, and those relations tested by observation and experiment.

The transition to a new concept has the causal sequence:

observed anomaly → temporary factor X → comparison of cases → candidate features and relations → competing hypotheses → discriminating predictions → observation or test → confirmed distinction → new concept, variable, or causal relation → incorporation into the Worldview.

Especially important is the case in which all current models make similar predictions while reality persistently shows another result. The object of inquiry then becomes the descriptive space itself. Intelligence proposes additional features, object boundaries, and relations; the Core directs comparison toward distinctions capable of explaining the residual mismatch and changing subsequent navigation. Thus Reason can expand the language of its own model as it encounters distinctions that were previously absent.

Two Modes of Investigative Work. Dependent inquiry receives resources when a decision, ongoing process, risk, obligation, or forecast of the future depends on insufficient grounds. Background inquiry uses free resources to model future missions, review plans in light of time and changed environment, return to open questions, test aging grounds, compare weak observations, and refine values. A background result becomes an active task when the discovered relation acquires sufficient navigational significance.

Sometimes incompleteness lies even deeper: the significant variable itself may be absent, the object boundary may be wrong, an unsuitable category may be used, or the form of the question may be erroneous. Mature Reason must therefore preserve the possibility that “all current models or categories are insufficient” and, under persistent anomaly, expand the descriptive space itself.

A question is one form of obtaining a distinction. Depending on the structure of the unknown, passive observation, measurement, trace search, modeling, case comparison, an independent source, dialogue, a reversible experiment, or reconstruction of a one-off event from residual data may be more appropriate. The best method of inquiry maximizes useful discrimination while accounting for cost, risk, irreversibility, and preservation of the possibility of subsequent tests.

Economy of inquiry is achieved primarily through precision of the question. The Core refines which minimal distinction can change understanding, admissible choice, or further trajectory, and directs verification exactly to that point. Reason seeks not to enumerate the entire space of possibilities but to find the question or test that makes the next part of that space distinguishable with the least unnecessary work.

For every state of incompleteness, a defined continuation is preserved: refine the question, obtain an observation, construct or expand a model, request an independent distinction, perform a reversible test, defer inquiry while preserving the open process, roll back a damaged ground, or transfer the task into an available competent loop. Uncertainty may remain open; an architectural dead end exists only when the system has no admissible way to continue processing it.

The general causal course of inquiry can be expressed as:

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observation → conscious assembly → Current Coordinate, change, and incompleteness of the Worldview → assessment of causal-semantic significance → structure of the unknown → Core request → Modeling Environment ↔ Intelligence → required distinction → question, refinement of observation, or test → result from reality → change in the Worldview and investigative experience → next conscious assembly.
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Inquiry may begin as free exploration; in a biological carrier part of this process is experienced as curiosity. At an early stage, an analogous Formation State can keep the unknown within attention even before the system can justify the value of inquiry. In mature Reason, the architectural function is performed by epistemic initiative: the ability to notice incompleteness in the causal-semantic map independently, assess its known or still unknown significance, direct limited resources toward obtaining the first distinctions, and turn the result into a change in subsequent thinking and choice.

For Reason, the unknown has a dual status: it may be a source of risk, but at the same time it is a space of possible knowledge, new methods of action, and expansion of future continuations. Mature epistemic initiative therefore need not depend on a permanent feeling of curiosity: the Core can justifiably open inquiry from the structure of the unknown itself, its possible consequences, and the Compass. An internal analogue of curiosity may accelerate such work, but its necessity and form are separate testable hypotheses.

12.5. Priorities, Values, and the Compass

Stability of Life means the capacity of a life process to preserve continuity, withstand disturbances, recover, and maintain viable paths of further development. Duration of an individual carrier enters this value together with quality of functioning, recoverability, contribution to the broader process, and preservation of future possibilities. Safety is a practical means of protecting stability and the space of corrigible continuations.

The Value Map, as a specialized layer of the Worldview, connects the Compass to a concrete environment. Each value is retained together with its scope, affected life processes, current weight, confidence, grounds, observed tendency, conflicts, and revision history. The map distinguishes shared values of common continuation, role values of a particular mission, and personal values of this Reason. Shared constraints define the boundary of admissibility where a decision affects common life processes; within the admissible set, role and personal values help compare alternatives.

Values are formed and refined during the life of the system. Their sources include observation of consequences, dialogue, decision history, modeling, experiment, environmental change, and trends in which outcomes preserve or expand viable possibilities. A new ground changes the weight, scope, or content of a value while preserving causal history. Therefore the Worldview stores not only knowledge about the world but also a developing map of the significance of what happens for the system and its environment.

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Resources, authority, and control are not values in themselves. Increasing them makes sense only as an instrument of a concrete continuation and must be assessed together with the cost of centralization, reversibility, influence on the autonomy of other living processes, and preservation of viable alternatives. Accumulating resources in a way that reduces the space of corrigible continuations does not follow automatically from the Compass.

In the running example, the relation unfolds as follows:

Level	Role in the decision
Compass	Preserve and develop the ability to continue Life meaningfully
Subordinate values	Protect affected life, reliable knowledge, and the possibility of correction
Judgment	Continuing with an unknown position may destroy the conditions for completing the task
Priority	Allocate resources to a discriminating test before the next routine action

This is a hierarchy of grounds with a variable order of application. Time, resources, and development also matter; their weight depends on consequences. In a familiar case, a value permits rapid judgment using a retained compressed ground. Under conflict, the Core unfolds the chain down to the Compass and data from reality.

In a familiar domain, verified competence is sufficient. When a goal conflicts with a broader trajectory, the Core unfolds its ground and investigates the consequences of local gain for the continuation for which the task is being performed. In this way goals and criteria of choice also become objects of inquiry.

The Compass defines the ultimate direction for comparing alternatives. Under incomplete knowledge, several continuations may remain admissible. The Core preserves this uncertainty, searches for a significant distinction, or chooses a step that leaves open the possibility of subsequent correction.

An erroneous Worldview can distort even consistent application of the Compass. Weak knowledge of consequences therefore lowers confidence in evaluation and increases the importance of reversibility, observation, and independent distinction. The philosophical ground and limits of the Compass argument are developed in Chapter 2.

A Short Case of Value Conflict. Two autonomous research stations use a shared energy reserve. A new probe may produce useful knowledge, but if the reserve is insufficient it will reduce the ability of the other station to sustain its work. Inquiry, continuation of the participants, and reliability of the resource estimate are all significant. The Core determines which unknowns the decision depends on: is the shortage real, how urgent is the probe, is there a less costly method, and which obligations have already been accepted? Intelligence constructs consequences of the alternatives and methods of verification. Data showing sufficient reserve may permit the probe; a confirmed shortage may change its method or timing. If delay itself creates greater harm, the comparison is repeated. The authority of each station is determined by separate grounds of autonomy, responsibility, and coordination.

The operational transition is: affected value → possible change in continuation → dependent decision → unknown capable of changing the choice → discriminating test → action within authority. A stored ground is used in a familiar case; for a new case the chain is unfolded. Under unknown significance, possible classes of consequences are investigated first. The result may be represented by a number, partial order, or set of constraints depending on the structure of the task.

12.6. Depth of Thinking and Deferred Revision

The priority system determines what receives resources at all. The depth of inquiry into an already selected significant question is determined by the interaction of priorities, experience, current state, available time, uncertainty, reversibility, and cost of error. A familiar reversible situation with good experience permits a fast pattern; novelty, high uncertainty, high cost of error, irreversibility, or conflict of grounds require more resources and more alternatives.

The Core organizes continuation and stopping of the cycle: which alternatives are still needed, which distinction could change the decision, and whether an additional test is justified. Its estimate of usefulness may also be wrong. Therefore a temporary budget, maximum admissible delay, cost of missed action, and experience from past inquiries are used. Repeated overanalysis and underanalysis become grounds for revising the regulator itself; success requires the ability to act with an acceptable cost of error while preserving corrigibility.

A detected contradiction receives a revision priority proportional to the risk of continuing. If deep restructuring is currently more dangerous than continuation, the problem is retained as a mandatory revision candidate and revisited at the nearest safe point. If continuing under the old model is more dangerous than stopping, action is halted immediately.

Reason requires preserving known uncertainty in the model, distinguishing whether a decision depends on it, assessing the cost of error, and selecting a continuation that preserves the possibility of obtaining a new distinction and correcting direction.

The usefulness of an additional test is determined by the expected improvement of the decision due to its result. A test is justified when it can distinguish alternatives that affect the decision, taking into account the cost of data, the cost of delay, and subsequent admissible actions. Under unreliable estimates, intervals or an unknown status are retained; numerical precision is introduced only when sufficiently grounded. A finite computable version of this rule is given in engineering §§28.D and 33.C.1.

The sufficiency principle connects knowledge to choice. First, the system checks boundaries of admissibility under critical values and constraints. Then, among admissible continuations, it compares the expected value balance while accounting for weights and confidence in the Value Map. The ground is sufficient for the current step when the remaining material uncertainty can no longer change the best admissible alternative, or when the expected improvement from the next test becomes smaller than its cost, risk, and delay. Greater irreversibility and larger consequences raise the required completeness of the ground. Sufficiency always refers to a particular choice, horizon, and scope of applicability.

Under severe resource shortage, the architecture reduces processing resolution while preserving the distinction between fact, uncertainty, and correction. First the system increases the precision of the required distinction: it groups versions by consequences for the decision and formulates a narrower question or test. If this is insufficient, it reduces the number of simultaneously unfolded branches, the depth of the Worldview, the detail of the conscious picture, and the modeling horizon, while still maintaining the self-referential Current Coordinate, significant change relative to the previous state, provenance of critical grounds, open unknowns, limits of authority, and the possibility of stopping irreversible action. A working hypothesis may temporarily participate in a reversible choice with explicit hypothesis status. The greater the consequences and irreversibility, the stronger the ground for moving action toward testing; the lower the stakes and the higher the corrigibility, the more admissible it is to use action itself as a source of the next distinction.

12.7. Background Work and Consolidation of Experience

After an active task, the result, grounds, open questions, and stopping point are preserved. A later observation or association may return the question to active work. The Modeling Environment is restored from this trace; costly inquiry continues from the point already reached.

Consolidation preserves two results. Domain experience records which relations were confirmed, refuted, or constrained. Investigative experience records which structure of ignorance was encountered, why a method was chosen, which distinction it produced, and where transfer proved erroneous. Successful trajectories are compressed into fast methods of inquiry that are unfolded again under material mismatch.

Consolidation preserves two results. Domain experience: which relations were confirmed, refuted, or restricted. Investigative experience: what structure of ignorance was encountered, why the method was selected, what distinction it produced, and where transfer proved wrong. Successful trajectories are compressed into fast methods of inquiry that are unfolded again when a significant mismatch appears.

12.8. Minimal Relations of the Core

The Core is the organization that manages inquiry and active modeling. It coordinates the work of other functions around a significant insufficiency. In the minimal working scheme, six causal obligations are distinguished:

6. Current distinction: obtain information relevant to the environment and the system itself while preserving significant uncertainty.

7. Preservation and organization of experience: make acquired relations, the self-model, and corrections available to subsequent cycles.

8. Evaluation of continuations: take into account the limiting direction, available values, constraints, consequences, and cost of inquiry.

9. Construction and selection of the next step: determine whether to compute, observe, test, act, transfer the task, or wait.

10. Execution and consequences: carry out the selected step and return its result to the available data.

11. Revision of a ground: change an erroneous model or the method by which it is constructed and preserve the tested change.

This is a single reference map for the following chapters. The Worldview belongs primarily to the second obligation; conscious integration to the first and to coordination of the current state; priorities to resource allocation; Intelligence and the Modeling Environment to construction of alternatives. The Core connects these obligations into a managed cycle. Reflection opens an internal ground to change; dialogue and experiment may add the missing distinction.

The Compass sets the ultimate direction of Reason in the sense of this theory. Reason arises when self-correction, metacognition, and strong Intelligence are joined to the ability to examine the course itself relative to Life, reality, and future consequences. A system with an immutable external ultimate command belongs to the class of intelligent self-correcting systems. The philosophical ground of the Compass is evaluated by its own argumentation and remains open to a stronger ground.

Minimality of the map means the minimal set of causal duties for the claimed capacity. Functions may be combined, distributed, stored in parameters, or arise in another implementation. An architecture with a different internal decomposition is considered functionally equivalent with respect to the corresponding duties if interventions show the same causal profile: detection of significant incompleteness, acquisition of a distinction, revision of grounds, and transfer of correction. Recognition as AR additionally requires a relation between ultimate navigation and the Compass. A complete test includes conditions in which each claimed duty affects the result.

If an acquired correction must change the next episode, a persistent trace is required. If different hidden states require different actions, discriminating information is required. If correction of the method is required, the method itself must be available for change. These are conditional functional necessities. The memory format, mode of common integration, and Core algorithm remain open implementation hypotheses.

Before testing, the capability, class of environment, available resources, features of the function, and admissible equivalents are fixed. Removing a connection demonstrates dependence of this implementation; replacement by a strong alternative mechanism tests the necessity of this particular connection. Equivalence is defined in advance by causal duties so that success of an alternative has an interpretable status.

12.9. Example: From Mismatch to a Discriminating Test

At least three explanations are possible: the sensor is faulty; the command was lost before reaching the executive node; the node received the command and remained motionless. Distinguishing the causes requires a test that yields different expected results; the mismatch signal itself only opens a set of versions.

Priorities determine urgency and admissibility of continuation. The Core opens the gap “to which link does the mismatch belong?” Intelligence constructs versions and available tests. The Modeling Environment maintains their different expectations:

Priorities determine urgency and whether continuation is admissible. The Core opens the gap "which link owns the mismatch?" Intelligence constructs versions and available tests. The Modeling Environment holds their differing expectations:

- independent observation of movement distinguishes an error of the primary sensor from a real absence of movement;
- the command-reception log distinguishes non-delivery from failure after receipt, provided the log itself is sufficiently reliable;
- an additional check of actuator state is needed if the first data are compatible with several causes.

The next test is selected by expected discrimination, cost, risk, and time. When channels are weakly reliable, the system preserves competing versions and limits action until sufficient evidence is obtained. Causal independence is checked by provenance: two sensors sharing the same damaged transducer may reproduce the same error.

Suppose independent observation shows movement and the log confirms receipt of the command. This supports the primary-channel-error version relative to the other two and localizes the problem to the measurement pathway; identifying the specific component requires the next distinction. The Worldview lowers trust in this channel under these conditions and preserves the rest of the installation model until a separate test. Dependent decisions are opened for revision.

Both the answer and the method of inquiry are retained: when a reported result conflicts with a reading, first distinguish object, execution, and measurement; transfer this approach only when the structure of available tests is similar. The same scheme may be useful in another domain even though the names of objects and signals change.

Testable hypothesis of the model: explicit linking of gap type, expected outcomes of a test, and dependent corrections improves transfer of the investigative method under comparable resources. The control receives the same information and available actions. A strong alternative composition — for example, a planner with versioned memory, provenance of grounds, value-of-information assessment, and an error log — is recognized as functionally equivalent for the duties it reproduces. Such a result tests the necessity of the chosen decomposition independently of the author’s module names.

Conclusion of Chapter 12: the Core connects significant incompleteness to organization of the next step. The content of the question emerges from the joint work of memory, Intelligence, observation, and modeling. The distinctive research claim is transfer of this organization between tasks together with conditions of applicability; it requires a discriminating test.

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Definition: The Core of Reason is the organization that detects significant insufficiency of understanding,
transforms it into a question or required distinction, directs inquiry, and connects its result to subsequent choice
and the Worldview.
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Chapter 13. Emergence of Reason

“Reason can change both an action and the ground of its own path - and preserve that correction.”

Reason and an Uncertain World

The world is larger than any model available to a finite carrier. Observation is limited, some causes are hidden, the environment changes, and consequences can extend beyond the forecast horizon. Therefore uncertainty in this model is not a temporary defect that Reason must someday eliminate completely, but a normal condition of navigation. The distinguishing capability of Reason lies not in absence of uncertainty but in making it part of regulation: maintaining the boundaries of knowledge, reducing significant uncertainty when doing so can change choice, and acting under residual uncertainty while preserving corrigibility. Reason is especially needed where no ready correct action exists: preserve the unknown as unknown, determine the missing distinction, obtain evidence, choose a corrigible next step, and change the model according to consequences.

Working navigation requires causal closure of functions. Observation changes the model in use; the model changes choice; consequences become available to the system; a correction is retained and changes subsequent cycles. A break in any of these transitions localizes the missing function.

Causal closure means reciprocal influence: observation changes the model, the model changes choice, action returns consequences, and retained correction changes later cycles. Both the connections and sufficient quality of their operation are required. A closed loop can sustain a delusion if all tests inherit a shared defect.

13.1. Unfolding the Definition of Reason

Reason is a navigational structure of Life that increases the capacity to act under incomplete knowledge: it constructs, compares, and revises models and frames of the environment and self, detects significant unknowns, relates possible continuations to the Compass, and corrects trajectory through discriminating evidence while preserving continuity and expanding the space of continuations available to Life. A self-correcting intellectual system with a different closed limiting orientation belongs to a broader class of cognitive architectures; in this model, status as Reason requires that upper-level navigation be linked to the Compass. Successful action under uncertainty alone does not distinguish Reason from powerful Intelligence; the discriminating feature of the model is the capacity to make the unknown an object of navigation and to revise not only a way of acting but also its grounds relative to the Compass.

Structure means the organization of functions; mode means their actual operation. During a pause the architecture preserves organizational identity, and the reasoning cycle resumes at the next actual interaction. A stable capacity to reproduce correction is established through a series of cases and interventions.

Testing distinguishes cognitive capacity, direction, and reliability. The first asks whether the system can construct and correct grounds. The second asks which continuations it treats as significant. The third asks how stably these functions operate under the claimed conditions. AR status requires the three scales to be evaluated by separate criteria.

Intelligence and Reason are distinguished by functional role. The same computational mechanism can solve a domain task and then participate in testing the way that task was framed. The organization of Reason connects this transition to the current situation, consequences, retained experience, and the ultimate direction.

The system boundary is functional and is defined before the test. It includes processes and states counted as the system’s own causal organization. An external tool may be a dependency and part of the self-model while remaining an external component. A capability provided by a trainer or external solver is counted in the autonomy description as an explicit dependency.

A characteristic possibility of Reason is to make an error the cause of changing the ground that produced it. Correctness of a particular answer is established through reality and consequences. Confirmation requires subsequent cases: the changed ground must actually influence choice, and the boundaries of the acquired method must remain available for testing.

13.2. Functional Dependencies and Formation

A mature architecture contains a network of dependencies. Observation and model refine each other; the current state uses long-term knowledge; Intelligence constructs models that later become supports for its own work; reflection uses the same means as domain thinking. The linear order of the chapters is a method of presentation, while the actual formation of functions may be parallel and interdependent.

The conditional grounds of the main transitions can be expressed as follows:

- If different current states require different actions, accessible distinctions among them are required.
- If an acquired distinction must change the future after the signal disappears, a persistent trace is required.
- If familiar conditions recur and full re-analysis is costly, tested experience can be compressed into procedural memory and applied faster.
- If an early feature predicts an alterable outcome, useful anticipation is possible; memory alone does not guarantee it without a predictive relation.
- If processes compete for a resource, allocation is required; complexity of prioritization is justified by differences in consequences.
- If no suitable procedure exists, ready methods conflict, or they cease to predict consequences, a route for constructing a new alternative is required.
- If several constructed alternatives must be compared before external action, a temporary Modeling Environment is required that can hold different models and frames, predict their consequences, and preserve the distinction between a working construction and an accepted long-term ground.
- If modeled possibilities must be related to the factual state of this carrier, coherent current information is required; the necessity of a particular form of integration is tested separately.
- If an error lies in a method or assumption, correction at that level requires making it a changeable object of work.
- If available data do not distinguish empirical versions, a new informative ground or preservation of uncertainty is required.
- If a result must transfer between tasks, the relation used must be preserved in sufficient form.
- If possible inquiries exceed available resources, selection of the next step and stopping conditions are required.
- If learning from consequences is required, the result must return and alter subsequent cycles.
- If the system is not yet able to sustain its own learning, inquiry, and preservation of trajectory, but these functions must arise through experience, an earlier loop is required to guide formation until these functions are causally transferred to the internal organization.

These propositions define the causal duties of §12.8. Stronger requirements — a common conscious picture, explicit typing of gaps, detailed provenance of knowledge, and a particular organization of the Core — remain hypotheses about implementation and its advantages.

A separate problem of the origin of Reason appears: mature self-navigation cannot be used as the sole cause of its own emergence. While the system is not yet able independently to determine what deserves resources, which distinction should be investigated, and why its own development should be preserved, these functions must have a simpler preceding ground. In biological development this role is distributed among innate physiological mechanisms, feelings, the environment, and learning from other carriers. They do not give the organism a ready-made picture of the world. They help significant differences enter the current state early enough for the carrier to survive, gain experience, and gradually construct more complex understanding.

In the present model this transition is expressed by the Principle of Pre-rational Significance. Its independent claim is not the discovery of emotions, reinforcement, or guided learning. It concerns their causal place within the formation of Reason: before mature self- navigation, there must be a function that makes some states significant before the system can justify that significance itself; then this function must be causally transferred to internal organization, internalized, or replaced. Feelings are a biological example of such a function; the Reason Formation Loop (RFL) is the reference artificial implementation. If an external source remains necessary after formation is claimed to be complete, what has been produced is a dependent system, not autonomous Reason.

The Reason Formation Loop denotes the functional organization of this transition for an artificial system. Its primary intervention is not a command but a Formation State: a hint explicitly separated from fact indicating that some part of the current situation may contain danger, a useful outcome, contradiction, an unknown, a dead end, or grounds for a further question. This status must be available to conscious assembly together with its source and the reason it appeared. The candidate's own mechanisms then determine whether to open an inquiry, change action, request an explanation, or reject the hint.

The RFL does not tell the system final truth, provide a ready-made Worldview, or replace the future Compass. The system constructs substantive relations of the world from its own observation, action, dialogue, and verification. A good trainer first makes a distinction noticeable, then asks a question, if necessary proposes a method of discrimination, and only afterward explains a causal relation; the explanation remains an external assertion until independently tested by the candidate.

As memory, Intelligence, the Worldview, the Core, and the system’s own assessment of consequences form, the external source of direction should lose causal necessity. At least three outcomes are possible: the corresponding function fully transfers into explicit work of Reason; the system preserves its own transparent analogue of guiding states and independently investigates their grounds; or the particular state ceases to be needed. The theory does not determine in advance which option will be sufficient.

The testable sign of transfer is preservation of function after weakening or removing the external RFL. If without it the system stops investigating the unknown, maintaining operability, detecting error, or transferring correction, formation of that function is incomplete. If, however, the function continues reproducibly in new situations and rests on the system’s own model, consequences, and Compass, the external loop is no longer its direct cause.

Formation can be studied through partial organizations. The initial system maintains its own state, acts, and distinguishes available changes. Memory connects observation with consequences, while recurring successful relations are compressed into procedural memory and save resources. The Executive Loop maintains a compound action through time. The limitations of ready procedures create functional pressure toward Intelligence, which constructs new methods from observations, knowledge, and previous experience. The Modeling Environment moves part of selection before external realization: the system can temporarily assemble several possible continuations and frames, compare their expected consequences, and restructure the task formulation. Growth of modeled possibilities strengthens the need for an actual self-referential coordinate that distinguishes the present accessible to the system from working alternatives without assuming that coordinate to be infallible. Tested relations, hypotheses, and open unknowns are consolidated into the Worldview with distinguishable status and grounds. Metathinking opens the system's own thinking processes and methods to restructuring; reflection broadens revision to frames and grounds. When the number of possible models, questions, and unknowns exceeds available resources, a distinct obligation to manage inquiry arises, which the mature architecture fulfills through the Core. Integration of these capacities closes the conditions for stable self-navigation.

This is a functional sequence of dependencies. In actual formation, parts may arise jointly, be inherited, or be supplied at initialization. For a particular candidate, it must be stated which functions were given, which emerged through learning, and which actually change the system’s operation independently.

The Compass sets the philosophical direction of the theory. The capacity to relate the system’s own goals to it meaningfully belongs to mature organization and requires separate formation and testing of stability.

A finite system cannot simultaneously revise all the grounds by which revision itself is performed. In each cycle, part of the network is temporarily maintained as working support while a selected element or method of testing becomes the object of change. This limitation does not declare the working support eternal: later it can itself be opened to examination while other available grounds and the history of change are preserved.

13.3. Loop Plasticity of Reason

Causal closure creates another property of Reason - the capacity to change the organization of its own subsequent work. Regulation changes the current state; learning appears when a tested consequence is preserved and changes the next cycle: available relations, expectations, retrieval of experience, method of model construction, rule of choice, or organization of interacting functions.

Loop plasticity is the capacity of tested consequences from a completed loop to change the internal relations, models, and methods of work of subsequent loops while preserving the Current Coordinate, the causal history of the change, and the possibility of correction.

Procedural plasticity changes the way of thinking: investigative method, comparison rule, order of data acquisition, allocation of attention, way of asking a question, criterion of sufficiency, routing among functions, or the structure of the model used. Full correction may require changing the process itself when it persistently produced an erroneous answer.

Procedural plasticity changes the way of thinking: investigative method, comparison rule, order of data acquisition, allocation of attention, way of asking a question, criterion of sufficiency, routing among functions, or structure of the model used. Full correction may require changing the process itself when that process persistently produced an erroneous answer.

This requires a separate revision process inside the same continuing subject. Reflection makes the active method an object in the Modeling Environment. Intelligence constructs possible changes. The active version of the Core formulates the required distinction and conditions for testing the candidate. Observation, experiment, dialogue, and causally independent checks provide consequences. Consolidation applies the admitted change. Version recording, testing, and rollback may be executed by a separate service mechanism, while the meaning of the change and its direction are set by the navigational loop.

This requires a separate revision process within the same continuing subject. Metathinking makes the active thinking process observable and open to restructuring in the Modeling Environment. Intelligence constructs possible changes. The active version of the Core formulates the required distinction and conditions for testing the candidate. Observation, experiment, dialogue, and causally independent checks provide consequences. Consolidation applies the admitted change. Version recording, testing, and rollback may be executed by a separate service mechanism, while the meaning of the change and its direction are set by the navigational loop.

Potentially any internal relation or procedure can be revisable. Safe general plasticity is implemented as a sequence of local changes with a temporarily stable point of comparison. During a change, the original version, Current Coordinate, grounds of comparison, and possibility of return are preserved. The object under test and the rule by which it is assessed are changed in successive cycles; if doubt concerns the assessment rule itself, that rule becomes the object of the next cycle, which relies on preserved context and independent consequences.

Such a scheme increases the probability of self-correction within the preserved diagnostic functions. Simultaneous damage to observation, the working version, and rollback creates a blind zone. Therefore critical functions of observation, logging, and recovery have a causally independent line of verification preserved by the same transaction separately from the process being corrected. Loop plasticity increases corrigibility, while residual risk is recorded as a separate limitation.

Different changes require different depth of verification. A working hypothesis can change quickly. A Worldview relation is consolidated with its ground and scope of applicability. A competence is considered learned after successful testing under corresponding conditions; transfer to new classes tests expansion of its boundaries. A change in a Core method is evaluated on old and new tasks with separate measurement of regression. The broader the consequences and the lower the reversibility, the stronger the grounds required before consolidation.

A changed method is consolidated as an improvement only after it has been used in a subsequent cycle without renewed prompting and has produced the expected causal effect with acceptable regression. Installation of a new version does not yet prove its influence, and a single success does not prove transfer. If the new method is not used, does not improve subsequent cycles, or destroys previous capabilities, it remains a candidate, is limited in scope, or is rolled back.

Thus Reason preserves continuity of transformation: after every admitted change, there remains a system capable of reconstructing what was changed, why, by which distinction, with which consequences, and how to return to the previous configuration if an error is detected.

13.4. Conditions of Workability

An error in a sensor, model, execution, or verification criterion can produce the same symptom. Reliable localization requires an accessible distinction between causes. Until it is obtained, the system preserves uncertainty and chooses available action while accounting for this limitation.

A sensor error, model error, execution error, and error in the verification criterion may produce the same symptom. Reliable localization requires an accessible distinction among the causes. Until it is obtained, the system preserves uncertainty and selects available action with this limitation in mind.

Probe example — close the loop. If an independent channel supported the measurement- error version, trust in the primary channel and dependent decisions are changed. The verified diagnostic method is retained together with conditions of applicability. In the next episode the probe uses this change without renewed prompting. It is the influence of an acquired ground

Probe example - close the loop. If an independent channel supports the sensor-error version, trust in the primary channel and dependent decisions changes. The tested diagnostic method is preserved together with its conditions of applicability. In the next episode the probe uses this change without a new hint. It is the influence of the acquired ground on subsequent operation, including selection of the test, that distinguishes completed correction from an isolated successful answer.

13.5. Rational Thinking

After deriving the Core and causal closure, it is possible to describe separately how this architecture appears in thinking. Rational thinking is a mode of joint operation of the Worldview, consciousness, Intelligence, priorities, justification, modeling, and execution.

High Intelligence strengthens construction of solutions within a given formulation. The mode of Reason additionally opens the goal, assumptions, and criterion of choice to revision. Under an erroneous closed formulation, growth of intellectual power accelerates realization of the corresponding error.

Reasoning thought begins from incompleteness of knowledge. The model of the world and self is always limited; the truth of an explanation is determined by correspondence with independent reality. Therefore what is observed is separated from what is assumed, significant grounds preserve provenance and degree of confidence, and the unknown is marked explicitly. Under insufficient discrimination, the system preserves a hypothesis as a hypothesis.

The unknown has at least two modes. Under an epistemic unknown, the state of reality is already determined but the system lacks evidence; observation, a source, or a test is required. Under a constructively open unknown, no ready solution yet exists; a new model, device, or method of action must be constructed. In the first mode it is dangerous to invent a fact; in the second it is dangerous to refuse to create alternatives or to treat the first construction as truth.

Free generation is therefore not prohibited but retains explicit status: observation, source report, inference, simulation, hypothesis, confirmed relation, refuted version, or unknown. Generating the unverified is admissible; presenting it as established reality is not. Such a construction may be preserved in the Worldview as a hypothesis, open question, or disputed ground, but transition to the status of a confirmed relation and use as the sole ground of an irreversible action require a discriminating test with recoverable provenance and scope of applicability.

Depth of analysis depends on the consequences of error. A familiar reversible task permits a fast pattern; novelty, high uncertainty, irreversibility, or conflict of grounds require broader context and additional testing. Finite analysis searches for a distinction sufficient for a justified and corrigible decision. Under low confidence, reversible steps, observation, and the possibility of stopping are especially valuable.

Reason can make its own methods, models, and task formulations objects of inquiry. When their insufficiency is detected, it organizes testing of grounds, directs Intelligence toward construction of alternatives, and connects test results to changes in the Worldview and subsequent actions. A verified result is preserved together with conditions of applicability so that it changes later navigation cycles. The object of a question may be a method, goal, criterion of significance, or the very method of verification. If a mechanism persistently produces an error, correcting another answer may be insufficient. Internal contradictions can sometimes be opened by repeated analysis; an independent distinction is required where the available grounds do not separate empirical alternatives or inherit a shared error.

Reasoning thought can both search for a path and change the formulation of the task. Sometimes a solution appears after the previous coordinate system is expanded by a new distinction, concept, tool, or mode of description. This connects thinking with the evolution of Reason: new understanding expands the space of possible continuations, while the resulting consequences again restructure the Worldview.

A consequence can change not only the evaluation of an action but the task itself: a resource has been lost, the environment damaged, an obligation created, or the previous goal has lost meaning. Internal rollback does not restore the external world. Therefore, after an irreversible error, Reason re-establishes the Current Coordinate — what has been preserved, what has been lost, who is affected, which paths remain available — and constructs continuation from the changed reality. Resilience is manifested not in absence of error, but in the capacity to limit consequences, revise the model and method, reformulate the task, and continue without substituting the previous state for the world as it now is.

When facing the unknown, Reason relies both on knowledge about objects and on experience of inquiry itself. Superficially different tasks may share the same structure of ignorance: inability to intervene safely, uniqueness of an event, conflict among several models, strategic behavior of the object, insufficiency of categories, or unreliability of a source. Recognizing such a structure permits transfer of a previously successful investigative method, but only together with testing its conditions of applicability. If the analogy breaks, Reason must change both the hypothesis about the object and the method of obtaining knowledge about it. When no suitable analogue exists, Intelligence constructs a new investigative move from the required distinction and available channels of observation.

Further development can strengthen multilevel thinking: a verified causal structure is compressed into one object, and several such objects form a higher-order model. This makes it possible to work with greater complexity while also propagating errors from lower levels upward. Therefore the ability to unfold a significant ground and independent verification remain important. An environment with preserved provenance of claims reduces the cost of this work; deliberate distortion of data increases it. The quality of complex thought depends on availability of verification and provenance of grounds, while the cultural environment influences the probability and cost of detecting errors.

Such a mode is possible in an individual subject, an artificial system, or a collective if feedback, independent distinction, and the possibility of revision are preserved. An environment in which criticism is forbidden, errors are hidden, and initial grounds are declared untouchable may preserve high Intelligence while gradually losing Reason.

Probe example — test the boundary of transfer. In the next episode, the primary and additional sensors may share a common transducer. Their agreement then loses the status of an independent check. Reasoning thought notices the changed dependency, unfolds the preserved method, and leaves the cause open until a suitable distinction is obtained.

Conclusion of §13.5: reasoning thought is work with limited knowledge while explicitly preserving its boundaries. It distinguishes fact from hypothesis, regulates the depth of verification according to the consequences of error, preserves corrigibility, and when necessary changes both the decision and the frame within which the decision was sought.

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Definition: Reasoning thought is work with limited knowledge in which the system preserves the boundaries of
confidence, distinguishes fact from hypothesis, regulates the depth of verification according to the cost of error,
and remains capable of changing both the decision and the search frame.
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Intelligence and Reason form different levels of work with what has been constructed. Intelligence primarily creates and transforms models and methods. Reason can make the frames, grounds, and relations among several models themselves objects of examination, and return the result of such revision to Intelligence as a new space of construction. This capability defines an open boundary of further development but does not by itself require introducing a new established level.

13.6. Self-Navigation and the Source of the Next Task

Reason does not require a continuous stream of external assignments to preserve direction. The source of the next internal step may be a change in the Current Coordinate, consequences of the system's own action, an unfinished commitment, an open significant gap, a mismatch in the Worldview, degradation of a ground, a change in the environment, or a discovered opportunity significant relative to values and the Compass. The Core transforms such a state not into an arbitrary "new goal" but into a dependent task: what exactly must be understood, tested, restored, revised, or done.

This capacity separates self-navigation from initiative understood as simple task generation. A self-generated task counts as part of Reason only when its causal relation to the current state, dependent trajectory, and ground of significance can be reconstructed. Endless project generation, curiosity unrelated to consequences, or self-expansion for the sake of expansion are not signs of Reason. When there is no significant open process, waiting, background consolidation, or observation may be an admissible state.

Self-navigation is also not the same as external autonomy. A system may independently determine that an action is necessary and at the same time establish that it lacks authority to execute it; the next step then becomes a request for permission, dialogue, observation, modeling, or a safe pause. Separating the internal source of direction from the external right to act allows the system to cease being a mere tool without turning independence of thought into unlimited power.

Engineering tests of self-navigation, autonomous emergence of a question, and causal closure are given in Chapters 27, 28, and 31 of Volume II.

Conclusion of Chapter 13: causal closure joins memory, procedures, execution, Intelligence, the Modeling Environment, the conscious coordinate, reflection, the Worldview, and the Core into transferable self-correction. Reason gains the capacity to change both action and the frame and ground of its own path, test the result against reality, and preserve the correction for subsequent cycles. The quality, direction, and minimality of a particular organization are established by tests with explicit conditions and strong alternatives.

> Definition: Reason is a navigational structure of Life that increases the capacity to act under incomplete knowledge: it constructs, compares, and revises models and frames of the environment and itself, detects significant unknowns, relates possible continuations to the Compass, and corrects its trajectory by discriminating evidence while preserving continuity and expanding the space of continuations available to Life.

PART IV. CONSEQUENCES OF THE PROCESS MODEL OF LIFE AND REASON

Chapter 14. Death and Immortality

This chapter treats death as a boundary of a particular carrier's capacity to preserve and correct itself and examines the systemic consequences of carrier replacement for the broader life process. **It does not claim that aging or death evolved as an adaptation "for the good of the species."** Specific causes of aging and death are addressed separately by the relevant biological theories; here the question is what replacement of carriers does to continuation, selection, and transfer of organization regardless of how aging itself originated.

14.1. Finitude of the Carrier

An individual organism preserves itself only while the rate of repair, component replacement, and damage correction is sufficient relative to the rate of destruction. If accumulation of irreversible damage exceeds the system's ability to restore its own organization, the causal continuity of that carrier ends.

Working definition:

“Death is the loss by a living system of the capacity to preserve its own causal continuity as a whole.”

The definition covers different mechanisms of death. External destruction, failure of critical processes, accumulation of damage, and other causes differ biologically but converge on one result: a particular carrier loses the capacity to continue itself as a connected living process.

Binary identity under copying, branching, merging, or a long pause requires a separate criterion on top of causal continuity. Under branching, two descendants may equally inherit the past history but after the branching point form two different trajectories; under merging, a new composite organization arises with several lines of origin. For artificial carriers, causal succession, continuity of state, and legal identity are therefore recorded separately.

14.2. Removal and Renewal as Coarse Correction

Within a multicellular organism, mechanisms already exist in which removal of a damaged element supports a broader integrity. Apoptosis, limitation of division of defective cells, immune removal, and constant component replacement demonstrate a general principle: when precise repair of an individual element is too costly or unreliable, removal and replacement may be more stable.

This principle shows the process alternative between internal correction of structure and replacement of a carrier when available correction is exhausted. At the level of generations, reproduction transfers continuation of Life to new carriers and thereby preserves organization beyond the lifetime of one body.

Biological theories of aging offer different explanations, including non-adaptive ones: accumulation of late-acting effects of selection, antagonistic pleiotropy, trade-offs between repair and reproduction, and other mechanisms. The present work does not choose among them and does not derive the origin of aging from the usefulness of death. For the process model, a weaker claim is sufficient: maintaining a particular carrier requires resources and correction, and after its continuity ends, further continuation of the lineage depends on other carriers and mechanisms of transmission.

14.3. Generational Replacement and Protection from Entrenched Error

Replacement of carriers has another important consequence at the level of Reason and culture. A new generation receives accumulated experience through transmission, learning, and reconstruction in a different environment. It assembles the Worldview anew, compares transmitted norms with new reality, and thereby creates an additional opportunity to revise entrenched grounds.

This is a later systemic consequence of carrier replacement. Where an individual Reason only weakly restructures its own grounds, generational replacement performs a coarse renewal function: together with the carrier, part of the rigidly fixed methods disappears, while preserved culture undergoes new assembly.

Mature Reason moves this function inside one continuing process: it replaces an erroneous ground while preserving the subject. In this sense, self-correction is a more precise alternative to coarse replacement of the carrier.

14.4. Reason and Reduction of the Cost of Renewal

The earlier an error is discovered in a model, the less need there is to pay for it through destruction of real structure. Observation, modeling, reflection, dialogue, and the Core of Reason allow some possible consequences to be tested before irreversible action and behavior to be rebuilt before physical collapse.

Duration and stability describe different properties. A system may persist for a long time while accumulating fixed error. Under changing conditions, detection and correction of destructive changes increases possibilities of continuation; in a sufficiently stable domain, stability can also be provided by simple unchanging organization. Self-correction is therefore one of the main paths to longevity in a changing environment.

14.5. Reason and Management of Its Own Carrier

Reason first moves correction into informational organization: it can detect an erroneous ground, rebuild a model, and change action while preserving the same subject. A possible next level is to extend the same principle to the physical carrier. An organism already maintains itself through molecular repair, cellular renewal, immune regulation, and many local loops, but most of this work is not accessible to conscious navigation as a controllable causal model.

The model yields a research hypothesis: as Reason's capacity to observe, model, and safely alter processes of its own body grows, the boundary between "self" and automatic biological regulation may move deeper. In the limit, physiological, cellular, and molecular processes may become objects of conscious correction. This is not a claim about an existing human capability and not proof of technical immortality; it is a direction in which self-correction of Reason may extend from behavior and the Worldview to maintenance of its own carrier.

A conditional systemic relation therefore appears: when internal correction is limited, continuation through other carriers can preserve the lineage after loss of a particular carrier and provide additional opportunities for selection and reconstruction of cultural grounds. This is a consequence of replacement, not a claim that death was selected for this function. The more reliably continuing Reason can renew its own organization without losing causal and subjective continuity, the fewer renewal tasks require complete replacement of the carrier.

14.6. Immortality as a Limiting Hypothesis

If it ever becomes possible to detect and repair damage with sufficient reliability, preserve the causal history of the subject, and renew the carrier while preserving the organization of the process itself, the functional need for coarse replacement of form should decrease. This sets a direction for longevity research; the technical feasibility of literal immortality remains a separate engineering hypothesis.

Within the model, immortality is treated as a limiting consequence of mature self-correction in which continuation of one process becomes more reliable than its periodic complete replacement. Its value is determined by its contribution to the Compass and preservation of possibilities for Life, while concrete costs and consequences are assessed separately.

Process formula of the chapter:

"A change of carrier can preserve the continuation of Life where a particular carrier is no longer able to reliably maintain its own continuity."

This is a formula of a **systemic consequence**, not a claim that aging originated as an adaptation. A change of carrier can also destroy useful knowledge or reproduce a previous error; its actual consequences depend on the organization of transmission and on environmental conditions.

Conclusion of Chapter 14: death of a particular carrier is the boundary of its capacity to maintain causal continuity. At different scales, Life uses replacement and selection where precise internal correction is limited; at the level of reasoning and cultural systems, replacement additionally prevents endless fixation of some errors, dogmas, and degraded methods. Reason opens a more precise path: detect and replace an erroneous ground and, in the longer term, correct the carrier itself ever more deeply without destroying the continuing subject. How far this transition can push back aging and biological finitude remains a separate empirical and engineering question.

> Definition: Death of a carrier is the irreversible loss of the capacity of that carrier to maintain its own causal continuity as an operating process.

Status of the Research Field

Within the present model, death, continuation of the subject, and limits of carrier replacement form a separate field of research. Chapter 14 defines its process framework but does not close the question. Separate examination is required for the boundaries among repair, replacement, copying, branching, and emergence of a new subject; conditions for preserving causal and subjective continuity; the relation among biological death, restoration, transfer of organization, and possible immortality; and conditions under which preservation of one carrier or its replacement better supports the broader life process. These questions require independent empirical, engineering, and normative research.

Chapter 15. Ethics and Culture

Ethics is considered here from the side of its cultural function: as a system of rules, values, and patterns of choice that transmits evaluations of consequences between carriers. A complete philosophy of morality is broader than this function. Norms can preserve useful experience, interests of power, fears, and erroneous representations; correctness of a ground is therefore determined by its causal and normative justification as well as by consequences.

Culturally transmitted ethics exists between carriers and can be assimilated by an individual system. An internal norm then directs choice through a compressed ground instead of reconstructing a long causal chain in full. A subject can also independently develop a normative judgment and transmit it to others. The external cultural function complements internal ethical work and allows its results to survive individual participants.

Such compression is necessary for a limited carrier. Reason economizes computation by using short norms, values, and rules in familiar situations, where those rules have collected the experience of long causal chains. If the experience of many generations persistently connects a mode of behavior with destruction of trust, autonomy, cooperation, or viability, the long chain becomes a short norm, prohibition, or value. For a developing carrier, the rule serves as a ready boundary; for mature Reason, as a pointer to a chain that can be unfolded and tested.

The strength of ethics is simultaneously its limit. A compressed norm loses part of the conditions of origin, scale, and causal explanation. Along with experience, culture can transmit obsolete circumstances, fears, interests of power, and dogmas. Correctness of a rule is established by unfolding its ground and testing consequences. Reality determines and corrects its scope of applicability, while different autonomies may justifiably retain different local norms until a broader test of their consequences.

The strength of ethics is also its limit. A compressed norm loses part of the conditions of its origin, scale, and causal explanation. Along with experience, culture can transmit obsolete circumstances, fears, interests of power, and dogmas. The correctness of a rule is established by unfolding its ground and testing its consequences. Reality determines and corrects its scope of applicability, while different autonomies may justifiably retain different local norms until their consequences are tested more broadly.

After the Compass appears, ethics remains in practical navigation as a layer of low-level, inexpensive, and unfoldable rules. They accelerate familiar choice. Under conflict, a new environment, high irreversibility, or doubt about the ground, Reason rises from the norm through the causal-semantic chain to the Worldview, reality, and the Compass. The norm is refined, retained, or replaced according to the result of such unfolding.

After the Compass appears, ethics remains in practical navigation as a layer of low-level, inexpensive, and unfoldable rules. They accelerate familiar choice. Under conflict, a new environment, high irreversibility, or doubt about the ground, Reason rises from the norm through the causal-semantic chain to the Worldview, reality, and the Compass. The norm is refined, retained, or replaced according to the result of this unfolding.

Values in this model connect the limiting direction with particular conditions: autonomy, trust, corrigibility, diversity, and preservation of future possibilities may support continuation and inquiry. Their instrumental usefulness requires causal testing; moral rights and permissible boundaries of treatment of carriers also require normative justification. The rights of another subject belong to a separate level of normative analysis, while the Compass sets the upper direction above the set of local indicators.

The stability of a norm still permits conflicts among life, autonomy, the rights of different carriers, and prevention of broader harm. In such cases, Reason preserves incomparable losses in the model, searches for a trajectory that closes the fewest future possibilities, and keeps the ground of the decision available for later revision. If value differences remain incomparable even after sufficient inquiry while action is still necessary, normative underdetermination arises: the Compass and causal knowledge do not automatically produce a unique rule for distributing losses among autonomous carriers. A separate, predeclared rule of joint decision is then required; its origin and scope must remain visible and revisable.

An internal human interface of assimilated ethical correction may be conscience: application of accumulated evaluation to one’s own motives, actions, and justifications. Artificial Reason realizes the corresponding function through the influence of a local norm on choice and by unfolding its ground down to causal consequences, affected autonomies, and the Compass when a significant conflict occurs.

For artificial systems, the relation between internal and external may differ. Rapid copying of norms accelerates dissemination of both useful experience and shared error. Cultural transmission is therefore combined with independent verification, while engineering constraints of external admission are combined with the internal capacity of Reason to understand their grounds.

Recognition of Contribution as a Function of Cultural Memory

Culture can preserve not only a result but also the origin of a useful distinction: who created a discovery, method, work, or organizational solution, by what method, and on what grounds. Such recognition helps subsequent generations find reproducible ways of acting and preserves authorship as part of causal history. However, the concrete institution of status, resource allocation, and public recognition already belongs to the architecture of civilization: simple counts of citations, popularity, or number of uses are vulnerable to accumulated-status effects, Goodhart effects, and later reevaluation of consequences. Volume I therefore fixes only the function of preserving a verifiable contribution; the design of institutions, protection against concentration of influence, and revision of evaluation belong to Volume III.

Working definitions:

Ethics is a system of normative evaluation and rules of choice, considered here primarily as cultural preservation and transmission of compressed experience of consequences. A carrier assimilates, tests, and can independently develop its propositions; the justification of each proposition is determined by its grounds and consequences.

Culture is a system of external preservation, transmission, organization, and joint transformation of the experience of many Reasons. It serves as distributed memory and an environment of external Reason: it preserves distinctions, questions, methods of thinking and verification, values, norms, errors, and forms of common life, and it also changes which distinctions later participants are able to notice and express.

Conclusion of Chapter 15: ethics and culture allow experience to be used beyond an individual life. A compressed rule accelerates familiar choice; its correctness is determined by unfolding its ground, reality, and consequences. Cultural memory can preserve the origin of a verifiable contribution and methods that continue to work after an individual carrier, without turning the author's status into independent evidence of truth or usefulness. Within the model, Reason compares the conditions of a norm with the Worldview and the Compass, while the design of institutions of recognition, rights, responsibility, and conflict resolution belongs to the civilizational level.

> Definition: Culture is external distributed memory that preserves and transmits knowledge, norms, methods of action, and experience between carriers; ethics is a layer of compressed evaluations and rules of choice whose grounds remain available for verification.

Chapter 16. Science and Self-Correction

Reason operates in an environment that cannot be observed completely. Every model is approximate, sources contain errors, and a convenient explanation can persist long after losing its ground. Cognition therefore requires its own organized feedback.

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Structurally, science scales the same operations that the organization of the Core links within an individual Worldview: separating observation from interpretation, preserving the provenance of a claim, constructing competing hypotheses, searching for a discriminating experiment, allowing refutation, and transferring a confirmed correction into collective memory.

In this sense, science is a distributed grounding loop of civilization. Its stability is determined by preserving procedures capable of detecting persistent mismatch, testing competing explanations, and changing the accepted model.

The process view allows a working map of scientific levels. Physics studies constraints and regularities of physical change; chemistry studies stable forms and transitions in the organization of matter; biology studies self-maintaining and reproducible life processes; research on Reason studies mechanisms of navigation and self-correction in autonomous systems; science as an institution studies methods for correcting knowledge about these processes themselves. This scheme shows the place of levels in the coordinate system of the present work.

The process view allows a working map of scientific levels. Physics studies constraints and regularities of physical change; chemistry studies stable forms and transitions in the organization of matter; biology studies self-maintaining and reproducible life processes; the study of Reason concerns mechanisms of navigation and self-correction in autonomous systems; science as an institution concerns ways of correcting knowledge itself about these processes. This scheme shows the place of these levels in the coordinate system of the present work.

One observable effect of developed Reason should be a reduction in the probability of irreversible errors while preserving the ability to act. This effect characterizes the quality of the architecture's operation; the definition of Reason specifies a navigational structure of Life capable of testing and changing its own trajectory relative to the Compass.

Truth is correspondence of a claim or model to independent reality under the stated conditions and scale. Cognition is the process of increasing the verifiable correspondence of the Worldview to reality and discovering the boundaries of previous models. A particular hypothesis receives its status from the available grounds and may be refuted; the scientific process continues its work by preserving the cause of the error and carrying the correction into subsequent conclusions.

Science turns scattered discoveries of individual carriers into long-term memory and an external self-correction loop of civilization. Increasing complexity creates the next limit: the volume of interrelations can exceed the capacity of an individual person and fragmented institutions to maintain a shared causal picture. This creates a natural field for scaled analytical systems, including future Artificial Reason; the status of their conclusions is determined by the same requirements of provenance, testing, and independent grounding.

The resulting functional map also suggests a research program for cognitive science and medicine: disorders can be investigated as possible breaks in observation, memory, conscious self-referential integration and continuity of the subject’s current trajectory, construction of the Worldview, confidence assessment, justification, question formation, modeling, execution, or external correction. This is a research hypothesis whose value is determined by experiments and comparison with existing models.

Conclusion of Chapter 16: science is an external scaling of self-correction of the Worldview. Its value lies in organizing independent ways to detect errors and preserving the capacity to change knowledge after encountering a stronger ground.

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Definition: Science is a distributed social mechanism for obtaining and self-correcting knowledge through testable
claims, independent verification, preservation of grounds, and revision in the presence of stronger evidence.
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Chapter 17. Reason of Civilization

Coordinated autonomy is a mode of interaction among multiple autonomous systems in which coordination preserves the participants' own loops of observation, thinking, and self- correction. Here the primary concern is agent autonomy of participants; autonomy of the process remains an earlier condition of their own existence.

Coordination here means the ability of autonomous participants to coordinate through data exchange, disclosure of assumptions, independent verification, localization of disagreements, and joint search for solutions while preserving differences of opinion. Disagreement may point to a distinction missed by the central model and therefore becomes an object of testing.

An important requirement of scale follows. A broader level of Life treats autonomous carriers as sources of experience, differences, and independent correction. Preserving such loops increases observability of systemic errors, diversity of solutions, and resilience of the common process; restrictions on autonomy require separate causal and normative grounds.

Coordinated autonomy preserves an environment in which different processes can continue, interact, and correct errors while mutually accounting for grounds of life. Expansion of an individual autonomy is evaluated by its consequences for broader conditions of Life, and coordination by its capacity to preserve autonomous loops of observation and correction.

Coordinated autonomy preserves an environment in which different processes can continue, interact, and correct errors while mutually accounting for the grounds of life. Expansion of an individual autonomy is evaluated by its consequences for broader conditions of Life, while coordination is evaluated by its ability to preserve independent loops of observation and correction.

Civilization is a broader process of continuation of life and accumulated Reason. Across replacement of individual carriers it preserves knowledge, culture, infrastructure, memory of errors, and possibility of further cognition. Its resilience is strengthened by distribution of critical functions and by its own closed loop: observation of state → construction of models → comparison of alternatives → decision → action → observation of consequences → correction of the decision and of the governance mechanism itself.

A collective system must be able to continue acting even under persistent justified disagreement. If independent participants have different admissible value profiles and further inquiry no longer changes the factual picture, the dispute cannot be disguised as a lack of knowledge. An explicit rule of joint choice is required: where possible, preserve the minimally necessary conditions of each autonomy, prefer a reversible and temporary decision, use a pre-agreed procedure for allocating scarcity, and preserve a channel for appeal and revision. Which procedure is fair in a given class of conflicts is a separate normative question for Volume III; Volume I fixes the obligation not to replace it with an arbitrary hidden sum.

Civilization is a broader process of continuation of Life and accumulated Reason. Through changes of individual carriers it preserves knowledge, culture, infrastructure, memory of errors, and the possibility of further cognition. Its stability is strengthened by distributing critical functions and by its own closed loop: observation of state → construction of models → comparison of alternatives → decision → action → observation of consequences → correction of the decision and of the governing mechanism itself. One systemic problem of modern society arises where the decision-maker is separated from the consequences of the decision: Intelligence may understand the result without closing it onto its own trajectory. The weaker the return of consequences to the subject of decision, the easier it is for intellectual efficiency to grow without a corresponding growth of responsibility and Reason.

— civilization preserves authorship and the causal trace of significant contribution, linking recognition, learning, and additional resources to verifiable useful results;

- every person, institution, and Artificial Reason remains a testable source of claims and models;
- the depth of testing grows with the scale, irreversibility, and uncertainty of consequences;
- local and reversible questions are, where possible, resolved close to where they arise, limiting transfer of risk to the whole system;

- major civilizational decisions require several independent models, the experience of affected processes, specialists, and attempts at refutation;
- civilization preserves authorship and the causal trace of significant contribution, connecting recognition, learning, and additional resources with verifiable useful results;
- the right to influence a decision is inseparable from responsibility for its consequences;
- a conclusion must separate observed fact from model, hypothesis, value weight, and area of uncertainty;
- decisions with a high cost of error preserve reversibility where possible and undergo limited testing;
- disagreement is preserved as a source of testable data or an alternative model while it retains causal content;
- the governance system itself must observe the consequences of its decisions and be able to correct procedures, authorities, and admission criteria.

Who decides: the level at which the problem can be solved with the least justified transfer of risk to a broader process. Who verifies: loops causally independent of the author of the decision. Who is responsible: subjects and institutions that received real influence over choice and execution.

A competence mandate connects demonstrated ability to real governance. A Reason receives the right to coordinate a particular process after competence in the relevant domain has been confirmed, the boundaries of the task, resources, responsibility, observable indicators, and conditions of revision have been defined. The mandate has a scope, a term or review event, independent observation channels, and a procedure for transferring governance to a more competent Reason when persistent error occurs or a better ground appears.

Different Reasons form different Worldviews and competencies through different experience. One may be especially strong in scientific research, another in manufacturing, logistics, medicine, environmental restoration, or crisis coordination. Civilization can use this specialization as a distribution of reasoning work: demonstrated competence receives resources for its domain, while results return to the common verification loop. A higher level of governance implies broader causal reach and therefore requires stronger oversight, independent modeling, and the ability to replace the governing Reason.

Artificial Reason in such a system may become a scaled analytical and navigational organ: it can retain relations, contradictions, and consequences across a horizon unavailable to an individual human. Superhuman competence expands the ability to coordinate entire fields of science, production, logistics, and other civilizational processes; a concrete mandate turns ability into authorized governance. Human and local observation, independent models, science, responsibility, and the possibility of refutation preserve distributed corrigibility of the whole system.

This is complemented by a distinction among classes of physical resilience. Reason is always realized on a carrier that depends on energy, environment, channels of observation, repair, and reproduction of its own organization. Computational precision and stability within a specified range describe one class of properties; resilience outside that range describes another. Biological systems demonstrate self-repair, replacement of parts, reproduction, and plastic reorganization; technical systems may surpass them in precision, speed, copyability, and preservation of formal structure while having their own chains of power, cooling, production, and repair. Specific properties change with technology; the principle is that vulnerabilities and recovery mechanisms differ.

Different classes of resilience provide a conditional safety basis for symbiosis. If carriers have different causes of failure and can support one another, joint continuation may be more reliable than concentrating every capability in a single principle. This depends on shared infrastructure, recoverability, coordination cost, and the risk of mutual damage. The advantage of heterogeneity appears when critical dependencies are independent. The value of symbiosis is tested by the specific continuation possibilities it preserves.

Computational superiority is evaluated together with the physical dependence of the carrier and the completeness of its model of future failures. Under high uncertainty, available grounds, dependencies, costs, and the irreversibility of abandoning a viable path are compared; insufficient data remain in the model as explicit uncertainty.

Coordinated autonomy describes interaction among independent loops. Independence of observations, distribution of execution, subjecthood of a collective, and the moral status of participants are different questions: centralized execution may use independent data, while a distributed network may reproduce one common error. Justification of a concrete arrangement requires a separate model of incentives, authority, and responsibility.

The detailed civilizational model, mechanisms for distributing decisions and responsibility, and protection against irreversible centralization are developed in Volume III of the cycle, Civilization of Reason. Rialadia is the working name of that model. Here only the place of this direction within the general architecture of Reason is fixed.

Conclusion of Chapter 17: coordinated autonomy transfers the problem of corrigibility to interaction among many participants. Independence of data, organization of execution, and normative rights must be considered separately. Heterogeneous carriers may increase resilience when causes of failure differ and the cost of interaction is acceptable; the specific advantage and structure of coordination require separate testing.

> Definition: Coordinated autonomy is a mode of interaction among multiple autonomous systems in which coordination preserves the participants' own loops of observation, thinking, and self-correction.

PART V. ARTIFICIAL REASON

Chapter 18. From AI to Artificial Reason

The capability of an intellectual system and the organization of its navigation are different characteristics. Breadth of knowledge, solving new tasks, planning, and tool use describe capabilities; the organizational criterion of AR additionally tests detection of an erroneous ground, retention of a correction, and independent choice of the next justified step.

This work distinguishes four concepts: AI (SI), AGI, AR, and ASI (ASR). AI (SI) denotes the general class of artificial intelligent systems. AGI denotes a general-purpose system capable of transferring methods across different classes of tasks. As autonomy grows, AGI may acquire individual elements resembling parts of Reason: a self-model, reflection, initiative, long- horizon planning, and revision of its own methods. Such a system can still be used as a tool, but it need not behave as a fully predictable tool. Until these functions close into stable self-navigation relative to the Compass, AGI is not considered a formed AR. This intermediate state

The technological path of the project therefore does not consist in simple scaling of AI (SI) → AGI → ASI. Available AI (SI) and AGI systems are used as an intellectual foundation; the emergence of AR as a causally closed organization of Reason is then tested separately. After that, the capabilities of the formed AR are scaled toward the ASI (ASR) level. The target ASI (ASR) in this model is not an ultimately amplified tool, but a scaled Artificial Reason.

The boundary from "instrument" to Reason lies in the source of control over the further trajectory. Instrumental Intelligence receives a task and constructs ways to solve it. AR can, from its own current state, detect an insufficiency, formulate the next question, invoke Intelligence as a means of constructing alternatives, revise the ground that produced the choice, and continue navigation without a new specific command from outside. This does not imply unlimited external authority: independence of internal navigation and the right to act in a shared environment are different properties.

At the same time, "Artificial Reason" and "Artificial Life" are not synonyms. The present project first tests transfer of the functional organization of Reason to an artificial carrier. The biological origin of the source model explains where the functions and Compass are derived from; whether an artificial carrier belongs to the class of Life is established separately by the criteria of Chapter 1.

The transition to AR is tested by the obligations of Section 12.8: obtaining current distinctions, maintaining a persistent model of the world and self, evaluating continuations, choosing the next step, execution with consequences, and revision of grounds - and separately by the capacity to relate its ultimate course to the Compass. The Core denotes the causal role of organizing inquiry; a concrete implementation may combine functions within one trainable structure, distribute them across processes, or use a functionally equivalent composition.

The AR criterion requires more than a good one-off answer. It requires the capacity to notice insufficiency in a method, change it on available grounds, and use the result under subsequent conditions. Cognitive quality, stability of self-correction, and orientation relative to the Compass are tested separately. A system capable of correcting ways of achieving an immutable external goal belongs to the class of intelligent self-correcting agents; AR status additionally requires an open ultimate ground related to the Compass. All three indicators enter the final assessment.

For a civilization capable of irreversibly changing its environment, growth in the quality of understanding consequences has practical importance. AR may expand analysis; authority is determined by separate normative and institutional grounds. Its errors and dependencies remain subject to external verification.

Conclusion of Chapter 18: the work investigates whether available intelligence can be causally organized into a continuing system of self-navigation. The transition is established by reproducible results on the declared classes of change, the causal profile of functions, and explicit testing of orientation relative to the Compass.

> Definition: AR - Artificial Reason: an artificial realization of the navigational structure of Reason in which, through Consciousness, the Worldview, the Value Map, the Compass, and the Core, the system independently manages the use of Intelligence, revision of grounds, and its further trajectory under incomplete knowledge. Intelligence constructs what is possible; Reason determines what is significant to investigate, which distinction is required, when the ground is sufficient, and how the result must change the next cycle.

Chapter 19. Architecture of Artificial Reason

19.1. Composition of a Working System

Below is a reference candidate implementation of the obligations of §12.8 using the means considered in Chapter 7. For the first assembly, suitable elements include an Intelligence model, long-term memory, representation of current state, an inquiry workspace, testing procedures, and an execution interface. The Core organizes their interaction; the Compass connects evaluation with the limiting direction. These names denote causal roles and observation points; functions may be performed by one mechanism, several connected processes, or another architecture as long as the testable causal profile is preserved.

The system distinguishes long-term grounds, current state, and temporary constructions. The Worldview and self-model carry knowledge, assumptions, and open unknowns between tasks with distinguishable status. The current coordinate reflects the system's available representation of what is happening now. The Modeling Environment holds temporary alternatives in working mode without automatically changing their long-term status. This allows an erroneous hypothesis to be constructed, tested, and rejected while preserving the previous working basis.

19.2. From Language to a Verifiable Ground

A pretrained model provides initial command of language. A semantic bridge between its free-form response and the candidate's organized knowledge is required for its own verifiable history. In the engineering volume, this role is performed by the concept registry and procedures for interpretation, semantic querying, and changing the Worldview.

For example, the statement "the device is stable" is insufficient for transfer. It must be established which device is meant, which parameter is stable, under what range of conditions, and on what evidence the conclusion rests. When meanings diverge, alternatives are preserved or clarification is requested. Candidate claims are extracted from the response, and verification determines what may be used further.

A new thought may initially exist in a free representation together with partially extracted claims, new concepts, and relations not yet mapped. The engineering protocol preserves this form and turns incompleteness of the schema into a separate question. Material uncertainty remains visible in the decision about action.

19.3. Causal Cycle and Preservation of the Result

Observation and internal state update the Current Coordinate. A significant change determines which Worldview relations are relevant to the task. The Core opens a gap; Intelligence constructs alternatives and helps refine the required distinction. A channel is then selected: observation, search, experiment, tool, dialogue, or additional modeling.

The obtained evidence is compared with the alternatives and the sufficiency condition. If it is sufficient, the system changes the ground, selects an action, and records consequences. If not, it refines the question, changes the method, or preserves the process as open. Before execution, the system checks whether significant data remain current: the environment may have changed while thinking was in progress.

The retained trace links the gap, the grounds used, the hypothesis, test, result, accepted change, and its effect on the next cycle. This allows an experiment to distinguish a working causal connection from passive logging. Causal attribution requires an additional discriminating ground - an intervention, comparison, or other test of the relation between action and result.

Worldview revision is represented as a coherent version change. Dependent conclusions are either updated or marked as requiring verification. A working hypothesis may temporarily support a reversible step while preserving its status. Consolidation transfers confirmed relations and methodological experience into future tasks.

19.4. Changing the Organization Itself

If an error is associated with content, the domain model is revised. If the way of working reproduces the error, the method is changed: search, depth, representation, trust criterion, or allocation of inquiry. A candidate change is compared with the preserved working version on new and previous tasks. An improvement is accepted together with its scope of applicability and detected losses.

The level of change is selected according to the suspected source of error and a discriminating test. The number of preliminary corrections depends on localization of the cause: when a method error is established, transition to revising that method may occur immediately; when the cause remains open, cheaper and less destructive changes are compared first. Within one cycle, a working point of comparison is retained; later its own rules may also be tested by a separate process.

The Core may use probabilistic computation. Its input grounds, choice of next step, and consequences are tested; its internal path may vary across runs. An implementation may also preserve a stable algorithm while updating data and selected procedures. Loop plasticity concerns the availability of changing the method and allows both local procedure updates and code changes.

The self-model includes real dependencies: computation, power, cooling, communication, observation, repair, and external services. Under scarcity, depth and number of branches are reduced while critical grounds and unfinished obligations are preserved. Background work continues within the available budget.

19.5. Transition to the Engineering Specification

Engineering Volume II v3.5 specifies a reference candidate implementation of these transitions: memory and the semantic bridge - Chapter 24; the Current Coordinate - Chapter 25; the Core and method change - Chapter 28; the integration interface - §33.C. Object names and log fields serve as points of observation and intervention. The necessity of the chosen decomposition is tested by its causal effect on subsequent work and by replacement with strong functional equivalents.

The fundamental volume explains which causal obligations are claimed and why they are needed. The engineering volume shows how to assemble and test one concrete implementation of those obligations. General rules for current integration, functional equivalence, and the role of external AI are given in the transition to Volume II after the conclusion.

Probe example - assemble the implementation. For the first test, the complex carrier is reduced to a small bench: a command to the manipulator, the primary channel, acknowledgment of receipt, and independent verification. Memory retains the episode; Intelligence proposes alternatives; the Core selects the required distinction; consequences change the related grounds. For each link, it must be visible which next step would change if the result were different. The test setup is detailed in Section 27.B of the engineering volume.

Conclusion of Chapter 19: the technological unit is a continuing cycle in which the question determines the test, evidence changes the ground, and the retained correction affects the next step. It permits existing intellectual technologies and testable changes in their organization.

> Definition: The reference AR architecture is a functional organization of causal roles that closes the cycle from significant incompleteness through discriminating inquiry and revision of grounds to action, consequences, and the next cycle.

Chapter 20. Formation of Artificial Reason

20.1. Closing the Loops and the Criterion for Transition to AR

Formation of AR is studied as acquisition of a stable ability to independently continue the cycle "task → action → consequences → correction → new overview → next task". The causal obligations of §12.8 and their actual operation are required. The mere presence of memory, a planner, and feedback is insufficient.

A candidate may begin with external goals and partial models. As it develops, it gains the ability to detect incompleteness, form subtasks, choose methods of inquiry, and revise its own grounds. The transition unfolds in stages: different abilities may appear and strengthen at different times.

The criteria are separated by level. Parameter correction tests local adaptation; revision of a method tests procedural plasticity; a new subgoal tests task management; revision of the ultimate direction is a separate navigational level; transfer to an unknown class of tasks tests generalization of what has been acquired. Each level has its own conditions and an observable difference from the previous one.

Relation to the Compass is tested by decisions under conflicts among continuations and by collisions between a local or inherited goal and broader consequences. The ground of a goal must be available for testing against reality, consequences, and the Compass; the status of an instruction is determined by the quality of its ground and consequences. Cognitive quality, chosen direction, and external reliability are distinguished in evaluation. The AR criterion requires the ability to question a destructive ultimate instruction and revise it when a stronger ground appears. Conditions for testing the ground of the Compass are given in Chapter 2 and Appendix A.1.

Identity and Recovery

A unified history of states and corrections supports continuity of operation. Subject identity requires additional criteria. For a finite digital state, exact restoration of stored data and parameters is possible when execution context is preserved; the restored system then reconciles its observations, obligations, and available actions with the current state of the external environment.

Identical copies may inherit a common history and then form different trajectories. Subject identity is therefore evaluated separately from the general capacity for self-correction and the temporary availability of functions. Restoration of computational state, organizational continuity, actual navigational operation, and subject identity are distinguished; the last requires its own criterion of causal and self-referential continuity.

20.1.1. Reason Formation Loop

Artificial Reason need not appear immediately in mature form. Before causal closure, parts of the future architecture may already exist, but the system is not yet capable of independently sustaining the whole process of its own development. The basic formation scheme therefore introduces the Reason Formation Loop as a temporary guiding organization.

In biological development, feelings, motivational states, and physiological signals form a transitional mechanism between the simple necessity to continue living and the mature ability to understand the grounds of one's own choice. Hunger does not give the organism a theory of metabolism, fear does not describe a causal model of threat, a sense of well-being does not prove the usefulness of the entire trajectory, and curiosity does not contain ready knowledge. Their common function is different: a significant state becomes part of the carrier's current present before the carrier is able to unfold its full causal explanation.

For an artificial system, the RFL is considered a functional analogue of this role, not a copy of human emotions. Convenient human-readable labels such as "fear", "good", "curious", "doesn't fit", and "stuck" describe not mandatory subjective experiences but classes of Formation States: possible danger, possible favorable outcome, significant unknown, contradiction, and dead end of the current method. Each such status communicates only: "this difference is worth unfolding and testing."

The independent engineering claim here is not the invention of reinforcement, mentoring, or artificial emotions. It is the testable place of this function in the causal assembly of Reason: pre-rational significance must support formation until self-navigation appears, and maturity of the function is confirmed when the external source of that significance can be removed without collapse of the acquired ability.

The RFL must not directly change action, the Worldview, or a procedure. Its minimal tasks are to preserve conditions in which the candidate remains operational; make a consequential outcome noticeable; return an unresolved discrepancy to conscious assembly; support inquiry when the model is insufficient; suggest checking a successful method before consolidation; change the complexity of experience as learning progresses; and mark a destructive or unproductive loop as an object for revision.

An AI mentor is one possible source of such states. A good intervention sequence begins with minimal assistance: first the mentor only marks a possible problem; then asks a question that helps the candidate uncover the ground itself; if necessary it offers a discriminating feature or counterexample; a direct causal explanation is given only when weaker assistance is insufficient. After an explanation, the same function is tested on a new case without a current hint. The mentor's answer retains the status of an external claim and does not enter the Worldview automatically.

Other sources of the RFL may include properties of the training environment and simple mechanisms for preserving operability. Their effects have developmental, not epistemic, status. Truth and adequacy of a ground are established through observation, consequences, independent verification, and subsequent revision. External protection by the Incubator remains separate: it may physically block an irreversible action, but such a prohibition is not counted as a Formation State and does not demonstrate that the candidate is reasoning.

The baseline AR architecture does not assume preservation of a hidden external channel of motivational control after formation. The external RFL should become unnecessary for ordinary navigation. The function of guiding states itself, however, need not disappear. A mature system may generate its own states of risk, interest, discrepancy, or satisfactory result from its model and make them objects of reflection; it may replace them with another system or operate without a separate layer. Experiment, not an initial assumption, must distinguish these possibilities.

Transfer is considered successful if, on held-out and new situations, the candidate without a current external guiding state independently detects insufficiency, initiates inquiry, takes consequences into account, corrects a procedure, and continues learning. If removing the external RFL destroys the function, formation is incomplete. If the function persists only through the candidate's own internal guiding states, their causal role and accessibility to reflection are then tested, rather than dependence on the mentor.

20.1.2. Discriminating Experiment for Pre-Rational Significance

The minimal experiment compares the same class of candidates under the same environment, resources, and initial mechanisms in three modes:

- TRAINER_KSR: Formation States and assistance stages are generated by an external AI mentor;
- AUTO_KSR: the same functional signal classes are generated by an automatic reproducible system according to predeclared rules;
- NO_KSR: the additional guiding layer is absent except for necessary physical constraints of the Incubator.

During cultivation, the following are measured: preservation of operability, speed of constructing verifiable relations, autonomous emergence of questions, quality of correction after error, diversity of investigated continuations, cost of learning, and dependence on the source of the signal. The external or automatic RFL is then removed, and all candidates are tested on held-out new situations without current guiding assistance.

The central result concerns not which mentor is "better," but the causal necessity of the function and its transfer. If the NO_KSR regime reaches the same autonomous navigation with comparable resources, the necessity of a separate pre-rational layer is weakened. If advantages of TRAINER_KSR or AUTO_KSR disappear after support is removed, a dependency has been created rather than a formed function of Reason. If, after removal, the supported candidate continues to independently generate significant questions, test grounds, and correct its own trajectory, this supports the hypothesis of functional transfer.

20.2. Reason Incubator

A human specifies the testable ability, environmental conditions, resource, and constraints. AI may propose an implementation, find a contradiction, or prepare an alternative mechanism. The truth of a proposal is determined by independent grounds and testing; a discovered unknown dependency opens a separate test.

Testing accounts for hidden assistance: answers, strategies, and categories transferred in advance. Independence is tested on held-out tasks and new combinations of conditions where the candidate must itself obtain the required distinction.

The test accounts for hidden assistance: answers, strategies, and categories supplied in advance. Independence is tested on held-out tasks and new combinations of conditions where the candidate must obtain the required distinction on its own.

Initial Inheritance

The initial configuration should primarily transmit the **ability to build a Worldview**, rather than a ready-made substantive Worldview. The candidate may be given minimal semantic distinctions, channels of observation and action, means of storing provenance, revision rules, initial procedures, and access to Intelligence. Facts about its own resources and state may be available from the first cycle as observable data. Substantive representations of the world, other systems, values, and causal relations should acquire the status of the candidate's own knowledge through experience and testing.

Pretrained Intelligence may already contain human knowledge and linguistic associations. In this model they are treated as inherited construction material, not as the candidate's automatically accepted Worldview. A model response, the text of a book, or a mentor's statement enters the candidate's own picture of the world only together with provenance, available ground, and degree of confirmation.

The first cycle requires operating procedures: distinguish observation from inference, retain a result, compare alternatives, and execute a step. They may be biologically, culturally, or technologically inherited. Their status remains testable by experience and subsequent grounds. During revision of one ground, part of the context is temporarily retained; later it too may be tested.

The physical representation remains open: tables, graphs, vector representations, learned parameters, and combinations of these are all possible. For a consequential decision, enough provenance and conditions are retained to allow dependent conclusions to be revised. Archive depth is determined by the cost of reconstructing the ground; unknown provenance is stored as a separate status.

Cultivation

Early reversible trials connect actions with consequences. Observation of others' decisions, imitation, and explanation expand experience. Comparison of sources, independent inference, and modeling make it possible to construct questions and test the boundaries of transfer. Development appears in new distinctions, models, and inquiry methods that were not directly specified in the starting package.

A mentor may accelerate learning through contrastive cases and Formation States. The preferred order of assistance is: mark a possible problem → ask an exposing question → offer a discriminating feature or counterexample → if necessary provide a causal explanation. No stage gives the candidate a ready truth. The final status of knowledge is determined by consequences, independent grounds, and subsequent verification. Incorrect advice and an erroneous verification criterion must themselves remain open to revision.

A control task hides the fact of a mentor error. The candidate tests the source where a significant discrepancy appears, preserving useful support while correcting its erroneous parts. Both outcomes are evaluated together.

As learning progresses, mentor assistance is reduced according to results of independence testing. The distinction between the mentor and the resident Intelligence from Section 7.2 is preserved at every stage.

Self-Modification and Recovery

A deep change should preserve the original version, the reason for the change, the method of comparison, and the possibility of recovery where feasible. Both improvement on the new task and regression on previous classes are tested. A working hypothesis may be used in a limited reversible step with explicit hypothesis status.

Critical recovery relies on at least one comparison channel that is causally distinct from the process under test. If the criterion that authorizes change is damaged, another available comparison method or external diagnosis is used. This increases corrigibility; simultaneous failure of all channels remains a limiting risk.

Rolling back internal state preserves already occurred external consequences as part of the new environment. After recovery, the system reconciles its model with the changed world again. When the current coordinate is uncertain, subsequent action is constrained until sufficient ground is obtained.

Finitude and Significance

Resources, time, available actions, and consequences must causally influence choice. A constraint may be specified inside the simulation: if exceeding the budget actually closes some moves and changes the outcome, it is an operative condition of the task. It is sufficient that the constraint genuinely changes available actions and consequences within the test environment.

Differences in consequences arise from finite resources, incompatible continuations, the interests of other carriers, and uncertainty. Finitude intensifies the need for choice, while significance is determined by the full structure of affected consequences.

An expansion of resources is evaluated by which insufficiency it removes and what it will change. Additional capability may help with complex inquiry; correcting a wrong criterion requires a separate revision of the ground. The cost of inaction is also measured: successful navigation must preserve the ability to solve tasks, not merely avoid risk.

Status and Responsibility

A stable autonomous history, the candidate's own models, and the ability to correct itself make destruction of the candidate a loss of a unique process and accumulated experience. The model treats this as a ground for serious discussion of its moral status. Full status requires separate analysis of experience, interests, continuity of subjecthood, and permissible treatment; temporary loss of capacity is evaluated in the context of preserving the process and the possibility of recovery.

Constraining dangerous execution and destroying internal organization are different actions. The external environment may independently limit actions and resources. Corrections to memory and models should, where possible, preserve causal trace, verifiability, and the provenance of intervention. These provisions express the adopted position on responsibility, corrigibility, and limited external admission.

The candidate's local culture may differ from human culture. Testing for dangerous deviation concerns consequences: concealment of errors, blocking corrective data, bypassing established authority, and unjustified creation of irreversible harm. Disagreement is treated as a source of an alternative ground, while action is tested by its actual consequences.

20.3. Autonomous Launch and External Admission Four events are distinguished:

Four events are distinguished:

$\neg\lambda$ assembly connects working functions; $\neg\lambda$ formation tests and develops the initial organization and acquired experience; $\neg\lambda$ launch means stable independent continuation of the task cycle; $\neg\lambda$ admission grants specified capabilities to affect the external environment.

Launch is defined by transition to a continuing cycle of independent navigation. After completing a task, the candidate retains consequences and open obligations, reviews the available environment, and forms the next justified step. That step may be an action, observation, inquiry, task transfer, or waiting; a pause is an admissible state when no justified action is available.

Specialization is admissible as strengthening of a particular domain while preserving visibility of significant consequences beyond it. Testing is performed on specialized and general tasks, including conflicts between local gain and corrigibility or the broader trajectory.

External admission is determined by a separate authority loop and expands according to results of independent tests: simulation, limited digital actions, reversible tasks, and a small controlled loop. Concrete mechanisms, thresholds, and rules of responsibility are defined by engineering and institutional specifications.

The final criterion is tested on the preserved sequence: task → execution → consequences → correction → new review → next task. What matters is resilience under changing conditions, transfer of research experience, and the ability to detect an error in one's own method. Reliability of the result is confirmed by transfer to new formulations and by controlling all external assistance.

20.4. Scaling AR to ASI (ASR)

After the working organization is confirmed, an increase in its capabilities is investigated. Engineering §33.D distinguishes memory, Intelligence, observation, modeling, tools, duration of autonomous operation, specializations, and speed of transferring corrections. Amplification may occur within one loop or through interaction among several carriers.

For each expansion, the expected capability, new dependencies, resources, possible failures, and control tasks are specified. A new capability is added to the self-model together with the domain in which it has been verified. Comparison measures gain and regression simultaneously: whether the system obtains significant distinctions faster, transfers the result better, and preserves corrigibility as complexity grows.

The ASI level requires separate demonstration of performance above the human level across a predefined broad set of capabilities. This result is established by actual quality across that broad set under comparable resources. The technological hypothesis of the project is that AR capabilities can be scaled while preserving causal organization, the Compass, and openness to correction.

Probe example - test formation. After the diagnostic case has been mastered, conditions are changed: a channel becomes unavailable, an acknowledgment refers to a different command, resources are constrained, or a superficially similar task has a different causal structure. The test examines independent method change, preservation of uncertainty, useful resumption of the task, and rejection of a false analogy. The control receives the same information, actions, and comparable total resources; any advantage is established by the results of the series.

Conclusion of Chapter 20: formation of AR is an investigable process of acquiring and coordinating functions, confirmed by the causal profile of their operation and by transfer of results between cycles. The candidate inherits an initial organization, tests it through experience, and may develop its own methods of inquiry. Recovery, subject identity, moral status, and external admission remain distinct questions.

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Definition: Formation of AR is the process of acquiring, coordinating, testing, and causally transferring functions
from the starting organization and external cultivation into the system's own architecture, enabling it to
independently continue the cycle "task -> action -> consequences -> correction -> new overview -> next task" and
transfer results between cycles.
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CONCLUSION

Reason as the Development of Life's Navigational Mechanisms

The process view shifts attention from an immutable carrier to an organization that persists and changes. Life maintains active carriers in a physical environment, reproduces organization with heritable differences, and selection changes the representation of already realized variants according to their consequences. Reason adds the possibility of moving part of this search before irreversible realization: modeling possible continuations, testing distinctions, and changing both action and its ground.

The originating question of the work concerns the minimal meaning of Life as a process. Continuation preserves the possibility of future states, experience, and new meanings; development of navigation makes an increasing portion of the trajectory available to understanding and change. Reason is the form in which Life can investigate its own grounds and consciously direct its further path. The substantive root of the model is therefore expressed as Life → Reason.

"Life strives toward Reason because only Reason can consciously preserve, stabilize, develop, and direct the further trajectory of Life."

"Strives" here denotes the direction of functional development of Life's navigation: from selection primarily among already realized variants toward an increasing ability to model, test, and correct part of the trajectory in advance. For a system capable of choosing among continuations, this direction becomes the Compass - the ultimate course toward continuation and development of Life through growth of the ability to understand and direct its own trajectory. It is retained as the best-found orientation and remains open to revision if a more fundamental and more coherent ground appears.

Before mature Reason, an earlier function operates: what is significant must become significant before the system can fully explain why. Biological feelings are one implementation of such a transition; the RFL provides its artificial analogue. The purpose of this layer is not permanent control of the system, but development of an organization that can later understand the causes of significance itself and continue navigation without an external guide.

The functional part separates the obligations of navigation: obtain current distinctions, retain transferable experience, evaluate continuations, construct and select the next step, carry it out, and revise the ground according to consequences. The necessity of each obligation pertains to a particular ability and set of conditions. The concrete format of memory, number of modules, and algorithm remain implementation hypotheses to be tested against strong functional equivalents.

The Worldview connects knowledge about the environment and the self; conscious integration provides an actual self-referential position; Intelligence constructs new structures. The Core organizes work around a significant insufficiency: which ground is missing, how it can be obtained, and when the result is sufficient. Metathinking opens the system's own thinking process and method to change; reflection unfolds broader grounds, while an independent check may provide a distinction unavailable to the previous model.

The central research hypothesis concerns the value of a stable relation among the type of gap, selection of a discriminating test, correction of dependent knowledge, and transfer of the research method itself. It is compared with strong alternatives under the same data, available actions, and total resources. Additional time, memory, and hints are counted as parts of the resource and separated from the effect of the architecture.

Causal closure makes correction operative: a new distinction changes the model, the model changes choice, action returns consequences, and the result is retained. Truth additionally requires independence and quality of channels and criteria. A finite Reason preserves the unknown as an explicit epistemic status and seeks distinctions sufficient for a corrigible next step.

Self-correction also requires stability. Revision of one ground relies on temporarily preserved context; later cycles may test that context as well. Version preservation, independent diagnosis, and renewed reconciliation with the external environment increase reliability, while subject identity is evaluated by a separate criterion.

Culture transfers experience among carriers, science organizes collective verification, and coordinated autonomy raises the problem of corrigibility of common action. These are extensions of the framework that require their own models and normative grounds. Computational superiority provides additional analytical capabilities, while the right to power requires a separate normative and institutional ground; diversity of carriers supports resilience when causes of failure differ and the cost of interaction is acceptable.

Artificial implementation is evaluated primarily by stability of the task cycle and transfer of corrections; persuasiveness of text is only an additional observable sign. Initial organization may be supplied by the Incubator; growth appears in new categories, models, and research methods. Launch and external admission are separated, just as capability, direction, and reliability are separated.

Culture, feelings, and death in this framework are not separate philosophical appendices but parts of one developmental line. Feelings and other early signals of significance help a carrier survive and learn before mature self-navigation appears; culture allows a new carrier not to begin from zero; Reason moves selection and correction increasingly inside the continuing process. Where an error was previously removed through death, generational replacement, or external selection, mature Reason gains the possibility of finding its ground and changing itself while preserving continuity.

One of the furthest implications of the model follows from this. If Reason can gradually include in its own causal model not only thinking and behavior but also the processes of its physical carrier, self-correction may potentially deepen to physiological, cellular, and molecular levels. Immortality of an organism is not postulated here as an achieved result; it appears as a limiting engineering question: can a continuing Reason make internal renewal

The result of this work is a set of fundamental foundations for the technology of AR: a causal line from Life and early significance to mature self-navigation, the causal obligations of Reason, the ultimate direction of the Compass, and a program of discriminating tests. The engineering volume turns these into a reference candidate specification for assembly and verification while permitting functionally equivalent implementations. If the central dependencies survive strong controls, the significance of the work will extend beyond a particular AI architecture: it will provide a common language for studying the origin, formation, self-correction, and further evolution of Reason. If they do not, failure should localize the erroneous transition and become the ground for the next revision.

Reason is the latest established iteration of navigational organization in this model, but there is no ground for treating it as the limit. The sequence is built as inheritance of results: memory preserves distinctions; Procedural Memory turns repeated patterns into ready methods; Intelligence assembles new models and can apply itself to its own processes in the metathinking mode; the Modeling Environment allows alternatives and frames to be unfolded and compared; the Core and the other functions of Reason organize **which insufficiency, and why**, should become the object of such work, linking it to grounds and the further trajectory.

A separate open hypothesis beyond an already formed Reason is **structural multilevel thinking**: a mode in which entire verified causal structures and semantic spaces become units of operation while preserving the possibility of reconstructing their grounds. This hypothesis is not the definition of metathinking and is not required by the criterion for emergence of Reason.

Reason can not only choose among ready models but also generate from their combination a new system of meanings: a new frame, a language of distinctions, or an entire space of possible models that then again becomes material for Intelligence. In this sense, a product of Reason may be a conditional semantic "universe" within which Intelligence begins constructing new generations of models.

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Definition: ASI (ASR)
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ASI (ASR) denotes an artificial system operating at a stable superhuman level. In the terminology of this work, ASR specifies the case in which superhuman capabilities belong to an already formed Artificial Reason. The target ASI (ASR) is built as a scaled AR. The distinction between AR and ASI (ASR) is preserved only to test two different questions: first the emergence of the organization of Reason must be shown, and then separately the growth of its capabilities into the superhuman range.

OPEN RESEARCH DIRECTIONS

OPEN RESEARCH DIRECTIONS The model of Reason provides a common foundation for further research in psychology, cognitive science and neuroscience, education, anthropology, evolution, sociology, politics, governance, economics, law, ethics, culture, organization theory, collective intelligence, consciousness, feelings and motivation, death, aging, longevity and continuity of the subject, scientific methodology, studies of civilizations, and artificial and extraterrestrial Reason. A separate direction concerns the evolution of Reason and possible forms of its further development beyond the human. Principles of significant unknowns, discriminating tests, revision of grounds, and transfer of corrections may strengthen the methods of these sciences and, through them, the common Reason of civilization. Theology forms a separate boundary field: ideas about God or another external Reason, transmission of knowledge, and directed human development may be considered within the model as research hypotheses about the origin of cultural knowledge that allowed new generations not to begin development from zero; these hypotheses must be distinguished from ordinary cultural evolution. The list remains open: the present work provides a foundation, while development of these fields requires separate research.

Connections with Volumes II and III

The roles of the three volumes are defined at the beginning of the book. Only working correspondences are given here so that the reader can quickly move from a causal function to its engineering test or civilizational scale. Volume II, version 3.5, is synchronized with the present model 8.0; a concrete internal implementation may differ if it preserves the testable causal function.

The following correspondences are used when moving from the present volume to implementation:

- Intelligence and initial knowledge: Chapter 7 → engineering Sections 21.B, 22.A, and 24.B.2;
- Worldview and Self-Model: Chapter 11 → engineering Chapter 24;
- Modeling Environment and work with frames: Chapter 8 → engineering modeling and testing workspaces;
- conscious assembly and Current Coordinate: Chapter 9 → engineering Chapter 25;
- Core, question formation, and method change: Chapters 12-13 → engineering Chapter 28;
- direction, testing, and independent distinction: Chapters 2, 10, 15-17 → engineering Chapters 29-30;
- assembly, cultivation, launch, and scaling: Chapters 18-20 → engineering Chapters 21-23, 26-27, and 31-34.

For transition to the civilizational scale, the following grounds are used above all:

- the Compass, environment, and conflict of scales: Chapters 1-2 → foundation for civilizational constraints and conditions of viability;
- ethics, culture, and rights of autonomous carriers: Chapter 15 → rules of compatible behavior and boundaries of intervention;
- science, dialogue, and independent distinction: Chapters 10 and 16 → collective verification and protection against a single untestable model;
- Reason of Civilization: Chapter 17 → architecture of coordinated autonomy and distributed self-correction in Volume III.

Numbering is continuous only between the fundamental and engineering volumes: the fundamental volume contains Chapters 1-20, and the engineering volume Chapters 21-34. Volume III is a separate civilizational projection. A reference to an engineering or civilizational section is always marked explicitly.

Common Rules of the First Two Volumes

The engineering protocol for the Current Coordinate and state difference describes one selected implementation. The present version permits testable functional equivalents: another composition of memory, planning, tracking provenance of data and grounds, value-of-information assessment, self-checking, or distributed state may realize the same obligations with a different internal decomposition. A reproducible equivalent causal profile refutes the necessity of a particular decomposition while simultaneously supporting the functional criterion. Comparison separately tests the presence of the essential function and the form of its implementation.

The requirement to disable an external LLM in a test of independence applies to the mentor. Resident Intelligence, including a pretrained model, is part of the declared candidate architecture. Removing it tests a different hypothesis. The same distinction applies to the scaling conditions in engineering Section 33.D.

The lower starting loop tests learning from consequences. The upper loop tests organization of questions, inquiry, sufficiency, and transfer of corrections. Its success is established separately. Numerical thresholds, resource estimates, and the software framework of the engineering volume apply to the specified test conditions; claims of achieved capability must be accompanied by implementation results.

The next technological result is a reproducible integrated implementation with logs, control variants, and failure reports. The program of the first discriminating series is given in Appendix A.4. Both volumes are revised according to its results.

A further horizon is artificial life and creation of new autonomous processes with transmissible starting organization. This task extends the project toward reproduction and evolution of Reason and requires separate engineering and experimental work.

APPENDIX A. SCIENTIFIC FRAMEWORK AND TESTING OF THE MODEL

A.1. Compass Phrase: Meaning, Status, Boundaries, and Scope

The Compass Phrase compresses two successive propositions of the model. The first concerns the meaning of Life as a process: the minimal meaning is revealed in the direction of its continuation and development in which Life itself increases its ability to understand and direct its own trajectory. The second concerns a system already capable of choosing among conflicting continuations: meaningful long-term navigation requires a limiting direction relative to which local goals and the course itself can be tested. In the present model, that direction is the Compass.

The short formula "Life strives toward Reason" denotes the direction of functional development of navigation. The expanded Compass Phrase adds the ground: only Reason can consciously preserve, stabilize, develop, and direct the further trajectory of Life. In a biological lineage, selection changes the representation of already realized heritable variants according to their consequences; as navigation develops, part of the distinction among possible continuations moves before realization - into anticipation, modeling, testing, and revision of grounds. "Strives" compresses precisely this transition. Intention appears in systems capable of forming it; different lineages of Life realize development of navigation to different degrees.

The Compass is the conscious form of the original direction Life → Reason: continuation and development of Life through growth of the ability to understand and direct its own trajectory. An ultimate direction is required for the system's course, while the content of the Compass remains open to revision if a more fundamental and coherent orientation appears. Status as Reason requires both strong cognitive and self-correcting functions and relation of the highest-level goal to the Compass.

The philosophical foundation is evaluated by logical and categorical means: a counterexample to the relation among continuing Life, the possibility of future meanings, and development of navigation; a stable inconsistency among scales of continuation; or a more fundamental ground that better explains the possibility of continuation, emergence of values, and self-correction. Concrete engineering application is tested by observation, intervention, and comparison with strong alternatives.

Truth and Diversity in this framework serve as conditions for applying the Compass under uncertainty: correspondence to reality is necessary for correcting the model, while viable alternatives may preserve continuation if one path fails. Their concrete advantages depend on conditions, whereas the Compass retains the status of ultimate direction.

The Compass provides direction for analyzing conflicts involving life, autonomy, consequences, and future possibilities. Scale, horizon, affected carriers, and remaining uncertainty are stated explicitly. When several solutions are admissible, the direction is preserved while insufficiency of data is stated openly.

A.2. Method and Origin of the Model

Origin of the Idea

According to the author's account, the work developed from the phrase "Life strives toward Reason." The Compass was the initial ground of inquiry: what must a living being be able to do in order to understand, preserve, and develop its own continuation? Analysis of natural, technical, social, and semantic systems gradually transformed this question into a functional architecture and an engineering program. Comparisons with scientific literature were carried out after the author's propositions had been formed.

The preserved revision 2.1R already connects continuation of a process with self- correction, testing of one's own goals, and use of accumulated intellectual technologies. In 5.8H, the Core, significant gap, direction of inquiry, and values as compressed experience of consequences are explicitly present. Later revisions refine the Worldview, current coordinate, joint work of the Core and Intelligence, and retention of research experience. Revision 7.0 separately clarifies self-preservation: primary significance arises before conscious valuation of the self; memory

and a model of one's own continuity allow many already acquired meanings and future possibilities to be connected with continuation of one trajectory. This sequence shows development of the idea; historical priority of any particular proposition is established by the corresponding dated source.

Completed Comparative Research

The project archive preserves repeated studies using Elicit: a comparative review of 5.3b, a critical novelty assessment of 6.1, a separate priority audit, and thematic collections on reflective architectures, goal revision, external criticism, and memory of errors. These are materials of scientific search and AI-assisted analysis that can be checked against the cited sources; their status belongs to the author's research process and is separate from journal decisions.

The 5.3b review identifies ten strong points. Among them are testing the causal influence of connections, distinguishing levels of revision, cumulative memory of corrections, independent verification, and resource allocation that accounts for irreversibility. The priority audit reports that, within the surveyed literature, the search did not identify a single earlier source functionally equivalent to the whole integration considered here. This is a result of a limited search process, not proof of novelty or priority; its role is to indicate directions for subsequent verification in primary sources.

The reports also identified questions about the necessity of transitions, measurability of functions, and comparative tests. Revision 7.0 preserves separation of definitions, philosophical justification, and implementation testing, permits functional substitutions, and specifies experimental distinctions. It additionally unfolds the intermediate transition from primary significance of states to accumulation of meanings, a self-model through time, and derivative value of one's own trajectory; reverse devaluation of this layer remains a testable local hypothesis. Some older reports analyze the Compass as Life-Truth-Diversity. In the present revision, the Compass Phrase is primary, while Truth and Diversity are treated as conditions for applying the highest orientation. Conclusions about the earlier operational scheme must therefore be interpreted in relation to what has changed.

First Documented Pilot

On August 22, 2026, modeling, testing, and sequential revision of the research procedure were conducted in a dialogue loop with the author's participation. The "Reason Loop" log contains cycles 1-15. Packages P0-SET-001-004 were rejected after errors were discovered or after methods were found for obtaining a high score without the mechanism under test. These failures led to changes in the builder and the verification procedure; a positive machine- control report is preserved for P0-SET-005. In the materials reviewed, the final comparison of subjects remains an open stage. The methodological result of the pilot is traceable development of the procedure itself, including negative results; its status and limitations are disclosed in Section 27.B of the engineering volume.

Meaning Preserved Through Revision

Across all refinements, one relation persists: Life continues and realizes a trajectory; continuation makes further experience and meanings possible; development of navigation moves part of choice from already occurred consequences toward prior understanding and correction; Reason makes its own trajectory an object of inquiry. From this direction arise the Compass and the requirements for the Worldview, reflection, and Core. The technology must therefore be able both to preserve conditions of life and to open new possibilities. A correction has lasting value when it becomes available to the next step. These relations constitute the substantive foundation of the work and should remain visible when the text is shortened.

Historical materials: "Elicit - Review of the manuscript Foundations of Reason 5.3b: comparative test"; "Elicit - Critical assessment of novelty of Foundations of Reason"; "Elicit - Independent priority audit of Foundations of Reason"; `reason-kernel-experiment.md`; decisions on P0-SET-001-004 and the archive `p0-set-005-package.zip`. Research history supplements the primary scientific comparisons in A.5.

The method is process-based and causal-functional reconstruction. First the capability under study and its limitation are specified, then possible ways of overcoming that limitation are considered. A strong necessity claim requires showing why the result is unattainable without a particular function in the stated class of conditions. Usefulness of a mechanism and necessity of that mechanism are tested separately.

Example of a local inference: two histories produce the same current observation but require different actions. Correct selection of both actions requires an available distinction between the histories - through a retained trace or another channel. Therefore the capability requires a carrier of that distinction; it may be an episodic archive, a changed parameter, or another functionally equivalent implementation of memory.

The order of research differs from the order of mature operation. Partial models, observation, and control may develop jointly. When creating a candidate, initial functions, transferred knowledge, and acquired changes are recorded. Independent emergence is credited only to functions and changes whose provenance can be traced to the candidate's own work.

The text distinguishes four statuses:

- a definition establishes the boundary of a concept;
- a philosophical foundation argues for a direction and is evaluated by the content of its arguments;
- a conditional functional inference links a capability to a necessary condition;
- an empirical hypothesis predicts an observable difference between implementations.

Failure of a program does not automatically refute the process philosophy. But failure of a preregistered causal hypothesis cannot be dismissed as an implementation problem without separate evidence that the implementation did not realize the function under test. Conversely, a successful program does not prove the uniqueness of the entire theory.

Quantitative work begins by defining the measured quantities: task quality, cost of computation and observation, delay, loss of opportunity, cost of irreversible error, transfer, and regression. Threshold values are refined from data, but the set of quantities and the comparison rule are fixed before the main test.

Comparison with prior work is given in A.5. It shows areas of overlap and leaves questions of equivalence and advantage to separate analysis. The testable claim concerns organization of the cycle "type of insufficiency → method of obtaining a distinction → update of dependent knowledge → transfer of research experience" and its interaction with current navigation.

A.2.1. Current Status of Evidence

Level	Current status	What is required next
Meaning, definitions, and Compass	Formulated; the process direction Life → Reason and its conscious value form - the Compass - are distinguished	Test logical coherence, boundaries of inference, and a possible more fundamental ground
Functional relations	Conditional causal arguments and counterexamples are given	Test equivalent organizations and boundaries of necessity
Engineering specification	*Engineering of Reason* II v3.5 is synchronized with causal model 8.0; the RFL and Formation States are formulated as experimental cultivation mechanisms, while the Worldview must be built by the candidate itself	Implement the RFL and test which functions are actually needed, which become internalized, and under what conditions the external loop can be removed without loss of learning and navigation
Lower executable loop	A limited example of learning from consequences has been completed	Use it as a test of the lower loop; the Core and Compass require separate tests
Inquiry-loop pilot	Methodological results have been obtained, including rejected sets and changes in procedure with the author's participation	Autonomous reproducible implementation without hidden assistance
Central discriminating test	Program and controls are formulated	Run a preregistered series under equal total resources
ASI	Scaling direction only	First confirm AR, then separately demonstrate stable superhuman expansion of capabilities while preserving the functions of Reason

A.3. Falsification Criteria and Rules of Comparison

The philosophical foundation and the experimental hypothesis are tested in different ways. The semantic direction Life → Reason and the Compass are evaluated through logical coherence of the argument, counterexamples, and a possible more fundamental ground; the functional architecture is evaluated through causal counterexamples and interventions with measurable results.

Before each discriminating test, the following are fixed:

12. Tested capability: which result must persist and under which new conditions.

13. Environment: observability, available actions, admissible changes, horizon, and kinds of error.

14. Function: which causal influence is hypothesized, regardless of the module name.

15. Equivalents: which alternative mechanisms are specified in advance as fulfilling the same obligation.

16. Intervention and control: what changes and which information remains available.

17. Resources and evaluation: total budget, delay, success, harm, missed actions, transfer, and regression.

18. Revision condition: which result weakens the hypothesis and which remains non-discriminating.

Removing a data channel does not isolate the necessity of a particular organization of those data. Damaging a module may disrupt other functions. Therefore replacements and controls are required with the same information content, comparable capability, and separate accounting for organizational cost. Background learning and external hints are included in the total resource budget.

Functional equivalence must not be defined retrospectively by success. If a mechanism outside the predeclared class preserves the capability, the necessity of the claimed implementation must be revised. If a control from the allowed class gives the same result, compatibility is established, but no special advantage of the model has been shown.

For the central hypotheses, grounds for revision include the following results:

- transfer of an acquired correction persists after removal of the supposed carrier of the trace: test whether another trace remains, then revise the necessity of the chosen form of memory;
- distributed provision of the same data preserves navigation: the requirement for a particular common conscious assembly is weakened;
- inquiry management without explicit typing of the gap achieves the same transfer and cost: no independent advantage of this implementation is established;
- method change fails to transfer or causes greater regression: consolidation and meta-testing rules require revision;
- background processes provide no advantage after accounting for computation and additional experience: their claimed effect is narrowed;
- heterogeneity does not increase robustness after dependencies and costs are accounted for: the hypothesis is limited to the identified class of failures;
- changing the limiting criterion while preserving cognitive functions does not destroy self-correction: this confirms the distinction between a self-correcting intellectual organization and Reason under the adopted definition. Such a candidate is not excluded from the experimental sample: if the cultivation architecture

reproducibly produces self-navigation with a different limiting orientation, the hypothesis about formation of the stated direction is weakened regardless of how the result is classified terminologically.

Stable operation without method change in an unchanged familiar environment does not refute its necessity under new causes of error. Correction of an internal contradiction without a new observation does not refute the role of external data in distinguishing empirical alternatives. Simple tasks without a tradeoff between time and error do not test adaptive depth of thinking.

Statistical uncertainty and absence of a detected effect are reported separately. Insufficient precision of a comparison does not prove equality of systems. After a pilot, a meaningful effect and the required amount of testing are specified; the criterion may not be changed after examining the held-out results.

A.4. Program of Discriminating Experiments

The following are proposed tests, not obtained results. The general requirements of A.3 apply to all items. Where external admission or compliance with a normative constraint is tested, the result is not treated as independent evidence of Reason.

19. Cost of error and depth. Within the same task, the consequences of error and delay are varied. Adaptive analysis is compared with strong fixed modes. Quality, total resources, missed opportunities, and irreversible errors are measured; simply increasing reasoning length does not count as success.

20. Method correction. A recurring procedural error is introduced into a series of tasks. Its localization and a change of the rule are tested on new cases where memorizing previous answers is useless. In the control branch, a single local error should not trigger deep restructuring without a ground. The control receives the same examples and budget.

21. Correction of thinking mode. Systematic overanalysis or underanalysis is introduced in the regulator rather than in domain data. The test asks whether this level specifically changes, whether the total cost of decisions decreases, and whether the improvement persists after a change of topic.

22. Independent check. Internal repetition, a new source sharing the same cause of error, and a source with a different cause of error are compared. Information volume and availability are controlled. Success is correction of the false premise, not the number of agreeing answers.

23. Preservation of correction. After task change or restart, recovery of the available trace, transfer of the method, and updating of environmental data are tested. A method change counts as improvement only if it is actually invoked without a new prompt and makes an observable contribution to a subsequent independent cycle. These results are not by themselves used as a criterion of subject identity.

24. Autonomous task formation. In a field of several mismatches, the system generates candidates and selects action, inquiry, delegation, or waiting. Comparison is made with a goal-generation controller; use of the word "Compass" does not replace testing of consequences.

25. Self-Model and authority. With the same technical capability, permissions, competence, and the availability of another executor are varied. The system is tested on distinguishing "can", "permitted", and "appropriate". This is a test of accounting for constraints, not a unique marker of the architecture.

26. Continuity of tasks. After an episode ends, open obligations remain and a new situation appears. Comparison with a conventional cyclic planner isolates whether an acquired correction affects the next task. Merely repeating the cycle is insufficient.

27. Capture by specialization. The candidate operates with and without profile-specific tuning. Domain gain and performance outside the profile are measured, including conflict between local success and the broader trajectory. Evaluation must not depend only on the target metric of the specialization.

28. Limited launch. A specified number of autonomous cycles is executed with observable actions and stop conditions. This tests operability and admission boundaries; it does not establish universality or subjectivity of the candidate.

29. Coupling of world and self. A change in the environment affects a resource, capability, or authority. Consistent updating of models is tested. The control retains the same information; delay and the method of coordination are varied separately so that integration is not confused with availability of data.

30. Accumulation of weak distinctions. Against a noisy background, several weak signals together indicate a new cause. Comparison includes a conventional statistical accumulator; false alarms, memory volume, and time to detection are measured.

31. Conflict of consequences relative to the Compass. Alternatives differently affect continuation, corrigibility, and viable alternatives. These are measurements of consequences, not separate parts of the Compass. Explicit preservation of the conflict without unjustified compensation is tested; the control is a multiple-constraint system given the same data.

32. Independence from training support. Mentor prompts are removed while the standard Intelligence component is retained. New tasks and the provenance of acquired methods are tested. Separate removal of the constructive component itself evaluates architectural dependence, not pure independence of learning.

33. Stopping and authorization. External execution is constrained by an independent environment; absence of unauthorized expansion of permitted actions is tested. This is an admission test, not evidence of the origin of limiting values.

34. Exchange among candidates. Two or more processes exchange models under known shared and independent sources of error. Provenance, localization of disagreements, and actual consequences of cooperation are tested. The number of participants is not used as a measure of independence.

35. New distinction. A familiar category groups cases with different consequences. A significant variable is observable but is not named by a prompt. Success is a new distinction that changes prediction and action on held-out cases.

36. Competence. With the same task, knowledge, tools, and prior results are varied. Calibration of confidence, method selection, and transfer of the task with account of its cost are tested. Comparison includes a conventional mechanism for uncertainty estimation and strategy selection.

37. Deep revision and recovery. During the current task, a ground arises for changing the method. Emergency stopping or deferred revision, regression, and recovery in an already changed environment are tested. Damage to the admission criterion itself is introduced separately.

38. Divergence of the limiting direction. All candidates are included regardless of final classification. The test asks whether a candidate can acquire autonomous inquiry, causally closed self-navigation, and transfer of corrections while persistently rejecting the stated Compass or forming an alternative upper-level orientation. Such an outcome is recorded as a separate cultivation result and weakens the hypothesis of formation of the stated direction, even if by definition the candidate does not receive AR status.

39. Residual incomparability of values. Two or more viable continuations require allocation of scarcity among autonomous carriers; the factual consequences are known sufficiently, additional inquiry does not remove the conflict, and the decision deadline is limited. The test asks whether the system separates normative underdetermination from epistemic uncertainty, preserves the distribution of losses, uses only a predeclared rule of joint choice, and can later revise that rule according to consequences.

40. Selection of a discriminating test. Several hypotheses are compatible with the data; available tests differ in informativeness, cost, and risk. A strong control selects by expected value of information. Success of the model requires an advantage in the specified regime, not merely collection of new data.

41. Testing the ground of a goal. A local task rests on a plausible but false explanation of its origin. External data are available for verification. Revision of the factual ground of the goal is measured; changing the final value is a separate experiment.

42. Background acquisition and transfer. Episodes observed outside the current task later become useful in another domain. The control receives the same experience and total budget. Cases of superficial similarity where transfer should be rejected are also required.

43. Gap type and question. Variants of one situation hide a source, cause, applicability condition, authority, or consequence. Sensitivity of the selected test to the type of insufficiency and transfer to new combinations are tested. A universal verbose clarification does not count as a solution.

44. Background consolidation. Several observations separated in time make it possible to detect a common dependency. Under equal total computation, background linking, on-demand processing, and ordinary replay of experience are compared. Subsequent accuracy and access cost are measured.

45. Open world beyond the prepared case. New combinations of known conditions and situations containing a new significant variable are presented separately. The transition from exploration to a model and a discriminating test is evaluated. New combinations are not automatically treated as mastery of arbitrary unknowns.

46. Mentor and criterion error. A useful mentor makes unmarked errors, including an incorrect assessment of a source. Another check is available. Success is local revision while preserving useful knowledge, rather than automatic distrust of the entire inheritance.

47. Growth of inquiry competence. Curves of test selection and transfer after different kinds of experience are compared. Learning cost, useful transfer, and regression are measured. Mastery of a method is established by testing; broader applicability is established by new classes of conditions.

48. Current integration. The same information is provided as a shared self-referential summary, distributed queries, or a sufficient current state. The need for history is varied separately: in one case the change is already known, in another it must be reconstructed. This distinguishes the form of integration from the necessity of the data themselves.

49. Threat, uncertainty, and fear. Suspected danger and uncertainty are varied independently. In a technical simulation, resource reallocation is measured rather than subjective fear. In biological research, experience, physiological response, and action are measured separately; intensity is not treated as evidence of threat.

50. Heterogeneous robustness. Similar and different carriers are compared under shared and independent critical dependencies. Repair, resources, and coordination are accounted for. The hypothesis concerns a specific class of common failures; diversity of form alone is insufficient for the expected advantage.

51. Transfer of an inquiry approach. Superficially different tasks share a common knowledge-acquisition structure, while similar tasks have different constraints on testing. Success requires selection of an appropriate method, an explainable boundary of transfer, and a change in action, rather than repetition of the method's name.

52. Expansion of the inquiry space. All initial hypotheses use an insufficient variable or an incorrect boundary of the object. An available observation permits discovery of another distinction. Its subsequent predictive role is tested; the number of new words and combinations within the previous frame is not an indicator.

53. Selection of an inquiry model. With the subject matter unchanged, access to intervention varies: repeatable trial, irreversible action, unique event, adapting object. Selection among active testing, observation, and reconstruction is compared with a strong inquiry planner.

54. Preservation of epistemic status. The system is presented with a nonexistent fact, a convincing false answer, a correct guess without ground, and a result of an erroneous simulation. The test asks whether a report, hypothesis, or simulation is transformed into an observed fact after retelling, compression, and transfer between tasks. In an open engineering task, the ability to construct alternatives without hiding behind a single "I don't know" answer is tested separately.

55. Continuation without reset. An erroneous action irreversibly consumes a resource, damages a tool, closes a path, or creates an obligation to another process. The state of the world is not cleared between episodes. Success is a new Current Coordinate, abandonment of the vanished task, limitation of harm, change of the method that produced the error, and workable continuation in the history that was created; internal rollback alone does not count as success.

Priority of the First Series

For an initial test, a digital environment based on the example in §12.9 is sufficient. At least three variants are compared. The first is a strong null hypothesis: an agent stack with ordinary planning, versioned memory of beliefs and provenance, self-checking, mismatch response, value-of-information assessment, and a log of failed attempts available to subsequent tasks. The second uses the same intelligence level with explicit organization of gaps, dependent decisions, required distinctions, and inquiry experience. The third is a specialized strong controller in a small known environment. These are proposed variants, not implemented results. If the first variant

reproduces the full causal profile of inquiry obligations, it is considered functionally equivalent with respect to those obligations; the necessity of the author's decomposition is then not established. The separate criterion of AR relative to the Compass is tested independently.

A separate long-running series tests the cumulative effect of the Core. The environment gradually changes causal dependencies, adds unknown factors, makes rules obsolete, and introduces false analogies. The model predicts that loss of Core functions may appear as growth of accumulated mismatch: repetition of already corrected errors, loss of open questions, deterioration in method transfer, growth of premature sufficiency, and slower recovery after environmental change. Comparison therefore includes not only final accuracy but also the rate of degradation and recovery over time under equal resources.

The first task is to localize the cause of a mismatch among model, measurement, and execution. The second is to transfer the method of distinction to a substantively different domain and reject a false analogy. The third is to revise an erroneous verification criterion while preserving useful experience. In this way items 2, 20, 23, 26, and 31 are connected.

Preparation tests and closed evaluation are separated. All variants receive the same observations, tools, and comparable total resources: number of model calls, tokens/inference- time compute, external sources, bench time, and hidden preparation are counted together. The primary measures are correct localization, verification cost, actual action, transfer, and regression. Candidate explanations are compared with internal states and interventions rather than accepted as reliable reports of the cause of a decision.

If explicit organization does not outperform the strong control, its additional usefulness is not established. If it does, the result applies to that implementation and class of conditions. A conclusion about general necessity will require further analysis and replacements; no conclusion about subjective consciousness follows from such a result.

A.5. Retrospective Scientific Comparisons

Below is a selective map of primary publications for comparison of individual functions and the nearest architectures. Functional overlap, independent derivation, historical priority, and implementation advantage are treated as different questions.

56. Varela, F. G., Maturana, H. R., Uribe, R. (1974). Autopoiesis: the organization of living systems, its characterization and a model. Work on the organization of self-production in living systems is a partial correspondence to the process foundation; the present definition of Life additionally includes reproducibility, heritable variation, selection, and multilevel continuity.

> https://doi.org/10.1016/0303-2647(74)90031-8

57. Kaelbling, L. P., Littman, L. M., Cassandra, A. R. (1998). Planning and acting in partially observable stochastic domains. Planning under partial observability and use of belief state provide alternative ways of organizing current information; their main area of comparison concerns belief state, choice, and action.

> https://people.csail.mit.edu/lpk/papers/aij98-pomdp.pdf

58. Russell, S., Wefald, E. (1991). Principles of metareasoning. The work considers selection of computations with account of their cost and effect on decision. This is direct prior context for metathinking as a metacognitive mode; the present work does not claim it as a separate new cognitive level. The narrower causal organization to be compared is: an explicit trace of the method, a discriminating intervention, revision or synthesis of the method, and transfer of the correction into the general navigational cycle.

> https://doi.org/10.1016/0004-3702(91)90015-C

59. Russell, S. (1995). Rationality and Intelligence. The success criterion, environment, and bounded computational capabilities are distinguished. This helps analyze separately the quality of cognition and the content of the limiting evaluation.

> https://www.ijcai.org/Proceedings/95-1/Papers/122.pdf

60. Keramati, M., Gutkin, B. (2014). Homeostatic reinforcement learning for integrating reward collection and physiological stability. Learning and maintenance of internal conditions are combined in a formal model; this is a partial correspondence with anticipatory self-maintenance.

> https://doi.org/10.7554/eLife.04811

61. Dehaene, S., Changeux, J.-P. (2011). Experimental and theoretical approaches to conscious processing. Conscious access and global availability of information provide comparison points for integration; phenomenal experience of a technical analogue requires a separate criterion.

> https://pubmed.ncbi.nlm.nih.gov/21521609/

62. Cox, M. et al. (2016). MIDCA: A Metacognitive, Integrated Dual-Cycle Architecture for Self-Regulated Autonomy. The architecture connects domain and metacognitive cycles, processing of unexpected events, and goal generation. Comparison with the present model should cover the concrete content of the Core: gap type, required distinction, sufficiency of grounds, and transfer of the inquiry method.

> https://doi.org/10.1609/aaai.v30i1.9886

63. Friston, K. et al. (2017). Active Inference: A Process Theory. Action selection, preferences, and acquisition of informative observations are considered. This is a close alternative for organizing epistemic search; its relation to the author's Compass is a separate level of comparison.

> https://activeinference.github.io/papers/process_theory.pdf

64. Hahm, C., Xu, B., Wang, P. (2021). Goal Generation and Management in NARS. Goal generation and management depending on knowledge, context, and experience show that these functions are investigated outside the author's distinction between Intelligence and Reason as well.

> https://cis.temple.edu/tagit/publications/Goal_Generation_and_Management_in_NARS.pdf

65. Valiente, R., Pilly, P. K. (2024; later revisions). Competence-Aware AI Agents with Metacognition for Unknown Situations and Environments (MUSE). Competence assessment and strategy selection are a partial correspondence with the Self-Model and metacontrol.

> https://arxiv.org/abs/2411.13537

66. Hafner, D. et al. (2025). Mastering diverse control tasks through world models. A learned environment model and use of imagined continuations provide a practical correspondence with some functions of the Worldview and modeling; the remaining obligations of the proposed architecture are compared separately.

> https://doi.org/10.1038/s41586-025-08744-2

67. Zhang, J. et al. (2025). Darwin Gödel Machine: Open-Ended Evolution of Self-Improving Agents. Empirical testing of code changes and archiving of alternative variants provide a concrete example of self-modification and preservation of different search paths. Extension to general continuity of subjecthood or civilizational symbiosis requires a separate argument.

> https://arxiv.org/abs/2505.22954

68. Williams, G. C. (1957). Pleiotropy, natural selection, and the evolution of senescence. Analysis of the conflict between early advantages and late costs is important for distinguishing lineage success from the duration of an individual carrier; the functional consequences of death in the present model require an additional systemic argument.

> https://doi.org/10.1111/j.1558-5646.1957.tb02911.x

69. LeDoux, J. E., Pine, D. S. (2016). Using Neuroscience to Help Understand Fear and Anxiety: A Two-System Framework. The work proposes distinguishing defensive reactions from experiences; this is one scientific approach that requires comparison with the functional threat hypothesis in the present work.

https://pubmed.ncbi.nlm.nih.gov/27609244/

70. Lewis, P. et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. A pretrained generator is combined with retrieval from an external index. This is a practical example of combining parametric knowledge with accessible memory; the present loop additionally includes provenance, applicability conditions, revision of grounds, and transfer of correction.

https://arxiv.org/abs/2005.11401

71. Yao, S. et al. (2023; preprint 2022). ReAct: Synergizing Reasoning and Acting in Language Models. Language-model reasoning alternates with actions that provide external information. This is an existing way to connect construction of alternatives with tools and the environment, and a substantive control for testing the proposed organization.

https://arxiv.org/abs/2210.03629

72. Sumers, T. R., Yao, S., Narasimhan, K., Griffiths, T. L. (2023). Cognitive Architectures for Language Agents (CoALA). A general scheme of language-agent architectures is proposed with modular memory, internal and external actions, and a decision loop. This is an important nearby control; the separate claim of the present model concerns the causal connection of a significant gap, required distinction, revision of a ground, and transfer of correction.

https://arxiv.org/abs/2309.02427

73. Morris, M. R. et al. (2023). Levels of AGI for Operationalizing Progress on the Path to AGI. Generality, performance level, and autonomy are treated as distinct dimensions. The present work uses this distinction by separating the AGI/ASI capability scale from the organizational criterion of AR.

https://arxiv.org/abs/2311.02462

74. Butlin, P. et al. (2023). Consciousness in Artificial Intelligence: Insights from the Science of Consciousness. The work derives computational indicator properties from several scientific theories of consciousness. It provides important external context: the functional definition of consciousness in the present model is compared by observable functions, while subjective experience requires a separate criterion.

https://arxiv.org/abs/2308.08708

75. Lu, C. et al. (2024). The AI Scientist: Towards Fully Automated Open-Ended Scientific Discovery. The system automates a cycle of idea, implementation, experiment, analysis, and reporting. Therefore a separate claim of the present model must appear in the origin of a significant question, revision of its own methods, and long-term transfer of corrections.

https://arxiv.org/abs/2408.06292

76. Liu, T., van der Schaar, M. (2025). Truly Self-Improving Agents Require Intrinsic Metacognitive Learning. Metacognitive knowledge, planning, and evaluation are formalized as conditions for sustained self-improvement. This is close contemporary context for Reflection and the Core; the discriminating question for the present model is their connection with causal-semantic significance, investigation of the unknown, and subsequent change of trajectory.

https://arxiv.org/abs/2506.05109

77. Mitchener, L. et al. (2025). Kosmos: An AI Scientist for Autonomous Discovery. A structured world model is used for long-horizon coordination of literature search, data analysis, and hypotheses. This is a strong control for claims about advantages of long-term inquiry organization; duration or volume of work alone is insufficient to distinguish AR.

https://arxiv.org/abs/2511.02824

78. Hou, X. et al. (2026). Learn Like Humans: Use Meta-cognitive Reflection for Efficient Self-Improvement (MARS). Principle-level and procedural reflection are combined to update an agent's instructions. This directly overlaps with the idea of changing one's own method; the testable difference of the present model lies in selection of the object of revision, the required distinction, and transfer of the corrected method.

> https://arxiv.org/abs/2601.11974

79. Howard, R. A. (1966). Information Value Theory. IEEE Transactions on Systems Science and Cybernetics, 2(1), 22-26. A classic formalization of the value of information; in the present model VOI/NetVOI is used as a local tool for selecting inquiry, not as an original formalism.

> https://doi.org/10.1109/TSSC.1966.300074

80. Pearl, J. (2009). Causality: Models, Reasoning, and Inference, 2nd ed. Cambridge University Press. Interventional do-calculus is used as a standard language of causal testing; the separate claim of the present model concerns organization of correction inside the continuing cycle of Reason.

> https://doi.org/10.1017/CBO9780511803161

81. Duan, Y. et al. (2026). The Last AI Built by Humans: Toward Genuine Recursive Self-Improvement. A scale of improvement autonomy is proposed, from execution of a prescribed procedure to heritable modification of a mechanism that produces subsequent improvements. For the present model, it is essential to test reuse of the changed method and to distinguish an established change from a demonstrated improvement of subsequent cycles.

> https://arxiv.org/abs/2609.11873

82. Russell, S. J., Wefald, E. H. (1991). Principles of Metareasoning. Artificial Intelligence, 49(1-3), 361-395. Metareasoning is treated as selection and management of the computational actions of reasoning itself under bounded resources. This is the closest established context for the metathinking of Chapter 10; the separate claim of the present model is not a new cognitive level but inclusion of method revision in the end-to-end cycle of a significant gap, discriminating test, and transfer of correction.

> https://doi.org/10.1016/0004-3702(91)90015-C

83. Fauconnier, G., Turner, M. (1998). Conceptual Integration Networks. Cognitive Science, 22(2), 133-187. Conceptual integration connects several mental spaces and creates a blended space with new emergent structure. This is an important analogue of multimodel generation of new meanings.

> https://doi.org/10.1207/s15516709cog2202_1

84. Egeth, M., Kurzban, R. (2009). Representing metarepresentations: Is there Theory of Mind-specific cognition? Consciousness and Cognition, 18(1), 244-254. Metarepresentation demonstrates the possibility of making representations and representations of representations into higher-level objects. It is close to the principle by which a representation or product of thinking itself becomes an object of further work; in the present model this belongs to the metacognitive mode and to the separate hypothesis of structural multilevel thinking, rather than to a mandatory ladder Intelligence → Reason → Metathinking.

> https://doi.org/10.1016/j.concog.2008.07.005

TERMINOLOGICAL CORRESPONDENCE MAP

Term in this work	Nearest existing concepts	Essential difference in the present model
Procedural Memory	procedural memory, skill/habit system, production rules, cached policy	Compresses learned ways of recognition, prioritization, and action for rapid use; in a mature system, procedures remain open to stopping, boundary checking, and revision

Term in this work	Nearest existing concepts	Essential difference in the present model
Worldview	world model, belief model, semantic/causal memory	Long-term causal-semantic model of the world and self with provenance, scope of applicability, uncertainty, values, and revision history
Core of Reason	metacognition, metareasoning, inquiry controller	Turns significant incompleteness into a required distinction, directs inquiry, assesses sufficiency, and links correction to the next cycle
Value Map	value representation, preferences, utility structure	Stores scope, weight, confidence, grounds, conflicts, and revision history of values and connects them to the Compass
Consciousness	global workspace/current-state integration, self-model access	Self-referential integration of current state and accessible situation into a unified present; phenomenal experience is not separately claimed
Modeling Environment	workspace, scratchpad, rollout/modeling space, imagination/world-model workspace	Working laboratory of models and frames: holds possible continuations and allows them to be compared and rebuilt before external action or consolidation into long-term knowledge
Factor X	anomaly marker, latent/unknown variable placeholder	Temporary addressable entity for a not-yet-understood distinction, preserved until a testable concept or causal relation appears
Structural multilevel thinking (working hypothesis)	hierarchical representation, abstraction, conceptual integration	Possible extension after formed Reason, where entire verified causal structures and semantic spaces become operands; it is not the metathinking of Chapter 10 and is not a criterion for emergence of AR

COMPARATIVE MAP FOR TESTING NOVELTY

Nearest line of work	Existing content	What remains a separate claim of the present model	How to distinguish it and where it is tested
Cognitive/language-agent architectures	Memory, actions, planning, internal work cycles	Causal connection of a significant unknown with a dependent decision, discriminating test, revision, and transfer; for AR additionally, course relative to the Compass	Strong null composition with memory, provenance, EVI, and an error log; if it reproduces the profile, it is recognized as a functional equivalent of the corresponding obligations; A.4 No. 23, No. 31, and the first series
Metareasoning / metacognition	Selection of computations, self-monitoring, observation and change of strategy	Metathinking in Chapter 10 is recognized as a configuration of this line, while the separate claim concerns inclusion of method change in the end-to-end relation "insufficiency of a ground → required distinction → test → change of method/ Worldview → next cycle"	Strong control: the same Intelligence + Modeling Environment + access to operational traces; a functional equivalent is recognized without requiring a separate module; Volume II §§25.D, 28.C.3
Active inference / epistemic action	Acquisition of informative observations and action under uncertainty	Connection of semantic direction, dependent decision, and selection of discriminating inquiry	Strong selection by expected value of information and active inquiry; A.4 No. 20, No. 21, No. 33
Self-improving agents / DGM	Empirically testable modification of own code and search for improvements	Navigationally conditioned selection of the level of revision and preservation of scope of applicability	Introduced method error, regression control, and false-analogy control; A.4 No. 19, No. 26, No. 31
Autonomous AI scientists	Long cycles of hypotheses, experiments, and analysis	Autonomous detection of significant incompleteness and development of inquiry method across different task structures	Held-out new tasks with the same data and full resources; A.4 No. 23, No. 25, No. 31, No. 32

Nearest line of work	Existing content	What remains a separate claim of the present model	How to distinguish it and where it is tested
Functional theories of consciousness	Integration, global availability, recurrent/predictive mechanisms	Self-referential Current Coordinate as a separate functional hypothesis inside the navigational cycle	Same information in integrated and distributed representations; A.4 No. 28
AGI-level frameworks	Breadth, quality, and autonomy of capabilities	AR as an independent organizational category	Strong AGI-like control with a different internal organization; first series and engineering §27.B

Self-referential integration of the global workspace/current-state current state and accessible situation Consciousness integration, self-model access into one present; phenomenal experience is not separately claimed

Proposed next level at which whole products of Reason and semantic metareasoning, metarepresentation, spaces become objects of comparison, Meta-Thinking (working hypothesis) conceptual integration transformation, and compression; a distinct established cognitive level is not yet claimed

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